



HEIDENHAIN



Rotary Encoders

October 2009

Rotary encoders from HEIDENHAIN

serve as measuring sensors for rotary motion, angular velocity and, when used in conjunction with mechanical measuring standards such as lead screws, for linear motion. Application areas include electrical motors, machine tools, printing machines, woodworking machines, textile machines, robots and handling devices, as well as various types of measuring, testing, and inspection devices.

The high quality of the sinusoidal incremental signals permits high interpolation factors for digital speed control.



Rotary Encoders for Separate Shaft Coupling



Rotary Encoders with Mounted Stator Coupling

This catalog supersedes all previous editions, which thereby become invalid. The basis for ordering from HEIDENHAIN is always the catalog edition valid when the contract is made.

Standards (ISO, EN, etc.) apply only where explicitly stated in the catalog.

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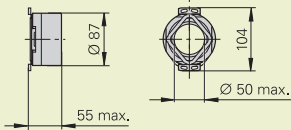
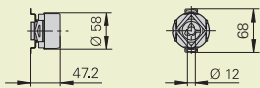
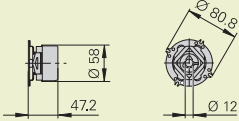
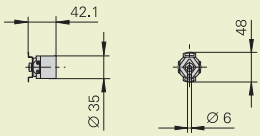
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Selection Guide

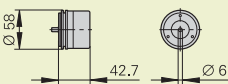
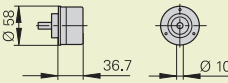
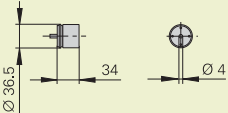
Rotary Encoders for Standard Applications

Rotary Encoders	Absolute Singleturn			Multiturn 4096 revolutions		
Interface	EnDat		SSI	PROFIBUS DP	EnDat	
Power supply	3.6 to 14 V	5 V	5 V or 10 to 30 V	9 to 36 V	3.6 to 14 V	5 V

With Mounted Stator Coupling

ECN/ERN 100 series 	ECN 125¹⁾ Positions/rev: 25 bits EnDat 2.2 / 22	ECN 113 Positions/rev: 13 bits EnDat 2.2 / 01	ECN 113⁵⁾ Positions/rev: 13 bits	–	–	–
ECN/EQN/ERN 400 series 	ECN 425 Positions/rev: 25 bits EnDat 2.2 / 22 ECN 413 Positions/rev: 13 bits EnDat 2.2 / 01	–	ECN 413 Positions/rev: 13 bits	ECN 413⁴⁾ Positions/rev: 13 bits	EQN 437 Positions/rev: 25 bits EnDat 2.2 / 22 EQN 425 Positions/rev: 13 bits EnDat 2.2 / 01	–
ECN/EQN/ERN 400 series with universal stator coupling 	ECN 425 Positions/rev: 25 bits EnDat 2.2 / 22 ECN 413 Positions/rev: 13 bits EnDat 2.2 / 01	–	ECN 413 Positions/rev: 13 bits	–	EQN 437 Positions/rev: 25 bits EnDat 2.2 / 22 EQN 425 Positions/rev: 13 bits EnDat 2.2 / 01	–
ECN/EQN/ERN 1000 series 	ECN 1023 Positions/rev: 23 bits EnDat 2.2 / 22 Functional Safety⁴⁾ ECN 1013 Positions/rev: 13 bits EnDat 2.2 / 01	–	–	–	EQN 1035 Positions/rev: 23 bits EnDat 2.2 / 22 Functional Safety⁴⁾ EQN 1025 Positions/rev: 13 bits EnDat 2.2 / 01	–

For Separate Shaft Coupling

ROC/ROQ/ROD 400 RIC/RIQ 400 series with synchro flange 	ROC 425 Positions/rev: 25 bits EnDat 2.2 / 22 ROC 413 Positions/rev: 13 bits EnDat 2.2 / 01	RIC 418 Positions/rev: 18 bits EnDat 2.1 / 01	ROC 413 Positions/rev: 13 bits	ROC 413 Positions/rev: 13 bits	ROQ 437 Positions/rev: 25 bits EnDat 2.2 / 22 ROQ 425 Positions/rev: 13 bits EnDat 2.2 / 01	RIQ 430 Positions/rev: 18 bits EnDat 2.1 / 01
ROC/ROQ/ROD 400 RIC/RIQ 400 series with clamping flange 	ROC 425 Positions/rev: 25 bits EnDat 2.2 / 22 ROC 413 Positions/rev: 13 bits EnDat 2.2 / 01	RIC 418 Positions/rev: 18 bits EnDat 2.1 / 01	ROC 413 Positions/rev: 13 bits	ROC 413 Positions/rev: 13 bits	ROQ 437 Positions/rev: 25 bits EnDat 2.2 / 22 ROQ 425 Positions/rev: 13 bits EnDat 2.2 / 01	RIQ 430 Positions/rev: 18 bits EnDat 2.1 / 01
ROC/ROQ/ROD 1000 series 	ROC 1023 Positions/rev: 23 bits EnDat 2.2 / 22 Functional Safety⁴⁾ ROC 1013 Positions/rev: 13 bits EnDat 2.2 / 01	–	–	–	ROQ 1035 Positions/rev: 23 bits EnDat 2.2 / 22 Functional Safety⁴⁾ ROQ 1025 Positions/rev: 13 bits EnDat 2.2 / 01	–

¹⁾ Power supply: 3.6 to 5.25 V

²⁾ Up to 10000 signal periods through integrated 2-fold interpolation

³⁾ Up to 36000 signal periods through integrated 5/10-fold interpolation (higher interpolation upon request)

			Incremental			
	SSI	PROFIBUS DP	 TTL	 TTL	 HTL	 1 V _{PP}
	5 V or 10 to 30 V	9 to 36 V	5 V	10 to 30 V	10 to 30 V	5 V

	–	–	ERN 120 1 000 to 5 000 lines	–	ERN 130 1 000 to 5 000 lines	ERN 180 1 000 to 5 000 lines
	EQN 425 Positions/rev: 13 bits	EQN 425 ⁴⁾ Positions/rev: 13 bits	ERN 420 250 to 5 000 lines	ERN 460 250 to 5 000 lines	ERN 430 250 to 5 000 lines	ERN 480 1 000 to 5 000 lines
	EQN 425 Positions/rev: 13 bits	–	ERN 420 250 to 5 000 lines	ERN 460 250 to 5 000 lines	ERN 430 250 to 5 000 lines	ERN 480 1 000 to 5 000 lines
	–	–	ERN 1020 100 to 3 600 lines ERN 1070 ³⁾ 1 000/2 500/ 3 600 lines	–	ERN 1030 100 to 3 600 lines	ERN 1080 100 to 3 600 lines

	ROQ 425 Positions/rev: 13 bits	ROQ 425 Positions/rev: 13 bits 4 096 revolutions	ROD 426 0 to 5 000 lines ²⁾	ROD 466 0 to 5 000 lines ²⁾	ROD 436 50 to 5 000 lines	ROD 486 1 000 to 5 000 lines
	ROQ 425 Positions/rev: 13 bits	ROQ 425 Positions/rev: 13 bits 4 096 revolutions	ROD 420 50 to 5 000 lines	–	ROD 430 50 to 5 000 lines	ROD 480 1 000 to 5 000 lines
	–	–	ROD 1020 100 to 3 600 lines ROD 1070 ³⁾ 1 000/2 500/ 3 600 lines	–	ROD 1030 100 to 3 600 lines	ROD 1080 100 to 3 600 lines



22



24



28



32



36



40



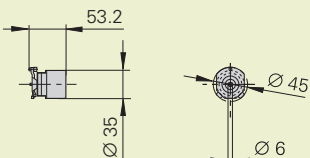
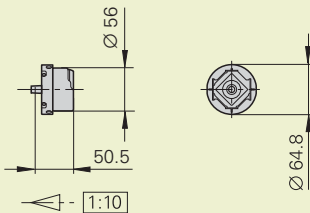
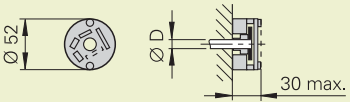
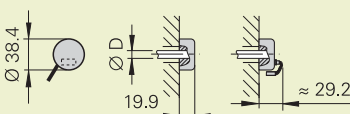
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⁴⁾ Upon request

⁵⁾ 10 to 30 V via connecting cable with voltage converter

Selection Guide

Rotary Encoders for Integration in Motors

Rotary Encoders		Absolute Singletum	Multiturn 4096 revolutions
With Integral Bearing and Mounted Stator Coupling			
Interface		EnDat	EnDat
Power supply		3.6 to 14 V	3.6 to 14 V
ECN/EQN/ERN 1100 series 		ECN 1123 Positions/rev: 23 bits EnDat 2.2 / 22 Functional safety upon request ECN 1113 Positions/rev: 13 bits EnDat 2.2 / 01	EQN 1135 Positions/rev: 23 bits EnDat 2.2 / 22 Functional safety upon request EQN 1125 Positions/rev: 13 bits EnDat 2.2 / 01
ECN/EQN/ERN 1300 series 		ECN 1325 Positions/rev: 25 bits EnDat 2.2 / 22 Functional safety upon request ECN 1313 Positions/rev: 13 bits EnDat 2.2 / 01	EQN 1337 Positions/rev: 25 bits EnDat 2.2 / 22 Functional safety upon request EQN 1325 Positions/rev: 13 bits EnDat 2.2 / 01
Without Integral Bearing			
Interface		EnDat	EnDat
Power supply		5 V	5 V
ECI/EQI 1100 series 		ECI 1118 Positions/rev: 18 bits EnDat 2.1 / 21 ECI 1118 Positions/rev: 18 bits EnDat 2.1 / 01	EQI 1130 Positions/rev: 18 bits EnDat 2.1 / 21 EQI 1130 Positions/rev: 18 bits EnDat 2.1 / 01
ECI/EQI 1300 series 		ECI 1319 Positions/rev: 19 bits EnDat 2.1 / 01	EQI 1331 Positions/rev: 19 bits EnDat 2.1 / 01
ERO 1200 series 		—	—
ERO 1400 series 		—	—

¹⁾ 8192 signal periods through integrated 2-fold interpolation

²⁾ 37500 signal periods through integrated 5/10/20/25-fold interpolation

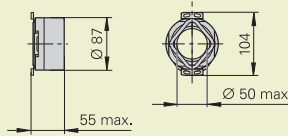
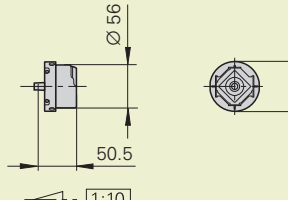
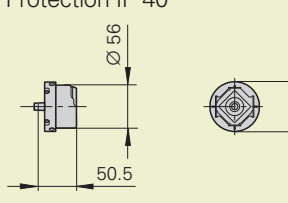
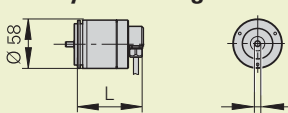
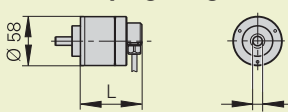
Incremental		
		
	5 V	5 V
ERN 1120 1 024 to 3 600 lines		ERN 1180 1 024 to 3 600 lines ERN 1185 512/2 048 lines Z1 track for sine commutation
ERN 1321 1 024 to 4 096 lines ERN 1326 1 024 to 4 096 lines ¹⁾ 3 TTL signals for block commutation		ERN 1381 512 to 4 096 lines ERN 1387 2 048 lines Z1 track for sine commutation
		
	5 V	5 V
–		–
–		–
ERO 1225 1 024/2 048 lines		ERO 1285 1 024/2 048 lines
ERO 1420 512 to 1 024 lines ERO 1470 1 000/1 500 ²⁾		ERO 1480 512 to 1 024 lines

These rotary encoders are described in the ***Position Encoders for Servo Drives*** catalog.



Selection Guide

Rotary Encoders for Special Applications

Rotary Encoders		Absolute Singleturn		Multiturn 4096 revolutions	
For Drive Control in Elevators					
Interface	EnDat		SSI	EnDat	SSI
Power supply	3.6 to 14 V 5 V		5 V or 10 to 30 V		
ECN/ERN 100 series Protection IP 64 	ECN 125¹⁾ Positions/rev: 25 bits EnDat 2.2 / 22	ECN 113 Positions/rev: 13 bits EnDat 2.2 / 01	ECN 113⁵⁾ Positions/rev: 13 bits	–	–
ECN/EQN/ERN 400 series Protection IP 64 	ECN 425 Positions/rev: 25 bits EnDat 2.2 / 22 ECN 413 Positions/rev: 13 bits EnDat 2.2 / 01	–	–	–	–
ECN/ERN 1300 series Protection IP 40 	ECN 1325 Positions/rev: 25 bits EnDat 2.2 / 22 Functional safety upon request ECN 1313 Positions/rev: 13 bits EnDat 2.2 / 01	–	–	–	–
For Potentially Explosive Atmospheres ⁴⁾ in zones 1, 2, 21 and 22					
Interface	EnDat		SSI	EnDat	SSI
Power supply	5 V		5 V	5 V	5 V
ROC/ROQ/ROD 400 series with synchro flange 	ROC 413 Positions/rev: 13 bits EnDat 2.1 / 01		ROC 413 Positions/rev: 13 bits	ROQ 425 Positions/rev: 13 bits EnDat 2.1 / 01	ROQ 425 Positions/rev: 13 bits
ROC/ROQ/ROD 400 series with clamping flange 	ROC 413 Positions/rev: 13 bits EnDat 2.1 / 01		ROC 413 Positions/rev: 13 bits	ROQ 425 Positions/rev: 13 bits EnDat 2.1 / 01	ROQ 425 Positions/rev: 13 bits

¹⁾ Power supply: 3.6 to 5.25 V

²⁾ Up to 10000 signal periods through integrated 2-fold interpolation

³⁾ 8192 signal periods through integrated 2-fold interpolation

⁴⁾ Versions with blind hollow shaft available upon request

⁵⁾ 10 to 30 V via connecting cable with voltage converter

Incremental				
	 TTL	 TTL	 HTL	 1 V _{PP}
	5 V	10 to 30 V	10 to 30 V	5 V
	ERN 120 1 000 to 5 000 lines	–	ERN 130 1 000 to 5 000 lines	ERN 180 1 000 to 5 000 lines
	ERN 421 1 024 to 5 000 lines ²⁾	–	–	ERN 487 2 048 lines Z1 track for sine commutation
	ERN 1321 1 024 to 5 000 lines ERN 1326 ³⁾ 1 024 to 4 096 lines 3 TTL signals for block commutation	–	–	ERN 1381 512 to 4 096 lines ERN 1387 2 048 lines Z1 track for sine commutation
	 TTL	 TTL	 HTL	 1 V _{PP}
	5 V	10 to 30 V	10 to 30 V	5 V
	ROD 426 1 000 to 5 000 lines	ROD 466 1 000 to 5 000 lines	ROD 436 1 000 to 5 000 lines	ROD 486 1 000 to 5 000 lines
	ROD 420 50 to 5 000 lines	–	ROD 430 50 to 5 000 lines	ROD 480 1 000 to 5 000 lines



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See product overview:
Rotary Encoders for the Elevator Industry



See catalog:
Encoders for Servo Drives



See product overview:
Rotary Encoders for Potentially Explosive Atmospheres



Measuring Principles

Measuring Standard

HEIDENHAIN encoders with **optical scanning** incorporate measuring standards of periodic structures known as graduations. These graduations are applied to a carrier substrate of glass or steel.

These precision graduations are manufactured in various photolithographic processes. Graduations are fabricated from:

- extremely hard chromium lines on glass,
- matte-etched lines on gold-plated steel tape, or
- three-dimensional structures on glass or steel substrates.

The photolithographic manufacturing processes developed by HEIDENHAIN produce grating periods of typically 50 μm to 4 μm .

These processes permit very fine grating periods and are characterized by a high definition and homogeneity of the line edges. Together with the photoelectric scanning method, this high edge definition is a precondition for the high quality of the output signals.

The master graduations are manufactured by HEIDENHAIN on custom-built high-precision ruling machines.

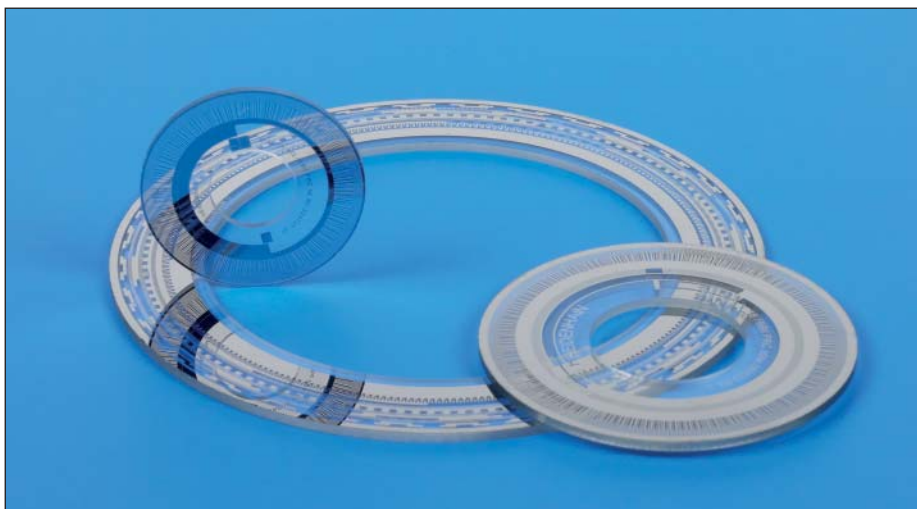
Encoders using the **inductive scanning principle** have graduation structures of copper. The graduation is applied to a carrier material for printed circuits.

Measurement Methods

With the **absolute measuring method**, the position value is available from the encoder immediately upon switch-on and can be called at any time by the subsequent electronics. There is no need to move the axes to find the reference position. The absolute position information is read from the **grating on the graduated disk**, which is designed as a serial code structure or—as on the ECN 100—consists of several parallel graduation tracks.

A separate incremental track (on the ECN 100 the track with the finest grating period) is interpolated for the position value and at the same time is used to generate an optional incremental signal.

In **singleturn encoders** the absolute position information repeats itself with every revolution. **Multiturn encoders** can also distinguish between revolutions.



Circular graduations of absolute rotary encoders

With the **incremental measuring method**, the graduation consists of a periodic grating structure. The position information is obtained **by counting** the individual increments (measuring steps) from some point of origin. Since an absolute reference is required to ascertain positions, the graduated disks are provided with an additional track that bears a **reference mark**.

The absolute position established by the reference mark is gated with exactly one measuring step.

The reference mark must therefore be scanned to establish an absolute reference or to find the last selected datum.



Circular graduations of incremental rotary encoders

Scanning Methods

Photoelectric scanning

Most HEIDENHAIN encoders operate using the principle of photoelectric scanning. Photoelectric scanning of a measuring standard is contact-free, and as such free of wear. This method detects even very fine lines, no more than a few microns wide, and generates output signals with very small signal periods.

The ECN, EQN, ERN and ROC, ROQ, ROD rotary encoders use the imaging scanning principle.

Put simply, the imaging scanning principle functions by means of projected-light signal generation: two graduations with equal grating periods are moved relative to each other—the scale and the scanning reticle. The carrier material of the scanning reticle is transparent, whereas the graduation on the measuring standard may be applied to a transparent or reflective surface.

When parallel light passes through a grating, light and dark surfaces are projected at a certain distance. An index grating with the same grating period is located here. When the two gratings move relative to each other, the incident light is modulated. If the gaps in the gratings are aligned, light passes through. If the lines of one grating coincide with the gaps of the other, no light passes through. Photovoltaic cells convert these variations in light intensity into nearly sinusoidal electrical signals. Practical mounting tolerances for encoders with the imaging scanning principle are achieved with grating periods of 10 µm and larger.

The ROC/ROQ 400/1000 and ECN/ EQN 400/1000 absolute rotary encoders with optimized scanning have a single large photosensor instead of a group of individual photoelements. Its structures have the same width as that of the measuring standard. This makes it possible to do without the scanning reticle with matching structure.

Other scanning principles

ECI/EQI and RIC/RIQ rotary encoders operate according to the inductive measuring principle. Here, graduation structures modulate a high-frequency signal in its amplitude and phase. The position value is always formed by sampling the signals of all receiver coils distributed evenly around the circumference.

Accuracy

The accuracy of position measurement with rotary encoders is mainly determined by

- the directional deviation of the radial grating,
- the eccentricity of the graduated disk to the bearing,
- the radial deviation of the bearing,
- the error resulting from the connection with a shaft coupling (on rotary encoders with stator coupling this error lies within the system accuracy),
- the interpolation error during signal processing in the integrated or external interpolation and digitizing electronics.

For **incremental rotary encoders** with line counts up to 5000:

The maximum directional deviation at 20 °C ambient temperature and slow speed (scanning frequency between 1 kHz and 2 kHz) lies within

$$\pm \frac{18^\circ \text{ mech.} \cdot 3600}{\text{Line count } z} \text{ [angular seconds]}$$

which equals

$$\pm \frac{1}{20} \text{ grating period.}$$

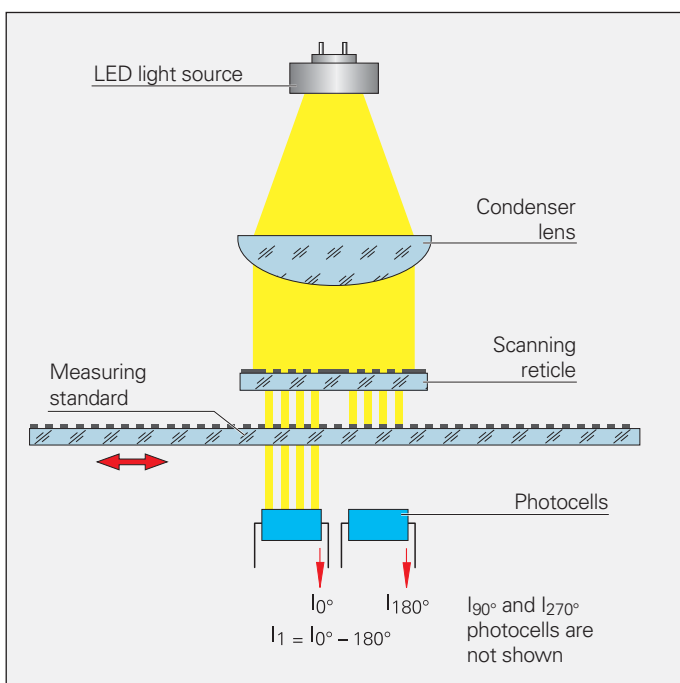
The ROD rotary encoders generate 6000 to 10000 signal periods per revolution through signal doubling. The line count is important for the system accuracy.

The accuracy of absolute position values from **absolute rotary encoders** is given in the specifications for each model.

For absolute rotary encoders with **complementary incremental signals**, the accuracy depends on the line count:

Line count	Accuracy
16	± 480 angular seconds
32	± 280 angular seconds
512	± 60 angular seconds
2048	± 20 angular seconds

The above accuracy data refer to incremental measuring signals at an ambient temperature of 20 °C and at slow speed.



Photoelectric scanning according to the imaging scanning principle

Mechanical Design Types and Mounting

Rotary Encoders with Stator Coupling

ECN/EQN/ERN rotary encoders have integrated bearings and a mounted stator coupling. They compensate radial runout and alignment errors without significantly reducing the accuracy. The encoder shaft is directly connected with the shaft to be measured. During angular acceleration of the shaft, the stator coupling must absorb only that torque caused by friction in the bearing. The stator coupling permits axial motion of the measured shaft:

ECN/EQN/ERN 400: $\pm 1 \text{ mm}$

ECN/EQN/ERN 1000: $\pm 0.5 \text{ mm}$

ECN/ERN 100: $\pm 1.5 \text{ mm}$

Mounting

The rotary encoder is slid by its hollow shaft onto the measured shaft, and the rotor is fastened by two screws or three eccentric clamps. For rotary encoders with hollow through shaft, the rotor can also be fastened at the end opposite to the flange. Rotary encoders of the ECN/EQN/ERN 1300 series with taper shaft are particularly well suited for repeated mounting (see the brochure *Position Encoders for Servo Drives*). The stator is connected without a centering collar on a flat surface. The **universal stator coupling** of the ECN/EQN/ERN 400 permits versatile mounting, e.g. by its thread provided for fastening it from outside to the motor cover.

Dynamic applications require the highest possible natural frequencies f_N of the system (also see *General Mechanical Information*). This is attained by connecting the shafts on the flange side and fastening the coupling by four cap screws or, on the ECN/EQN/ERN 1000, with special washers.

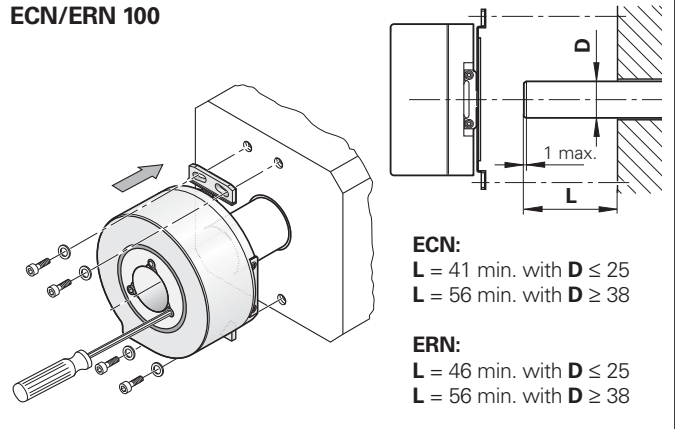
Natural frequency f_E with coupling fastened by 4 screws

	Stator coupling	Cable	Flange socket	
			Axial	Radial
ECN/EQN/ERN 400	Standard Universal	1550 Hz 1400 Hz ¹⁾	1500 Hz 1400 Hz	1000 Hz 900 Hz
ECN/ERN 100		1000 Hz	–	400 Hz
ECN/EQN/ERN 1000		1800 Hz ²⁾	–	–

¹⁾ Also when fastening with 2 screws

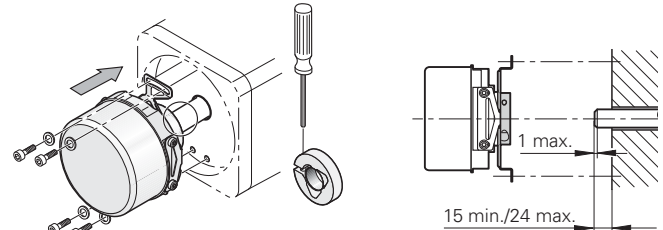
²⁾ Also when fastening with 2 screws and washers

ECN/ERN 100

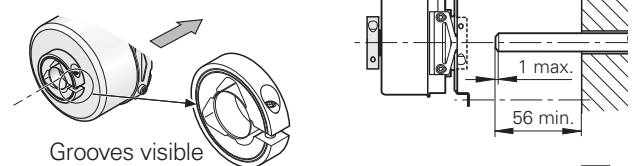


ECN/EQN/ERN 400 e.g. with standard coupling

Blind hollow shaft



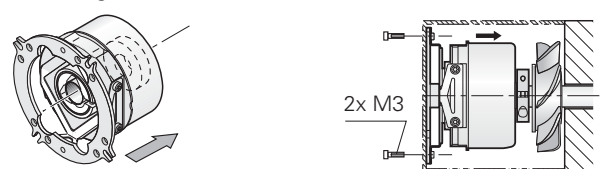
Hollow through shaft



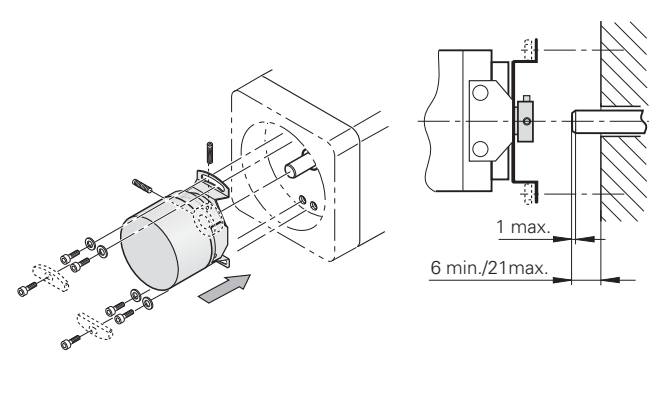
ECN/EQN/ERN 400

e.g. with universal stator coupling

Hollow through shaft



ECN/EQN/ERN 1000



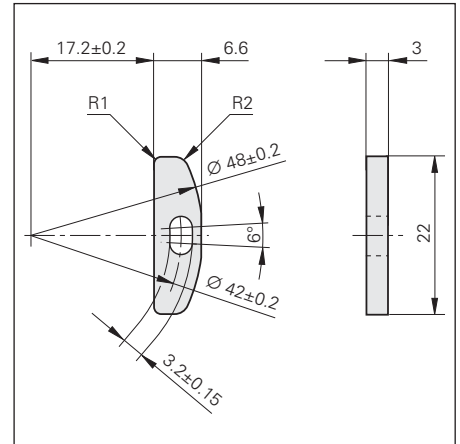
Mounting accessories

Washer

for ECN/EQN/ERN 1000

For increasing the natural frequency f_E and mounting with only two screws.

ID 334653-01

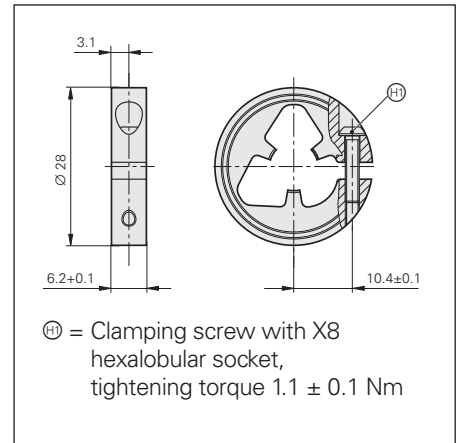


Shaft clamp ring

for ECN/EQN/ERN 400

By using a second shaft clamp ring, the mechanically permissible speed of rotary encoders with hollow through shaft can be increased to a maximum of 12 000 min⁻¹.

ID 540741-xx



If the encoder shaft is subject to high loads, for example from friction wheels, pulleys, or sprockets, HEIDENHAIN recommends mounting the ECN/EQN/ERN 400 with a bearing assembly.

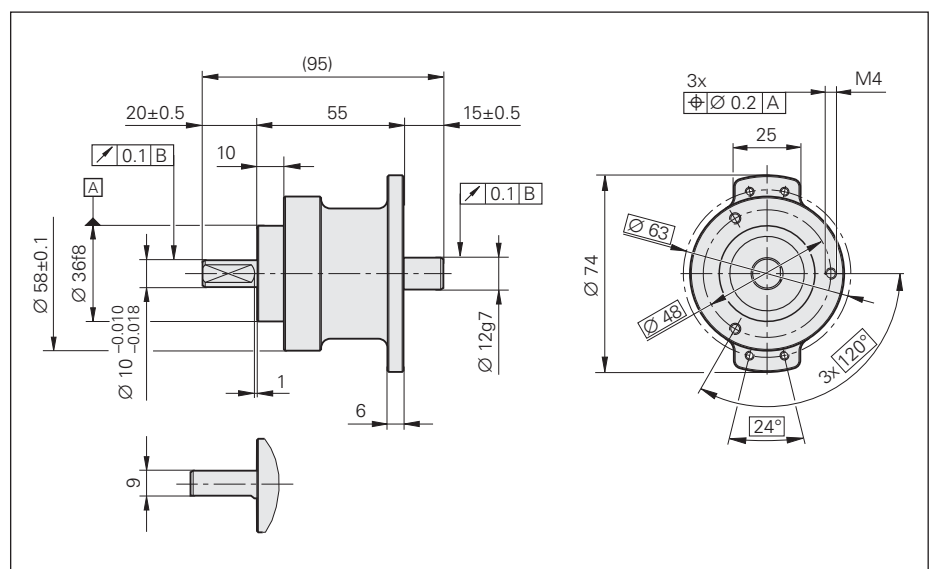
Bearing assembly

for ERN/ECN/EQN 400 with blind hollow shaft

ID 574185-03

	Bearing assembly
Permissible speed n	≤ 6000 min ⁻¹
Shaft load	Axial: 150 N; Radial: 350 N
Operating temperature	−40 °C to 100 °C

The bearing assembly is capable of absorbing large radial shaft loads. It prevents overload of the encoder bearing. On the encoder side, the bearing assembly has a stub shaft with 12 mm diameter and is well suited for the ERN/ECN/EQN 400 encoders with blind hollow shaft. Also, the threaded holes for fastening the stator coupling are already provided. The flange of the bearing assembly has the same dimensions as the clamping flange of the ROD 420/430 series. The bearing assembly can be fastened through the threaded holes on its face or with the aid of the mounting flange or the mounting bracket (see page 15).



Torque supports for the ERN/ECN/ EQN 400

For simple applications with the ERN/ECN/ EQN 400, the stator coupling can be replaced by torque supports. The following kits are available:

Wire torque support

The stator coupling is replaced by a flat metal ring to which the provided wire is fastened.

ID 510955-01

Pin torque support

Instead of a stator coupling, a “synchro flange” is fastened to the encoder. A pin serving as torque support is mounted either axially or radially on the flange. As an alternative, the pin can be pressed in on the customer's surface, and a guide can be inserted in the encoder flange for the pin.

ID 510861-01



General accessories

Screwdriver bits

For HEIDENHAIN shaft couplings

For ExN 100/400/1000 shaft couplings

For ERO shaft couplings

Screwdriver

Adjustable torque

0.2 Nm to 1.2 Nm

ID 350379-04

1 Nm to 5 Nm

ID 350379-05

Width across flats	Length	ID
1,5	70 mm	350378-01
1.5 (ball head)		350378-02
2		350378-03
2 (ball head)		350378-04
2.5		350378-05
3 (ball head)		350378-08
4		350378-07
4 (with dog point) ¹⁾		350378-14
TX8	89 mm	350378-11
	152 mm	350378-12



¹⁾ For screws as per DIN 6912 (low head screw with pilot recess)

Rotary Encoders for Separate Shaft Coupling

ROC/ROQ/ROD and **RIC/RIQ** rotary encoders have integrated bearings and a solid shaft. The encoder shaft is connected with the measured shaft through a separate rotor coupling. The coupling compensates axial motion and misalignment (radial and angular offset) between the encoder shaft and measured shaft. This relieves the encoder bearing of additional external loads that would otherwise shorten its service life. Diaphragm and metal bellows couplings designed to connect the rotor of the ROC/ROQ/ROD/RIC/RIQ encoders are available (see *Shaft Couplings*).

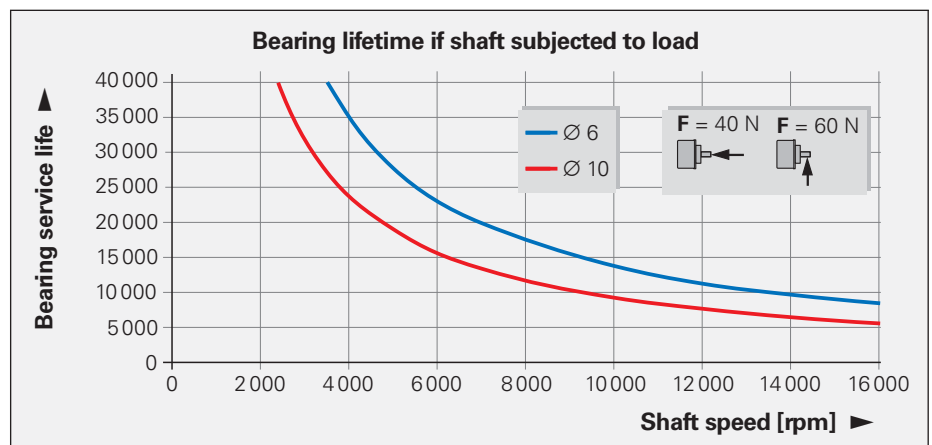
ROC/ROQ/ROD 400 and RIC/RIQ 400 series rotary encoders permit high bearing loads (see diagram). They can therefore also be mounted directly onto mechanical transfer elements such as gears or friction wheels.

If the encoder shaft is subject to relatively high loads, for example from friction wheels, pulleys, or sprockets, HEIDENHAIN recommends mounting the ECN/EQN/ERN 400 with a bearing assembly.



Bearing service life in the ROC/ROQ/ROD 400 and RIC/RIQ 400

The lifetime of the shaft bearing depends on the shaft load, the shaft speed, and the point of force application. The values given in the specifications for the shaft load are valid for all permissible speeds, and do not limit the bearing lifetime. The diagram shows an example of the different bearing lifetimes to be expected at further loads. The different points of force application of shafts with 6 mm and 10 mm diameters have an effect on the bearing lifetime.

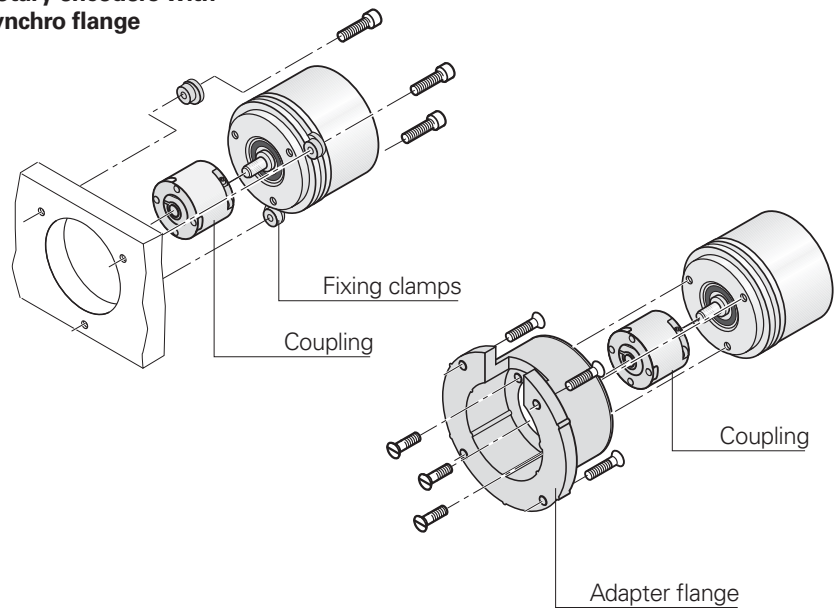


Rotary encoders with synchro flange

Mounting

- by the synchro flange with three fixing clamps or
- by fastening threads holes on the encoder flange to an adapter flange (for ROC/ROQ/ROD 400 or RIC/RIQ 400).

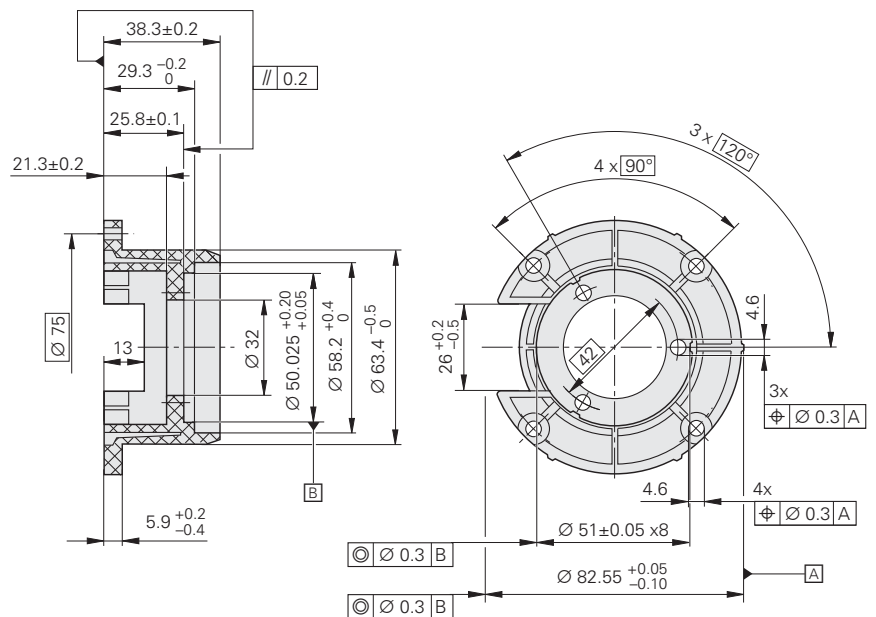
Rotary encoders with synchro flange



Mounting accessories

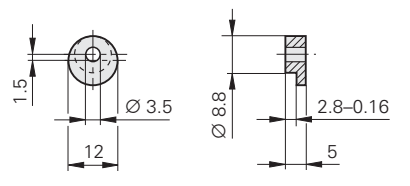
Adapter flange

(electrically nonconducting)
ID 257 044-01



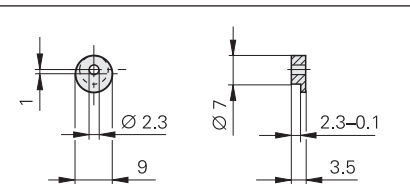
Fixing clamps

For ROC/ROQ/ROD 400 and
RIC/RIQ 400 series
(3 per encoder)
ID 200 032-01



Fixing clamps

For ROC/ROQ/ROD 1000 series
(3 per encoder)
ID 200 032-02



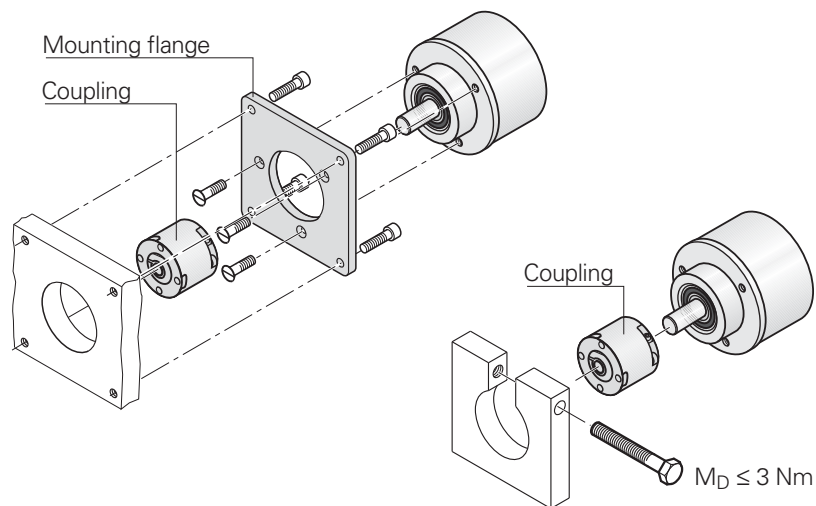
Rotary encoders with clamping flange

Mounting

- by fastening the threaded holes on the encoder flange to an adapter flange or
- by clamping at the clamping flange.

The centering collar on the synchro flange or clamping flange serves to center the encoder.

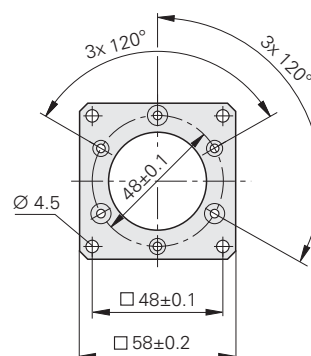
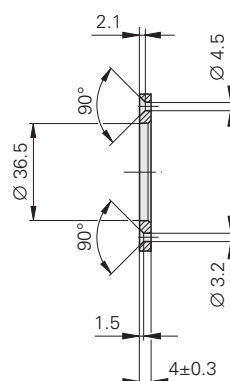
ROC/ROQ/ROD 400 with clamping flange



Mounting accessories

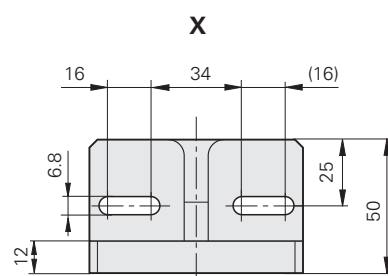
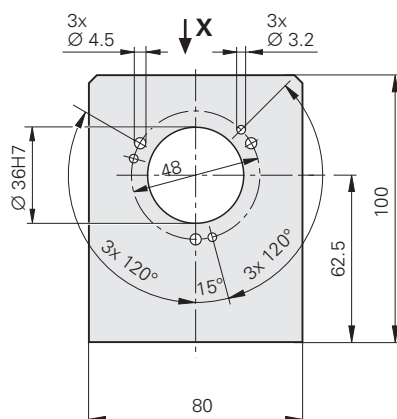
Mounting flange

ID 201 437-01



Mounting bracket

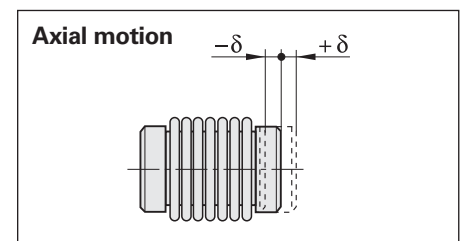
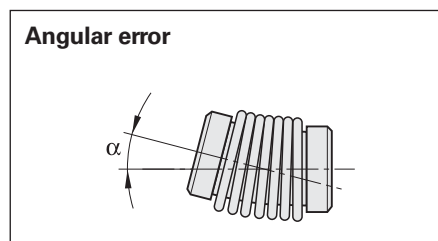
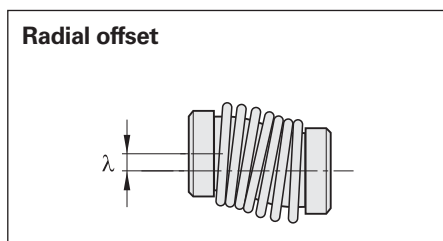
ID 581 296-01



Shaft Couplings

	ROC/ROQ/ROD 400				ROD 1000
	Diaphragm coupling				Metal bellows coupling
	K 14	with galvanic isolation K 17/01 K 17/06	K 17/02 K 17/04 K 17/05	K 17/03	18EBN3
Hub bore	6/6 mm	6/6 mm 6/5 mm	6/10 mm 10/10 mm 6/9.52 mm	10/10 mm	4/4 mm
Kinematic transfer error*	± 6"	± 10"			± 40"
Torsional rigidity	500 $\frac{\text{Nm}}{\text{rad}}$	150 $\frac{\text{Nm}}{\text{rad}}$	200 $\frac{\text{Nm}}{\text{rad}}$	300 $\frac{\text{Nm}}{\text{rad}}$	60 $\frac{\text{Nm}}{\text{rad}}$
Max. torque	0.2 Nm	0.1 Nm		0.2 Nm	0.1 Nm
Max. radial offset λ	≤ 0.2 mm	≤ 0.5 mm			≤ 0.2 mm
Max. angular error α	≤ 0.5°	≤ 1°			≤ 0.5°
Max. axial motion δ	≤ 0.3 mm	≤ 0.5 mm			≤ 0.3 mm
Moment of inertia (approx.)	6 · 10 ⁻⁶ kgm ²	3 · 10 ⁻⁶ kgm ²		4 · 10 ⁻⁶ kgm ²	0.3 · 10 ⁻⁶ kgm ²
Permissible speed	16000 min ⁻¹	16000 min ⁻¹			12000 min ⁻¹
Torque for locking screws (approx.)	1.2 Nm				0.8 Nm
Weight	35 g	24 g	23 g	27.5 g	9 g

*With radial offset $\lambda = 0.1 \text{ mm}$, angular error $\alpha = 0.15 \text{ mm over } 100 \text{ mm} \triangleq 0.09^\circ$ valid up to 50 °C



Mounting accessories

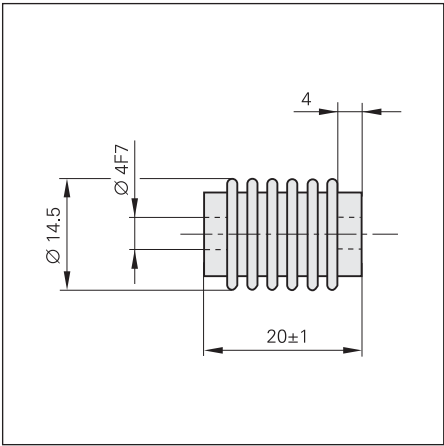
Screwdriver bits

Screwdriver

See page 14

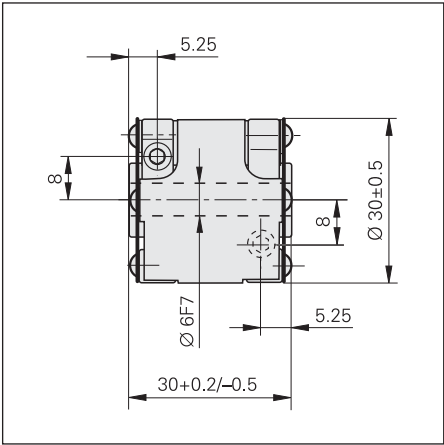
Metal bellows coupling 18 EBN 3

For ROC/ROQ/ROD 1000 series
With **4 mm shaft diameter**
ID 200393-02



Diaphragm coupling K 14

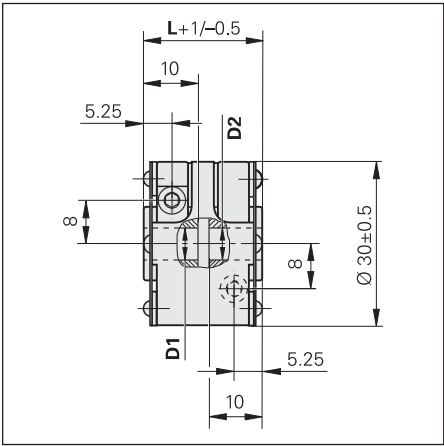
For ROC/ROQ/ROD 400 and RIC/RIQ 400 series
With **6 mm shaft diameter**
ID 293328-01



Recommended fit for
the mating shaft: h6

Diaphragm coupling K 17 with galvanic isolation

For ROC/ROQ/ROD 400 and RIC/RIQ 400 series
With **6 or 10 mm shaft diameter**
ID 296746-xx



K 17 Variant	D1	D2	L
01	Ø 6 F7	Ø 6 F7	22 mm
02	Ø 6 F7	Ø 10 F7	22 mm
03	Ø 10 F7	Ø 10 F7	30 mm
04	Ø 10 F7	Ø 10 F7	22 mm
05	Ø 6 F7	Ø 9.52 F7	22 mm
06	Ø 5 F7	Ø 6 F7	22 mm

Dimensions in mm

Tolerancing ISO 8015
ISO 2768 - m H
< 6 mm: ±0.2 mm

General Mechanical Information

UL certification

All rotary encoders and cables in this brochure comply with the -UL safety regulations "cULus" for the USA and the "CSA" safety regulations for Canada. They are listed under file no. E205635.

Acceleration

Encoders are subject to various types of acceleration during operation and mounting.

• Vibration

The encoders are qualified on a test stand to operate with the specified acceleration values from 55 to 2000 Hz in accordance with EN 60068-2-6. However, if the application or poor mounting cause long-lasting resonant vibration, it can limit performance or even damage the encoder. **Comprehensive tests of the entire system are required.**

• Shock

The encoders are qualified on a test stand to operate with the specified acceleration values and duration in accordance with EN 60068-2-27. This does not include **permanent shock loads, which must be tested in the application.**

- The **maximum angular acceleration** is 10^5 rad/s^2 (DIN 32878). This is the highest permissible acceleration at which the rotor will rotate without damage to the encoder.

Humidity

The max. permissible relative humidity is 75 %. 95 % is permissible temporarily. Condensation is not permissible.

Magnetic fields

Magnetic fields > 30 mT can impair the proper function of encoders. If required, please contact HEIDENHAIN, Traunreut.

RoHS

HEIDENHAIN has tested the products for harmlessness of the materials as per European Directives 2002/95/EC on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS) and 2002/96/EC (WEEE) on waste electrical and electronic equipment. For a Manufacturer Declaration on RoHS, please refer to your sales agency.

Natural frequencies

The rotor and the couplings of ROC/ROQ/ROD and RIC/RIQ rotary encoders, as also the stator and stator coupling of ECN/EQN/ERN rotary encoders, form a single vibrating spring-mass system.

The **natural frequency f_N** should be as high as possible. A prerequisite for the highest possible natural frequency on **ROC/ROQ/ROD rotary encoders** is the use of a diaphragm coupling with a high torsional rigidity C (see *Shaft Couplings*).

$$f_N = \frac{1}{2 \cdot \pi} \cdot \sqrt{\frac{C}{I}}$$

f_N : Natural frequency in Hz

C: Torsional rigidity of the coupling in Nm/rad

I: Moment of inertia of the rotor in kgm^2

ECN/EQN/ERN rotary encoders with their stator couplings form a vibrating spring-mass system whose **natural frequency f_N** should be as high as possible. If radial and/or axial acceleration forces are added, the stiffness of the encoder bearings and the encoder stators are also significant. If such loads occur in your application, HEIDENHAIN recommends consulting with the main facility in Traunreut.

Protection against contact (EN 60529)

After encoder installation, all rotating parts must be protected against accidental contact during operation.

Protection (EN 60529)

Unless otherwise indicated, all rotary encoders meet protection standard IP 64 (ExN/ROx 400: IP 67) according to EN 60529. This includes housings, cable outlets and flange sockets when the connector is fastened.

The **shaft inlet** provides protection to IP 64. Splash water should not contain any substances that would have harmful effects on the encoder parts. If the standard protection of the shaft inlet is not sufficient (such as when the encoders are mounted vertically), additional labyrinth seals should be provided.

Many encoders are also available with protection to class IP 66 for the shaft inlet. The sealing rings used to seal the shaft are subject to wear due to friction, the amount of which depends on the specific application.

Expendable parts

Encoders from HEIDENHAIN are designed for a long service life. Preventive maintenance is not required. They contain components that are subject to wear, depending on the application and manipulation. These include in particular moving cables.

On encoders with integral bearing such components are also the bearings, shaft sealing rings on rotary encoders and angle encoders and sealing lips on sealed linear encoders.

System tests

Encoders from HEIDENHAIN are usually integrated as components in larger systems. Such applications require **comprehensive tests of the entire system** regardless of the specifications of the encoder.

The specifications given in the brochure apply to the specific encoder, not to the complete system. Any operation of the encoder outside of the specified range or for any other than the intended applications is at the user's own risk.

In safety-oriented systems, the higher-level system must verify the position value of the encoder after switch-on.

Mounting

Work steps to be performed and dimensions to be maintained during mounting are specified solely in the mounting instructions supplied with the unit. All data in this catalog regarding mounting are therefore provisional and not binding; they do not become terms of a contract.

Changes to the encoder

The correct operation and accuracy of encoders from HEIDENHAIN is ensured only if they have not been modified. Any changes, even minor ones, can impair the operation and reliability of the encoders, and result in a loss of warranty. This also includes the use of additional retaining compounds, lubricants (e.g. for screws) or adhesives not explicitly prescribed. In case of doubt, we recommend contacting HEIDENHAIN in Traunreut.

Temperature ranges

For the unit in its packaging, the **storage temperature range** is –30 to 80 °C. The **operating temperature range** indicates the temperatures that the encoder may reach during operation in the actual installation environment. The function of the encoder is guaranteed within this range (DIN 32878). The operating temperature is measured on the face of the encoder flange (see dimension drawing) and must not be confused with the ambient temperature.

The temperature of the encoder is influenced by:

- Mounting conditions
- The ambient temperature
- Self-heating of the encoder

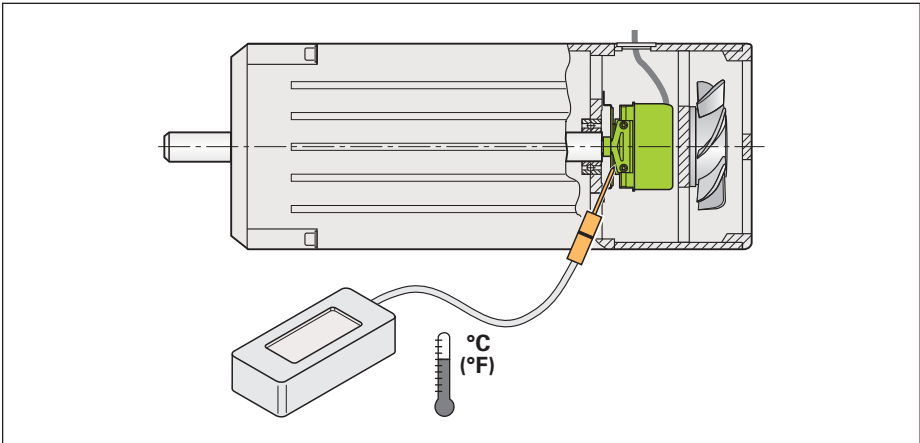
The self-heating of an encoder depends both on its design characteristics (stator coupling/solid shaft, shaft sealing ring, etc.) and on the operating parameters (rotational speed, power supply). Higher heat generation in the encoder means that a lower ambient temperature is required to keep the encoder within its permissible operating temperature range.

These tables show the approximate values of self-heating to be expected in the encoders. In the worst case, a combination of operating parameters can exacerbate self-heating, for example a 30V power supply and maximum rotational speed. Therefore, the actual operating temperature should be measured directly at the encoder if the encoder is operated near the limits of permissible parameters. Then suitable measures should be taken (fan, heat sinks, etc.) to reduce the ambient temperature far enough so that the maximum permissible operating temperature will not be exceeded during continuous operation. For high speeds at maximum permissible ambient temperature, special versions are available on request with reduced degree of protection (without shaft seal and its concomitant frictional heat).

Self-heating at supply voltage	≥ 15 V	30 V
ERN/ROD	Approx. 5 K	Approx. +10 K
ECN/EQN/ROC/ ROQ/RIC/RIQ	Approx. 5 K	Approx. +10 K

Heat generation at speed n _{max}		
Solid shaft	ROC/ROQ/ROD/ RIC/RIQ	Approx. + 5 K with IP 64 protection Approx. + 10 K with IP 66 protection
Blind hollow shaft	ECN/EQN/ERN 400	Approx. + 30 K with IP 64 protection Approx. + 40 K with IP 66 protection
	ECN/EQN/ERN 1000	Approx. +10 K
Hollow through shaft	ECN/ERN 100 ECN/EQN/ERN 400	Approx. + 40 K with IP 64 protection Approx. + 50 K with IP 66 protection

An encoder's typical self-heating values depend on its design characteristics at maximum permissible speed. The correlation between rotational speed and heat generation is nearly linear.



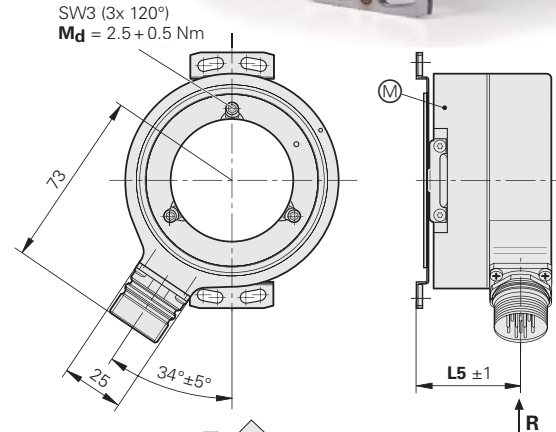
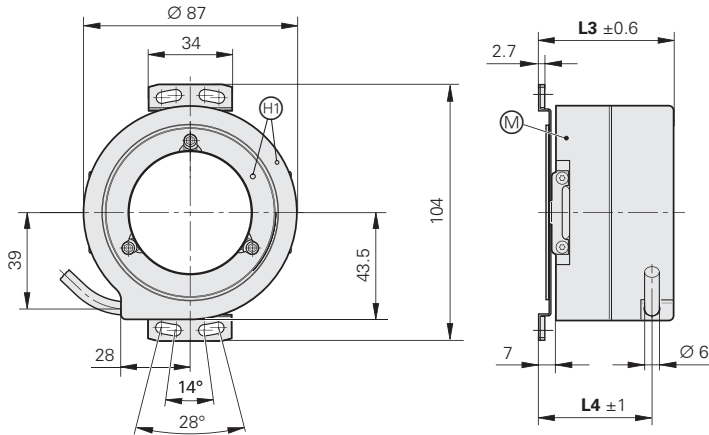
Measuring the actual operating temperature at the defined measuring point of the rotary encoder (see *Specifications*)

ECN/ERN 100 Series

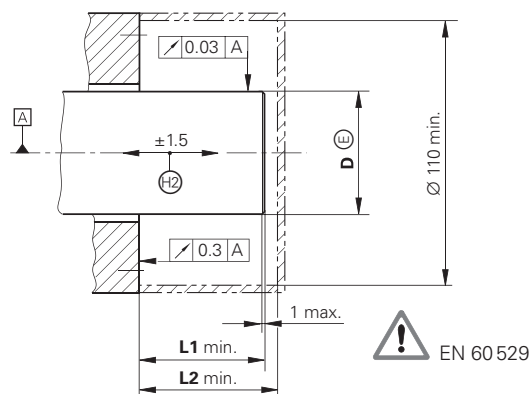
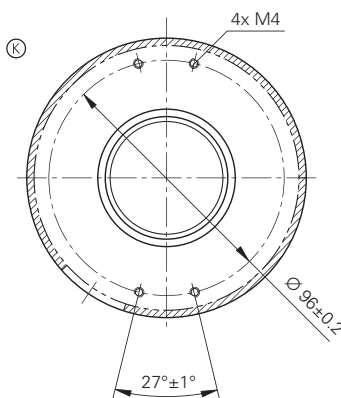
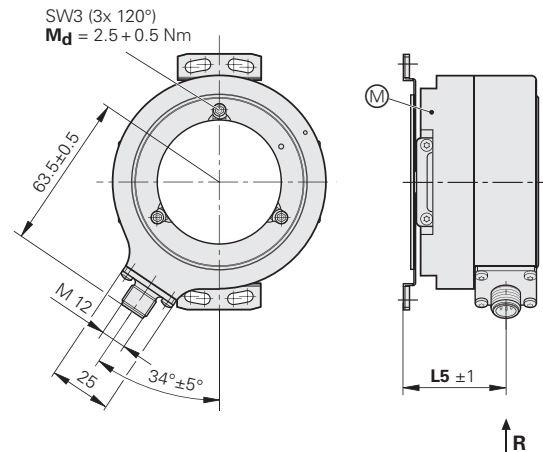
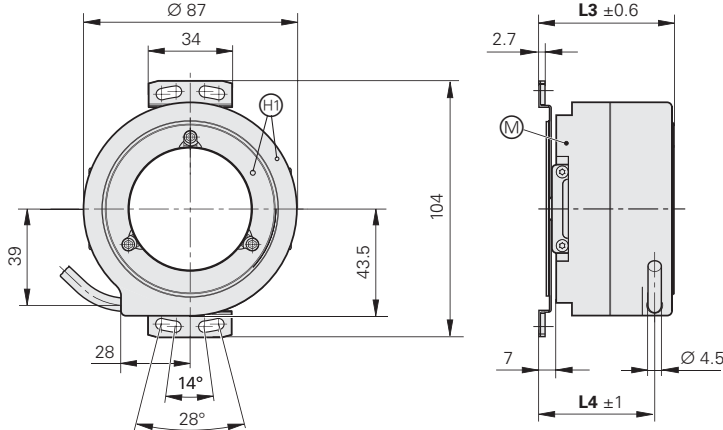
- Rotary encoders with mounted stator coupling
- Hollow through shaft up to $\varnothing 50$ mm



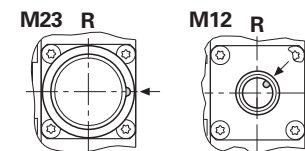
ERN 1x0/ECN 113



ECN 125



Connector coding
R = radial



Dimensions in mm



Tolerancing ISO 8015

ISO 2768 - m H

< 6 mm: ± 0.2 mm

Cable radial, also usable axially

$\square$ = Bearing

$\odot$ = Required mating dimensions

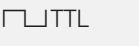
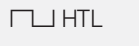
$\odot$ = Measuring point for operating temperature

$\oplus$ = ERN: reference-mark position $\pm 15^\circ$; ECN: zero position $\pm 15^\circ$

$\oplus$ = Compensation of mounting tolerances and thermal expansion, no dynamic motion

$\curvearrowright$ Direction of shaft rotation for output signals as per the interface description

D	L1	L2	L3	L4	L5
$\varnothing 20h7$	41	43.5	40	32	26.5
$\varnothing 25h7$	41	43.5	40	32	26.5
$\varnothing 38h7$	56	58.5	55	47	41.5
$\varnothing 50h7$	56	58.5	55	47	41.5

	Absolute Singleturn			Incremental		
	ECN 125	ECN 113	ECN 113	ERN 120	ERN 130	ERN 180
Absolute position values*	EnDat 2.2	EnDat 2.2	SSI	–		
Ordering designation	EnDat 22	EnDat 01	39n1			
Positions per revolution	33554432 (25 bits)	8192 (13 bits)		–		
Code	Pure binary		Gray	–		
Elec. permissible speed Deviations ¹⁾	n _{max} for continuous position value	≤ 600 min ^{–1} /n _{max} ± 1 LSB/± 50 LSB		–		
Calculation time t _{cal}	≤ 5 μs	≤ 0.25 μs	≤ 0.5 μs	–		
Incremental signals	None	~ 1 V _{PP} ²⁾				~ 1 V _{PP} ²⁾
Line counts*	–	2048		1000 1024 2048 2500 3600 5000		
Cutoff frequency –3 dB	–	Typically ≥ 200 kHz		–		Typ. ≥ 180 kHz
Scanning frequency	–	–		≤ 300 kHz		–
Edge separation a	–	–		≥ 0.39 μs		–
System accuracy	± 20″			1/20 of grating period		
Power supply Current consumption without load	3.6 to 5.25 V ≤ 200 mA	5 V ± 5 % ≤ 180 mA	5 V ± 5 % ³⁾ ≤ 180 mA	5 V ± 10 % ≤ 120 mA	10 to 30 V ≤ 150 mA	5 V ± 10 % ≤ 120 mA
Electrical connection*	• Flange socket M12, radial • Cable 1 m/5m, with M12 coupling	• Flange socket M23, radial • Cable 1 m/5 m, with or without coupling M23		• Flange socket M23, radial • Cable 1 m/5 m, with or without coupling M23		
Shaft*	Hollow through shaft D = 20 mm, 25 mm, 38 mm, 50 mm			Hollow through shaft D = 20 mm, 25 mm, 38 mm, 50 mm		
Mech. permissible speed n ⁴⁾	D > 30 mm: ≤ 4000 min ^{–1} D ≤ 30 mm: ≤ 6000 min ^{–1}			D > 30 mm: ≤ 4000 min ^{–1} D ≤ 30 mm: ≤ 6000 min ^{–1}		
Starting torque at 20 °C	D > 30 mm: ≤ 0.2 Nm D ≤ 30 mm: ≤ 0.15 Nm			D > 30 mm: ≤ 0.2 Nm D ≤ 30 mm: ≤ 0.15 Nm		
Moment of inertia of rotor	D = 50 mm 220 · 10 ^{–6} kgm ² D = 38 mm 350 · 10 ^{–6} kgm ² D = 25 mm 96 · 10 ^{–6} kgm ² D = 20 mm 100 · 10 ^{–6} kgm ²			D = 50 mm 220 · 10 ^{–6} kgm ² D = 38 mm 350 · 10 ^{–6} kgm ² D = 25 mm 95 · 10 ^{–6} kgm ² D = 20 mm 100 · 10 ^{–6} kgm ²		
Permissible axial motion of measured shaft	± 1.5 mm			± 1.5 mm		
Vibration 55 to 2000 Hz Shock 6 ms	≤ 200 m/s ^{2 5)} (EN 60068-2-6) ≤ 1000 m/s ² (EN 60068-2-27)			≤ 200 m/s ^{2 5)} (EN 60068-2-6) ≤ 1000 m/s ² (EN 60068-2-27)		
Max. operating temp. ⁴⁾	100 °C			100 °C	85 °C (100 °C at U _P < 15 V)	100 °C
Min. operating temp.	Flange socket or fixed cable: –40 °C For frequent flexing: –10 °C			Flange socket or fixed cable: –40 °C For frequent flexing: –10 °C		
Protection ⁴⁾ EN 60529	IP 64			IP 64		
Weight	0.6 kg to 0.9 kg depending on the hollow shaft version			0.6 kg to 0.9 kg depending on the hollow shaft version		

Bold: These preferred versions are available on short notice

* Please select when ordering

¹⁾ Velocity-dependent deviations between the absolute value and incremental signal

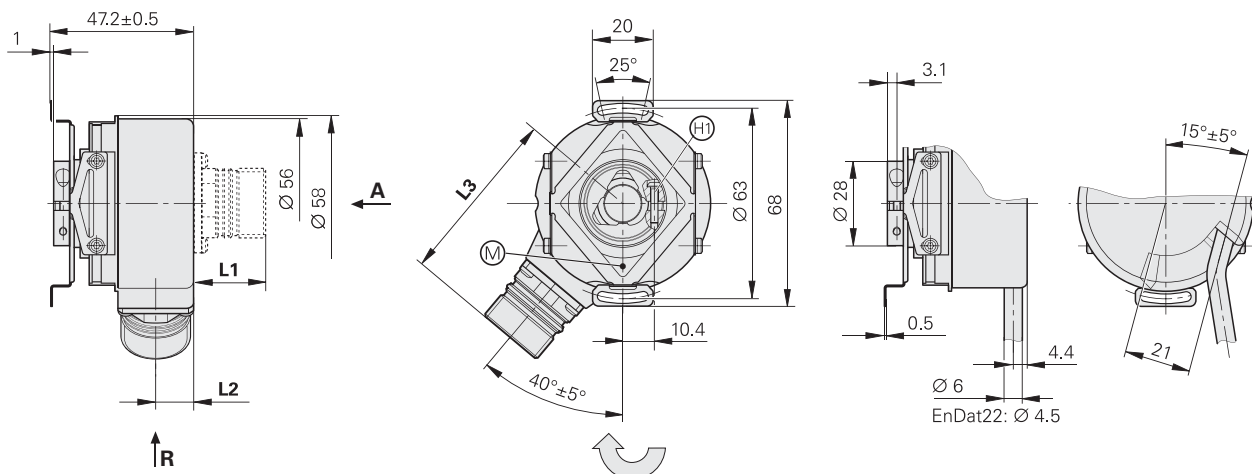
²⁾ Restricted tolerances: Signal amplitude 0.8 to 1.2 V_{PP}

³⁾ 10 to 30 V via connecting cable with voltage converter

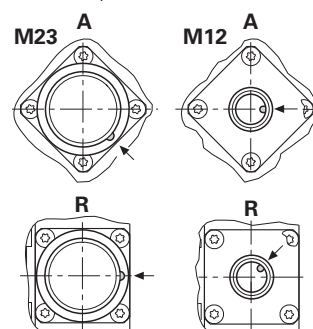
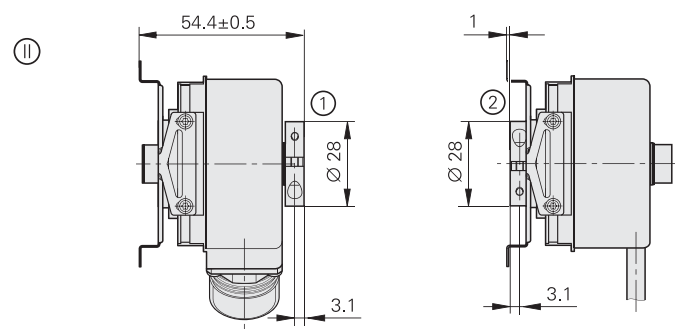
⁴⁾ For the correlation between the protection class, shaft speed and operating temperature, see *General Mechanical Information*

⁵⁾ 100 m/s^2 with flange socket version

- ### Blind hollow shaft

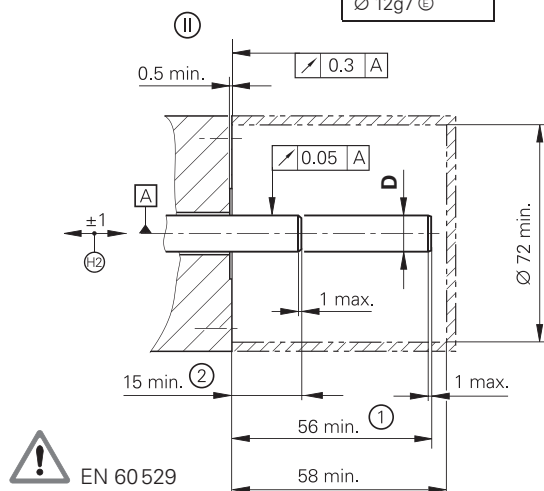
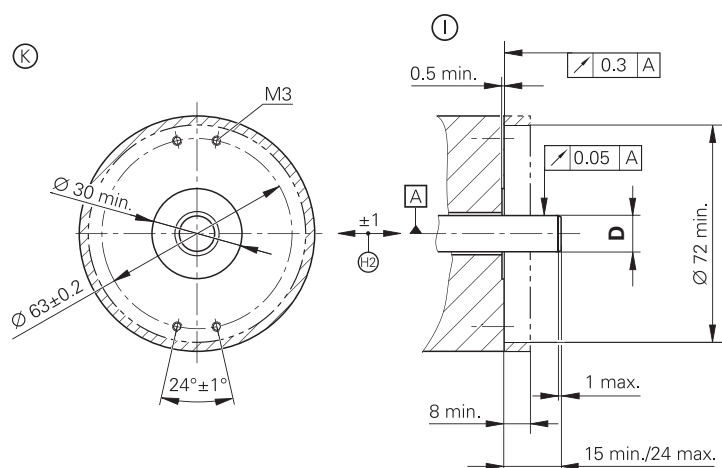


Connector coding
A = axial, **R** = radial



Flange socket		
	M12	M23
		
L1	14	23.6
L2	12.5	12.5
L3	48.5	58.1

D
Ø 8g7 ☹
Ø 12g7 ☹



 EN 60529

↻ Direction of shaft rotation for output signals as per the interface description

	Incremental									
	ERN 420			ERN 460			ERN 430		ERN 480	
Incremental signals	 TTL						 HTL		 1 V _{PP} ¹⁾	
Line counts*	250 500						–			
	1000	1024	1250	2000	2048	2500	3600	4096	5000	
Cutoff frequency –3 dB Scanning frequency Edge separation a	– ≤ 300 kHz ≥ 0.39 μs						≥ 180 kHz – –			
System accuracy	1/20 of grating period									
Power supply Current consumption without load	5 V ± 10 % 120 mA			10 to 30 V 100 mA			10 to 30 V 150 mA		5 V ± 10 % 120 mA	
Electrical connection*	• Flange socket M23, radial and axial (with blind hollow shaft) • Cable 1 m, without connecting element									
Shaft*	Blind hollow shaft or hollow through shaft ; D = 8 mm or D = 12 mm									
Mech. permissible speed n ²⁾	≤ 6000 min ^{–1} /≤ 12000 min ^{–1} ³⁾									
Starting torque at 20 °C below –20 °C	<i>Blind hollow shaft:</i> ≤ 0.01 Nm <i>Hollow through shaft:</i> ≤ 0.025 Nm ≤ 1 Nm									
Moment of inertia of rotor	≤ 4.3 · 10 ^{–6} kgm ²									
Permissible axial motion of measured shaft	± 1 mm									
Vibration 55 to 2000 Hz Shock 6 ms/2 ms	≤ 300 m/s ² ; <i>flange socket version:</i> 150 m/s ² (EN 60068-2-6) ≤ 1000 m/s ² /≤ 2000 m/s ² (EN 60068-2-27)									
Max. operating temp. ²⁾	100 °C			70 °C			100 °C ⁴⁾			
Min. operating temp.	<i>Flange socket or fixed cable:</i> –40 °C <i>For frequent flexing:</i> –10 °C									
Protection EN 60529	IP 67 at housing (IP 66 with hollow through shaft); IP 64 at shaft inlet									
Weight	Approx. 0.3 kg									

Bold: These preferred versions are available on short notice

* Please select when ordering

¹⁾ Restricted tolerances: Signal amplitude 0.8 to 1.2 V_{PP}

²⁾ For the correlation between the operating temperature and the shaft speed or supply voltage, see *General Mechanical Information*

³⁾ With two shaft clamps (only for hollow through shaft)

⁴⁾ 80° for ERN 480 with 4096 or 5000 lines



Absolute

Singleturn

ECN 425

ECN 413

ECN 413

Absolute position values*

EnDat 2.2

EnDat 2.2

SSI

Ordering designation

EnDat 22

EnDat 01

39r1

Positions per revolution

33554432 (25 bits)

8192 (13 bits)

Revolutions

–

Code

Pure binary

Gray

Elec. permissible speed
Deviations¹⁾

$\leq 12000 \text{ min}^{-1}$
for continuous position value

512 lines: $\leq 5000/12000 \text{ min}^{-1}$
 $\pm 1 \text{ LSB}/\pm 100 \text{ LSB}$
2048 lines: $\leq 1500/12000 \text{ min}^{-1}$
 $\pm 1 \text{ LSB}/\pm 50 \text{ LSB}$

$\leq 12000 \text{ min}^{-1}$
 $\pm 12 \text{ LSB}$

Calculation time t_{cal}

$\leq 5 \mu\text{s}$

$\leq 0.5 \mu\text{s}^{5)}$

Incremental signals

None

$\sim 1 V_{\text{PP}}^{2)}$

Line counts*

–

512 2048

512

Cutoff frequency –3 dB
Scanning frequency
Edge separation a

–
–
–

512 lines: $\geq 130 \text{ kHz}$; *2048 lines:* $\geq 400 \text{ kHz}$
–
–

System accuracy

$\pm 20''$

512 lines: $\pm 60''$; *2048 lines:* $\pm 20''$

Power supply*

3.6 to 14 V

3.6 to 14 V

5 V $\pm 5\%$ or 10 to 30 V

Current consumption
without load

$\leq 110 \text{ mA}^{6)}$

$\leq 110 \text{ mA}^{6)}$

$\leq 100 \text{ mA}$ or $\leq 45 \text{ mA}$

Electrical connection*

- **Flange socket** M12, radial
- Cable 1 m, with M12 coupling

- **Flange socket** M23, radial
- Cable 1 m, with M23 coupling or without connecting element

Shaft*

Blind hollow shaft or **hollow through shaft**; **D = 8 mm** or **D = 12 mm**

Mech. permissible speed $n^{3)}$

$\leq 6000 \text{ min}^{-1}/\leq 12000 \text{ min}^{-1}^{4)}$

Starting
torque

at 20 °C

below –20 °C

Blind hollow shaft: $\leq 0.01 \text{ Nm}$
Hollow through shaft: $\leq 0.025 \text{ Nm}$
 $\leq 1 \text{ Nm}$

Moment of inertia of rotor

$\leq 4.3 \cdot 10^{-6} \text{ kgm}^2$

Permissible axial motion
of measured shaft

$\pm 1 \text{ mm}$

Vibration 55 to 2000 Hz

Shock 6 ms/2 ms

$\leq 300 \text{ m/s}^2$; *flange socket version:* 150 m/s^2 (EN 60068-2-6)
 $\leq 1000 \text{ m/s}^2/\leq 2000 \text{ m/s}^2$ (EN 60068-2-27)

Max. operating temp.³⁾

100 °C

Min. operating temp.

Flange socket or fixed cable: –40 °C
For frequent flexing: –10 °C

Protection EN 60529

IP 67 at housing; IP 64 at shaft inlet

Weight

Approx. 0.3 kg

Bold: These preferred versions are available on short notice

* Please select when ordering

¹⁾ Velocity-dependent deviations between the absolute value and incremental signal

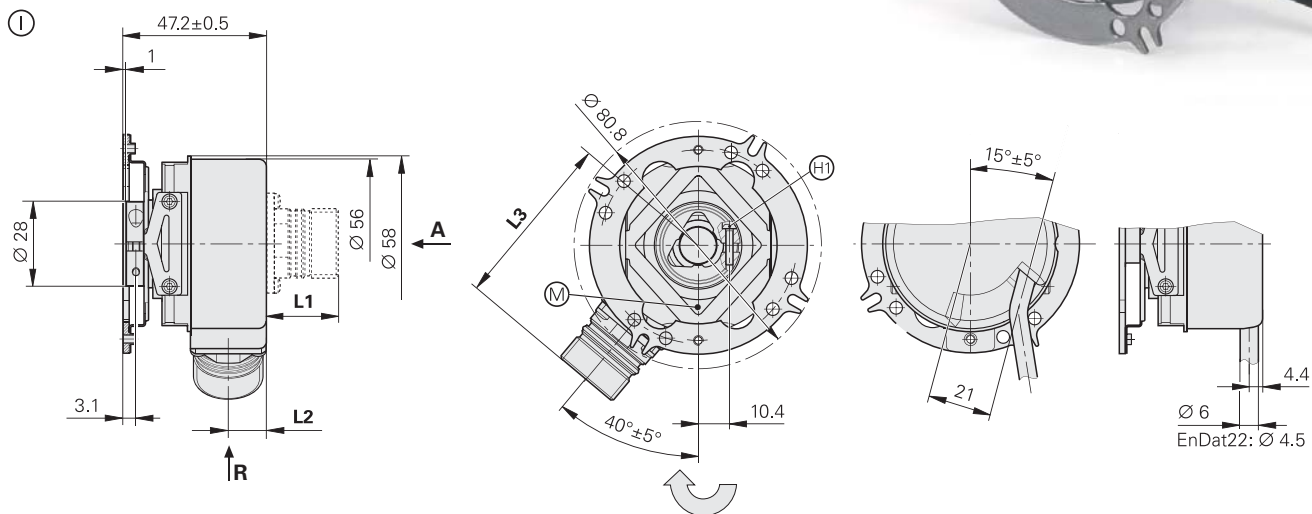
²⁾ Restricted tolerances: Signal amplitude 0.8 to 1.2 V_{PP}

ECN, EQN, ERN 400 Series

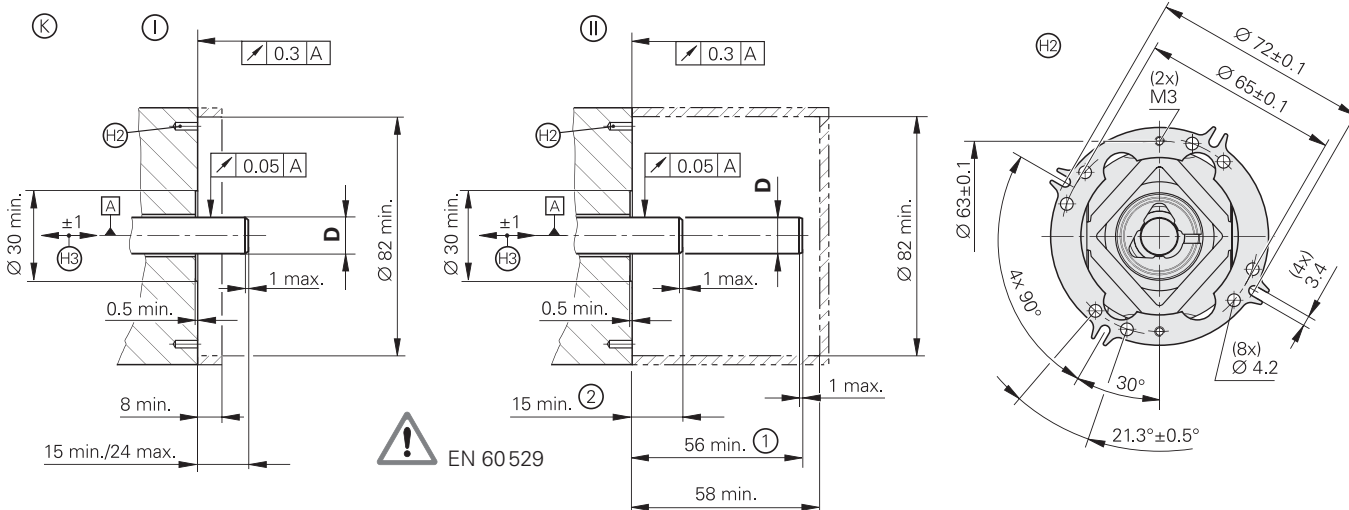
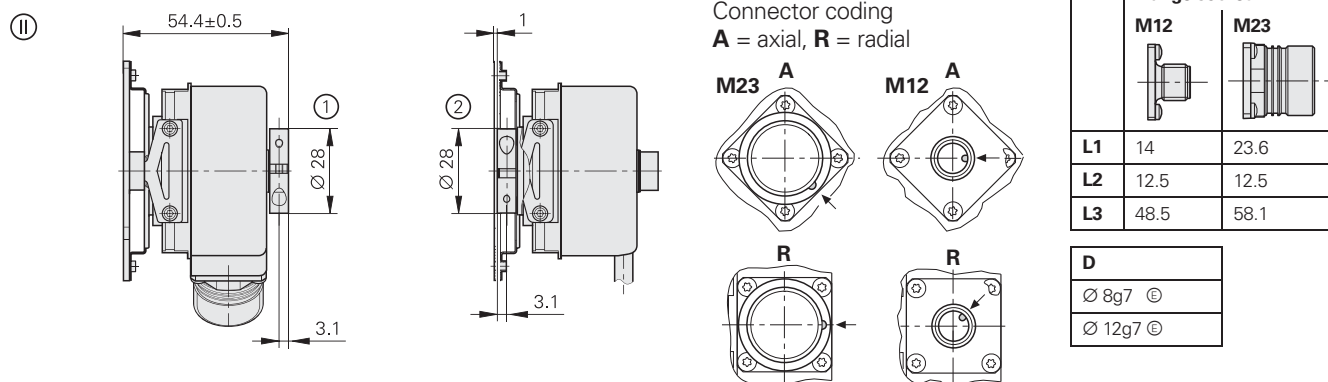
- Rotary encoders with mounted universal stator coupling
- Blind hollow shaft or hollow through shaft



Blind hollow shaft



Hollow through shaft



Dimensions in mm



Tolerancing ISO 8015
ISO 2768 - m H
< 6 mm: ±0.2 mm

Cable radial, also usable axially

- Ⓐ = Bearing
- Ⓑ = Bearing of encoder
- Ⓜ = Measuring point for operating temperature
- Ⓚ = Required mating dimensions
- Ⓜ = Clamping screw with X8 hexalobular socket
- Ⓜ = Hole circle for fastening, see coupling
- Ⓜ = Compensation of mounting tolerances and thermal expansion, no dynamic motion permitted
- ① = Clamping ring on housing side (condition upon delivery)
- ② = Clamping ring on coupling side (optionally mountable)
- ↻ = Direction of shaft rotation for output signals as per the interface description

	Incremental			
	ERN 420	ERN 460	ERN 430	ERN 480
Incremental signals	 TTL		 HTL	 1 V _{PP} ¹⁾
Line counts*	250 500			–
	1000 1024 1250 2000 2048 2500 3600 4096 5000			
Cutoff frequency –3 dB Scanning frequency Edge separation a	– ≤ 300 kHz ≥ 0.39 μs			≥ 180 kHz – –
System accuracy	1/20 of grating period			
Power supply Current consumption without load	5 V ± 10 % 120 mA	10 to 30 V 100 mA	10 to 30 V 150 mA	5 V ± 10 % 120 mA
Electrical connection*	• Flange socket M23, radial and axial (with blind hollow shaft) • Cable 1 m, without connecting element			
Shaft*	Blind hollow shaft or hollow through shaft ; D = 8 mm or D = 12 mm			
Mech. permissible speed n ²⁾	≤ 6000 min ^{–1} /≤ 12000 min ^{–1} ³⁾			
Starting torque at 20 °C below –20 °C	<i>Blind hollow shaft:</i> ≤ 0.01 Nm <i>Hollow through shaft:</i> ≤ 0.025 Nm ≤ 1 Nm			
Moment of inertia of rotor	≤ 4.3 · 10 ^{–6} kgm ²			
Permissible axial motion of measured shaft	± 1 mm			
Vibration 55 to 2000 Hz Shock 6 ms/2 ms	≤ 300 m/s ² ; <i>flange socket version:</i> 150 m/s ² (EN 60068-2-6) ≤ 1000 m/s ² /≤ 2000 m/s ² (EN 60068-2-27)			
Max. operating temp. ²⁾	100 °C	70 °C	100 °C ⁴⁾	
Min. operating temp.	<i>Flange socket or fixed cable:</i> –40 °C <i>For frequent flexing:</i> –10 °C			
Protection EN 60529	<i>At housing:</i> IP 67 (IP 66 for hollow through shaft) <i>At shaft inlet:</i> IP 64 (IP 66 upon request)			
Weight	Approx. 0.3 kg			

Bold: These preferred versions are available on short notice

* Please select when ordering

¹⁾ Restricted tolerances: Signal amplitude 0.8 to 1.2 V_{PP}

²⁾ For the correlation between the operating temperature and the shaft speed or supply voltage, see *General Mechanical Information*

³⁾ With two shaft clamps (only for hollow through shaft)

⁴⁾ 80° for ERN 480 with 4096 or 5000 lines

	Absolute		
	Singletum		
	ECN 425	ECN 413	ECN 413
Absolute position values*	EnDat 2.2	EnDat 2.2	SSI
Ordering designation	EnDat 22	EnDat 01	39r1
Positions per revolution	33554432 (25 bits)	8192 (13 bits)	
Revolutions	–		
Code	Pure binary		Gray
Elec. permissible speed Deviations ¹⁾	≤ 12000 min ^{–1} for continuous position value	512 lines: ≤ 5000/12000 min ^{–1} ± 1 LSB/± 100 LSB 2048 lines: ≤ 1500/12000 min ^{–1} ± 1 LSB/± 50 LSB	≤ 12000 min ^{–1} ± 12 LSB
Calculation time t _{cal}	≤ 5 μs		≤ 0.5 μs ⁵⁾
Incremental signals	None	~ 1 V _{PP} ²⁾	
Line counts*	–	512 2048	512
Cutoff frequency –3 dB Scanning frequency Edge separation a	– – –	512 lines: ≥ 130 kHz; 2048 lines: ≥ 400 kHz – –	
System accuracy	± 20"	512 lines: ± 60"; 2048 lines: ± 20"	
Power supply* Current consumption without load	3.6 to 14 V ≤ 110 mA ⁶⁾	3.6 to 14 V ≤ 110 mA ⁶⁾	5 V ± 5 % or 10 to 30 V ≤ 100 mA or ≤ 45 mA
Electrical connection*	• Flange socket M12, radial • Cable 1 m, with M12 coupling	• Flange socket M23, radial • Cable 1 m, with M23 coupling or without connecting element	
Shaft*	Blind hollow shaft or hollow through shaft; D = 8 mm or D = 12 mm		
Mech. permissible speed n ³⁾	≤ 6000 min ^{–1} /≤ 12000 min ^{–1} 4)		
Starting torque at 20 °C below –20 °C	Blind hollow shaft: ≤ 0.01 Nm Hollow through shaft: ≤ 0.025 Nm ≤ 1 Nm		
Moment of inertia of rotor	≤ 4.3 · 10 ^{–6} kgm ²		
Permissible axial motion of measured shaft	± 1 mm		
Vibration 55 to 2000 Hz Shock 6 ms/2 ms	≤ 300 m/s ² ; flange socket version: 150 m/s ² (EN 60068-2-6) ≤ 1000 m/s ² /≤ 2000 m/s ² (EN 60068-2-27)		
Max. operating temp. ³⁾	100 °C		
Min. operating temp.	Flange socket or fixed cable: –40 °C For frequent flexing: –10 °C		
Protection EN 60529	IP 67 at housing, IP 64 at shaft end (IP 66 available on request)		
Weight	Approx. 0.3 kg		

Bold: These preferred versions are available on short notice

* Please select when ordering

¹⁾ Velocity-dependent deviations between the absolute value and incremental signal

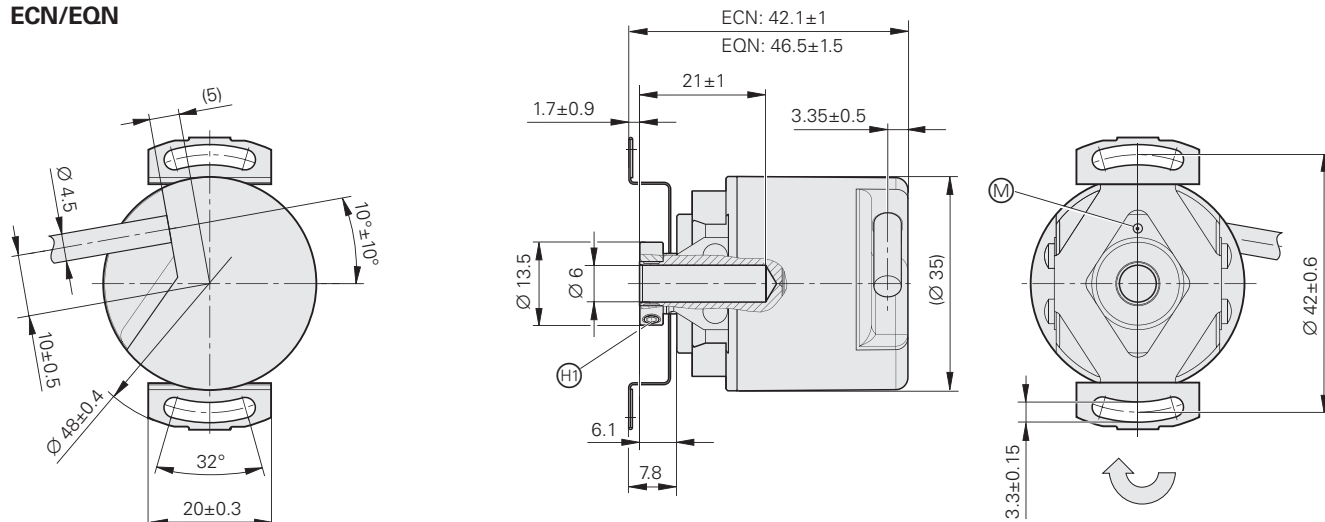
²⁾ Restricted tolerances: Signal amplitude 0.8 to 1.2 V_{PP}

ECN, EQN, ERN 1000 Series

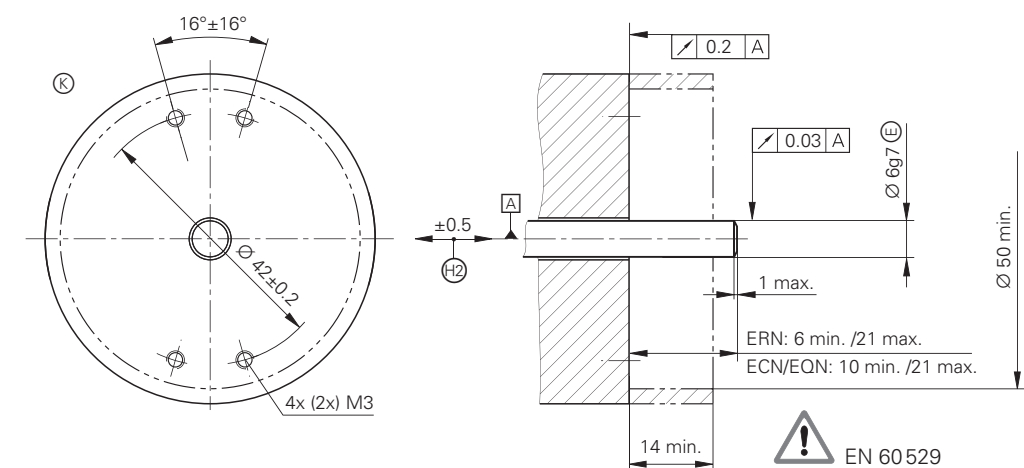
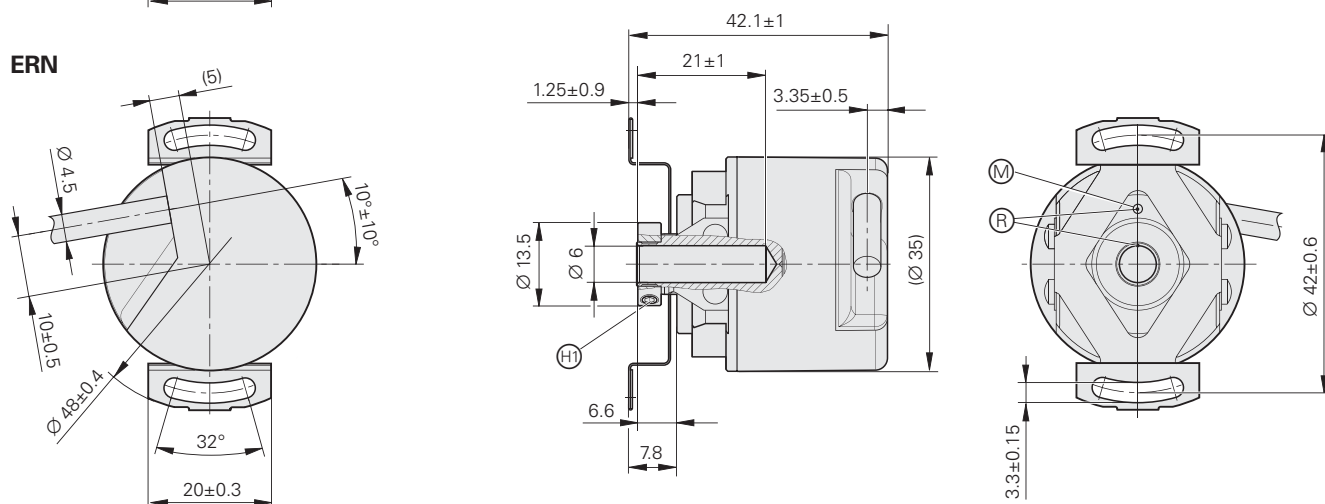
- Rotary encoders with mounted stator coupling
- Compact dimensions
- Blind hollow shaft $\varnothing 6$ mm
- Functional Safety for ECN 1023/EQN 1035 upon request



ECN/EQN



ERN



Dimensions in mm



Tolerancing ISO 8015
ISO 2768 - m H
< 6 mm: ± 0.2 mm

Ⓐ = Bearing of mating shaft

Ⓚ = Required mating dimensions

Ⓜ = Measuring point for operating temperature

Ⓡ = Reference mark position $\pm 20^\circ$

Ⓢ = 2 screws in clamping ring. Tightening torque 0.6 ± 0.1 Nm, width across flats 1.5

Ⓣ = Compensation of mounting tolerances and thermal expansion, no dynamic motion

↻ Direction of shaft rotation for output signals as per the interface description

	Incremental									
	ERN 1020			ERN 1030			ERN 1080		ERN 1070	
Incremental signals	 TTL			 HTL			 1 V _{PP} ¹⁾		 TTL	
Line counts*	100 1000	200 1024	250 1 250	360 1 500	400 2 000	500 2048	720 2500	900 3600	1000 2500 3600	
Integrated interpolation*	–							5-fold		10-fold
Cutoff frequency –3 dB Scanning frequency Edge separation a	– ≤ 300 kHz ≥ 0.39 μs			– ≤ 160 kHz ≥ 0.76 μs			≥ 180 kHz – –		– ≤ 100 kHz ≥ 0.47 μs	– ≤ 100 kHz ≥ 0.22 μs
Power supply Current consumption without load	5 V ± 10 % ≤ 120 mA			10 to 30 V ≤ 150 mA			5 V ± 10 % ≤ 120 mA		5 V ± 5 % ≤ 155 mA	
Electrical connection*	Cable 1 m/5 m, with or without coupling M23							Cable 5 m without M23 coupling		
Shaft	Blind hollow shaft D = 6 mm									
Mech. permissible speed n	≤ 12 000 min ^{–1}									
Starting torque	≤ 0.001 Nm (at 20 °C)									
Moment of inertia of rotor	≤ 0.5 · 10 ^{–6} kgm ²									
Permissible axial motion of measured shaft	± 0.5 mm									
Vibration 55 to 2000 Hz Shock 6 ms	≤ 100 m/s ² (EN 60068-2-6) ≤ 1 000 m/s ² (EN 60068-2-27)									
Max. operating temp. ²⁾	100 °C			70 °C			100 °C		70 °C	
Min. operating temp.	For fixed cable: –30 °C For frequent flexing: –10 °C									
Protection EN 60529	IP 64									
Weight	Approx. 0.1 kg									

Bold: These preferred versions are available on short notice

* Please select when ordering

¹⁾ Restricted tolerances: Signal amplitude 0.8 to 1.2 V_{PP}

²⁾ For the correlation between the operating temperature and the shaft speed or supply voltage, see *General Mechanical Information*



Absolute
Singleturn
ECN 1023

ECN 1013

Absolute position values	EnDat 2.2	
Ordering designation	EnDat 22	EnDat 01
Positions per revolution	8388608 (23 bits)	8192 (13 bits)
Revolutions	–	
Code	Pure binary	
Elec. permissible speed Deviations ¹⁾	12000 min ⁻¹ (for continuous position value)	4000 min ⁻¹ /12000 min ⁻¹ ± 1 LSB/± 16 LSB
Calculation time t_{cal}	≤ 5 μs	
Incremental signals	–	~ 1 V _{PP} ²⁾
Line count	–	512
Cutoff frequency –3 dB	–	≥ 190 kHz
System accuracy	± 60"	
Power supply Current consumption without load	3.6 V to 14 V ≤ 110 mA ³⁾	
Electrical connection	Cable 1 m, with M12 coupling	Cable 1 m, with M23 coupling
Shaft	Blind hollow shaft Ø 6 mm	
Mech. permissible speed n	12000 min ⁻¹	
Starting torque	≤ 0.001 Nm (at 20 °C)	
Moment of inertia of rotor	Approx. $0.5 \cdot 10^{-6}$ kgm ²	
Permissible axial motion of measured shaft	± 0.5 mm	
Vibration 55 to 2000 Hz Shock 6 ms	≤ 100 m/s ² (EN 60068-2-6) ≤ 1000 m/s ² (EN 60068-2-27)	
Max. operating temp.	100 °C	
Min. operating temp.	For fixed cable: –30 °C For frequent flexing: –10 °C	
Protection EN 60529	IP 64	
Weight	Approx. 0.1 kg	

¹⁾ Velocity-dependent deviations between the absolute and incremental signals

²⁾ Restricted tolerances: Signal amplitude 0.80 to 1.2 V_{PP}

³⁾ Depends on the power supply; see *General Electrical Information*

	Multiturn	
	EQN 1035	EQN 1025
	EnDat 22	EnDat 01
	8388608 (23 bits)	8192 (13 bits)
	4096 (12 bits)	
	12000 min ⁻¹ (for continuous position value)	4000 min ⁻¹ /12000 min ⁻¹ ± 1 LSB/± 16 LSB
	–	~ 1 V _{PP} ²⁾
	–	512
	–	≥ 190 kHz
	3.6 V to 14 V ≤ 140 mA ³⁾	
	Cable 1 m, with M12 coupling	Cable 1 m, with M23 coupling
	≤ 0.002 Nm (at 20 °C)	

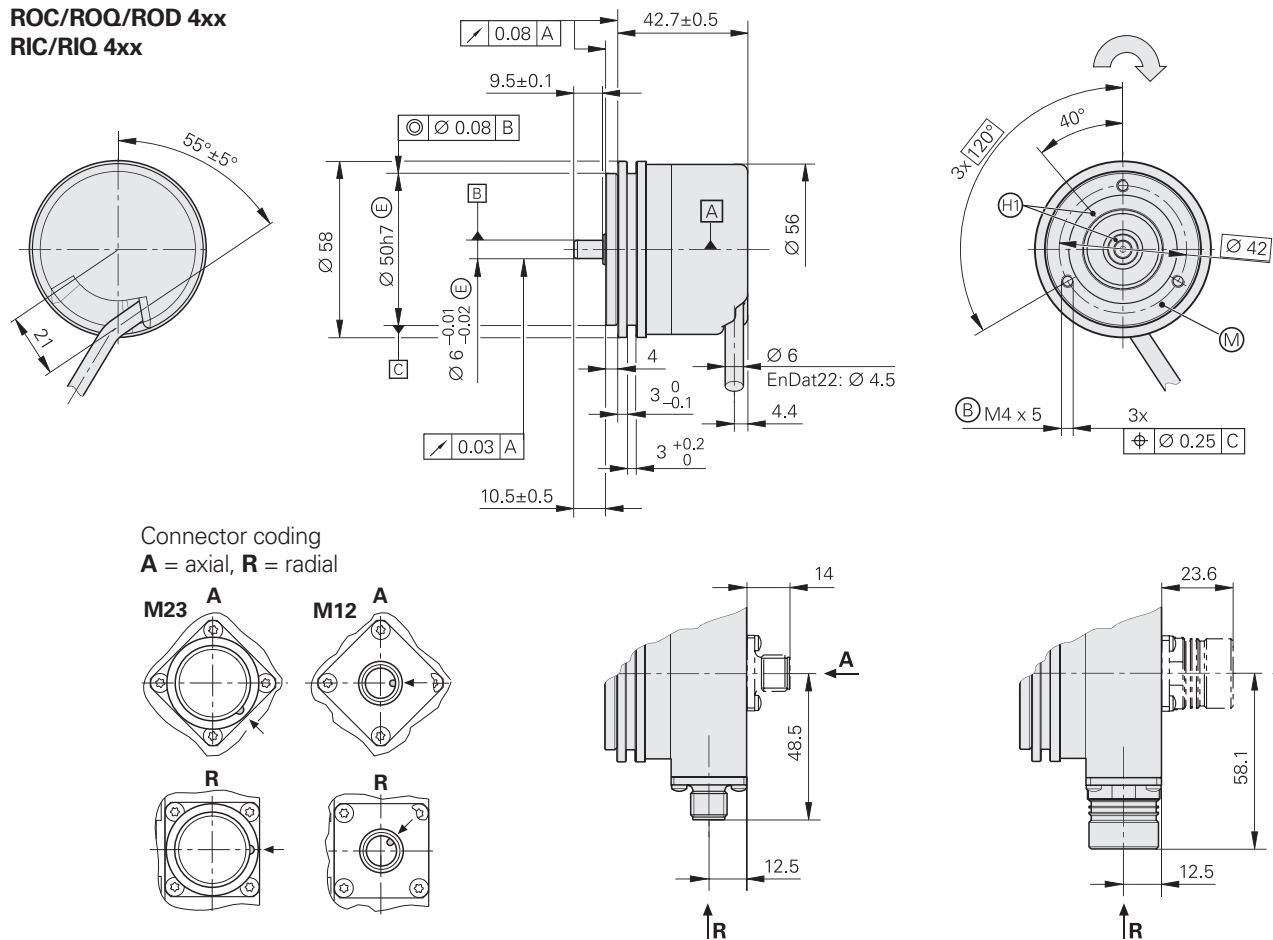
For ROC/ROQ/ROD 400 and RIC/RIQ 400 series

With Synchro Flange

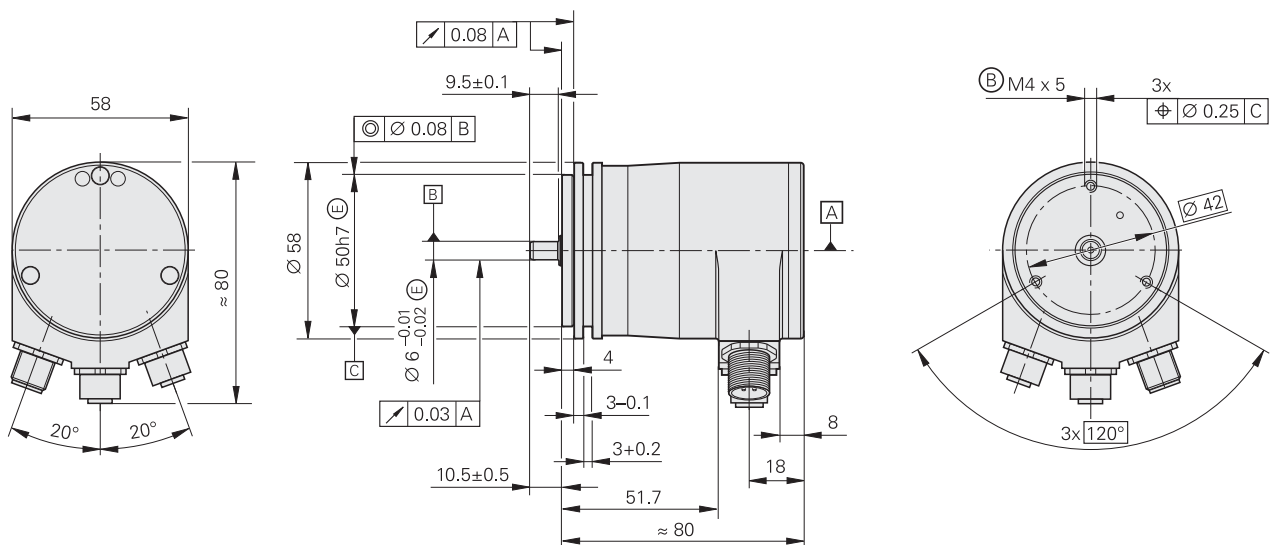
Rotary encoders for separate shaft coupling



ROC/ROQ/ROD 4xx RIC/RIQ 4xx



ROC 413/ROQ 425 with PROFIBUS DP



Dimensions in mm



Tolerancing ISO 8015
ISO 2768 - m H
< 6 mm: ±0.2 mm

Cable radial, also usable axially

□ = Bearing

⊗ = Threaded mounting hole

⊙ = Measuring point for operating temperature

⊕ = ROD reference mark position on shaft and flange ±30°

↻ Direction of shaft rotation for output signals as per the interface description

	Incremental			
	ROD 426	ROD 466	ROD 436	ROD 486
Incremental signals	□□ TTL		□□ HTL	~ 1 V _{PP} ¹⁾
Line counts*	50 100 150 200 250 360 500	512 720	–	
	1000 1024 1250	1500 1800 2000 2048	2500 3600 4096 5000	
	6000 ²⁾ 8192 ²⁾ 9000 ²⁾ 10000 ²⁾		–	
Cutoff frequency –3 dB	–			≥ 180 kHz
Scanning frequency	≤ 300 kHz/≤ 150 kHz ²⁾			–
Edge separation a	≥ 0.39 μs/≥ 0.25 μs ²⁾			–
System accuracy	1/20 of grating period			
Power supply	5 V ± 10 %	10 to 30 V	10 to 30 V	5 V ± 10 %
Current consumption without load	120 mA	100 mA	150 mA	120 mA
Electrical connection*	• Flange socket M23, radial and axial • Cable 1 m/5 m, with or without coupling M23			
Shaft	Solid shaft D = 6 mm			
Mech. permissible speed n	≤ 16000 min ^{–1}			
Starting torque	≤ 0.01 Nm (at 20 °C)			
Moment of inertia of rotor	≤ 2.7 · 10 ^{–6} kgm ²			
Shaft load ³⁾	Axial 10 N/radial 20 N at shaft end			
Vibration 55 to 2000 Hz	≤ 300 m/s ² (EN 60068-2-6)			
Shock 6 ms/2 ms	≤ 1000 m/s ² /≤ 2000 m/s ² (EN 60068-2-27)			
Max. operating temp. ⁴⁾	100 °C	70 °C	100 °C ⁵⁾	
Min. operating temp.	<i>Flange socket or fixed cable: –40 °C</i> <i>For frequent flexing: –10 °C</i>			
Protection EN 60529	IP 67 at housing; IP 64 at shaft end ⁶⁾			
Weight	Approx. 0.3 kg			

Bold: These preferred versions are available on short notice

* Please select when ordering

1) Restricted tolerances: Signal amplitude 0.8 to 1.2 V_{PP}

2) Signal periods; generated through integrated 2-fold interpolation (TTL x 2)

3) Also see *Mechanical Design and Installation*

4) For the correlation between the operating temperature and the shaft speed or supply voltage, see *General Mechanical Information*

5) 80° for ERN 80 with 4096 or 5000 lines

6) IP 66 upon request



Absolute

Singleturn

ROC 425

ROC 413

RIC 418

Absolute position values*	EnDat 2.2	EnDat 2.2	SSI	PROFIBUS DP	EnDat 2.1
Ordering designation	EnDat 22	EnDat 01	39r1		EnDat 01
Positions per revolution	33554432 (25 bits)	8192 (13 bits)		8192 (13 bits) ³⁾	262144 (18 bits)
Revolutions	–				
Code	Pure binary		Gray	Pure binary	
Elec. permissible speed Deviations ¹⁾	≤ 12000 min ^{–1} for continuous position value	512 lines: ≤ 5000/12000 min ^{–1} ± 1 LSB/± 100 LSB 2048 lines: ≤ 1500/12000 min ^{–1} ± 1 LSB/± 50 LSB	12000 min ^{–1} ± 12 LSB	≤ 5000/12000 min ^{–1} ± 1 LSB/± 100 LSB	≤ 4000/15000 min ^{–1} ± 400 LSB/± 800 LSB
Calculation time t _{cal}	≤ 5 μs		≤ 0.5 μs ⁶⁾	–	≤ 8 μs
Incremental signals	None	~ 1 V _{PP} ²⁾		None	~ 1 V _{PP}
Line counts*	–	512 2048	512	–	16
Cutoff frequency –3 dB	–	512 lines: ≥ 130 kHz; 2048 lines: ≥ 400 kHz		–	≥ 6 kHz
System accuracy	± 20″	512 lines: ± 60″; 2048 lines: ± 20″		± 60″	± 480″
Power supply*	3.6 to 14 V	3.6 to 14 V	5 V ± 5 % or 10 to 30 V	9 to 36 V	5 V ± 5 %
Current consumption without load	≤ 110 mA ⁸⁾	≤ 110 mA ⁸⁾	≤ 100 mA or ≤ 45 mA	≤ 150 mA at 24 V	≤ 160 mA
Electrical connection*	• Flange socket M12, radial • Cable 1 m, with M12 coupling	• Flange socket M23, axial or radial • Cable 1 m/5 m, with or without M23 coupling		Three flange sockets, M12 radial	• Flange socket M23, radial • Cable 1 m, with M23 coupling
Shaft	Solid shaft D = 6 mm				
Mech. permissible speed n	≤ 12000 min ^{–1}				
Starting torque	≤ 0.01 Nm (at 20 °C)				
Moment of inertia of rotor	≤ 2.7 · 10 ^{–6} kgm ²				
Shaft load ⁵⁾	Axial 10 N/radial 20 N at shaft end				
Vibration 55 to 2000 Hz Shock 6 ms/2 ms	≤ 300 m/s ² ; PROFIBUS-DP: ≤ 100 m/s ² (EN 60068-2-6) ≤ 1000 m/s ² /≤ 2000 m/s ² (EN 60068-2-27)				
Max. operating temp. ⁷⁾	100 °C			70 °C	100 °C
Min. operating temp.	Flange socket or fixed cable: –40 °C For frequent flexing: –10 °C			–40 °C	Flange socket or fixed cable: –40 °C For frequent flexing: –10 °C
Protection EN 60529	IP 67 at housing; IP 64 at shaft end ⁴⁾				
Weight	Approx. 0.35 kg				

Bold: These preferred versions are available on short notice

* Please select when ordering

¹⁾ Velocity-dependent deviations between the absolute value and incremental signal

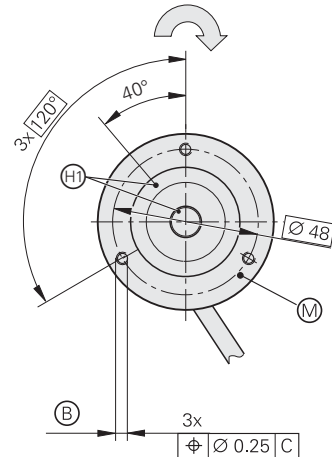
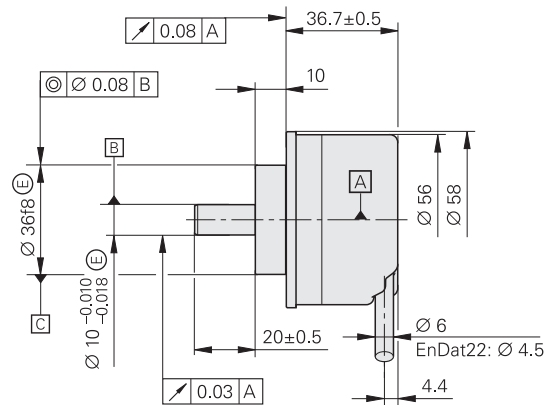
²⁾ Restricted tolerances: Signal amplitude 0.8 to 1.2 V_{PP}

³⁾ These functions are programmable

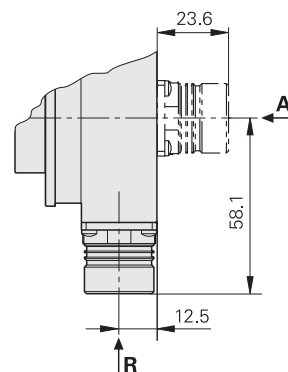
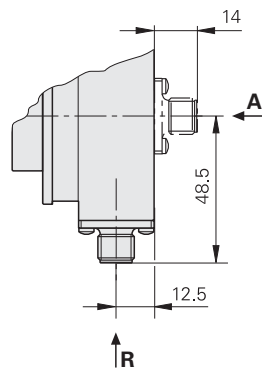
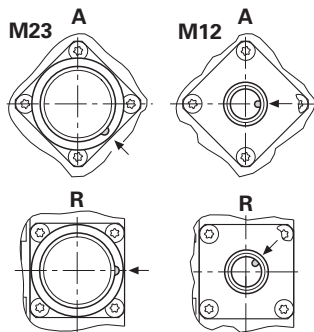
With Clamping Flange



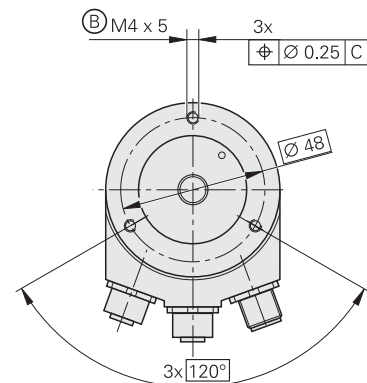
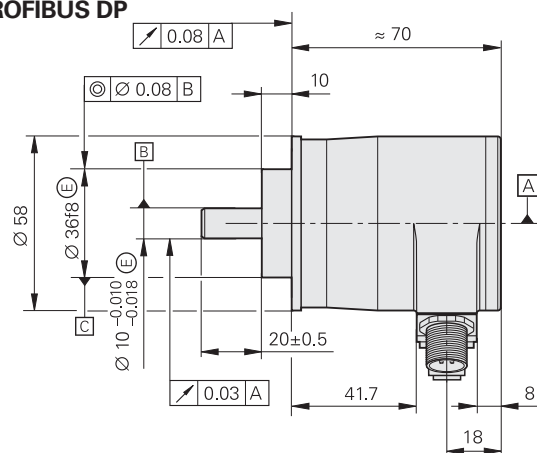
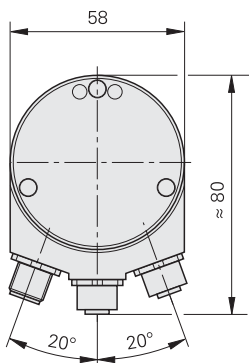
ROC/ROQ/ROD 4xx
RIC/RIQ 4xx



Connector coding
A = axial, **R** = radial



ROC 413/ROQ 425 with PROFIBUS DP



Dimensions in mm



Tolerancing ISO 8015
ISO 2768 - m H
< 6 mm: ± 0.2 mm

Cable radial, also usable axially

[A] = Bearing

Ⓢ = Threaded mounting hole M3x5 on ROD; M4x5 on ROC/ROQ/RIC/RIQ

Ⓜ = Measuring point for operating temperature

Ⓜ = Measuring point for operating temperature
Ⓜ① = ROD: Reference mark position on shaft and flange $\pm 15^\circ$

 Direction of shaft rotation for output signals as per the interface description

	Incremental											
	ROD 420					ROD 430				ROD 480		
Incremental signals	 TTL					 HTL				 1 V _{PP} ¹⁾		
Line counts*	50	100	150	200	250	360	500	512	720	–		
	1000	1024	1250	1 500	1 800	2000	2048	2500	3600	4096	5000	
Cutoff frequency –3 dB	–									≥ 180 kHz		
Scanning frequency	≤ 300 kHz									–		
Edge separation a	≥ 0.39 μs									–		
System accuracy	1/20 of grating period											
Power supply	5 V ± 10 %					10 to 30 V				5 V ± 10 %		
Current consumption without load	120 mA					150 mA				120 mA		
Electrical connection*	• Flange socket M23, radial and axial • Cable 1 m/5 m, with or without coupling M23											
Shaft	Solid shaft D = 10 mm											
Mech. permissible speed n	≤ 12 000 min ^{–1}											
Starting torque	≤ 0.01 Nm (at 20 °C)											
Moment of inertia of rotor	≤ 2.3 · 10 ^{–6} kgm ²											
Shaft load ²⁾	Axial 10 N/radial 20 N at shaft end											
Vibration 55 to 2000 Hz	≤ 300 m/s ² (EN 60 068-2-6)											
Shock 6 ms/2 ms	≤ 1 000 m/s ² /≤ 2 000 m/s ² (EN 60 068-2-27)											
Max. operating temp. ³⁾	100 °C ⁴⁾											
Min. operating temp.	Flange socket or fixed cable: –40 °C For frequent flexing: –10 °C											
Protection EN 60529	IP 67 at housing; IP 64 at shaft end ⁵⁾											
Weight	Approx. 0.3 kg											

Bold: These preferred versions are available on short notice

* Please select when ordering

¹⁾ Restricted tolerances: Signal amplitude 0.8 to 1.2 V_{PP}

²⁾ Also see *Mechanical Design and Installation*

³⁾ For the correlation between the operating temperature and the shaft speed or supply voltage, see *General Mechanical Information*

⁴⁾ 80 °C for ROD 480 with 4096 or 5000 lines

⁵⁾ IP 66 upon request

	Absolute				
	Singleturn				
	ROC 425	ROC 413			RIC 418
Absolute position values*	EnDat 2.2	EnDat 2.2	SSI	PROFIBUS DP	EnDat 2.1
Ordering designation	EnDat 22	EnDat 01	39r1		EnDat 01
Positions per revolution	33554432 (25 bits)	8192 (13 bits)		8192 (13 bits) ³⁾	262144 (18 bits)
Revolutions	–				
Code	Pure binary		Gray	Pure binary	
Elec. permissible speed Deviations ¹⁾	≤ 12000 min ^{–1} for continuous position value	512 lines: ≤ 5000/12000 min ^{–1} ± 1 LSB/± 100 LSB 2048 lines: ≤ 1500/12000 min ^{–1} ± 1 LSB/± 50 LSB	12000 min ^{–1} ± 12 LSB	≤ 5000/12000 min ^{–1} ± 1 LSB/± 100 LSB	≤ 4000/15000 min ^{–1} ± 400 LSB/± 800 LSB
Calculation time t _{cal}	≤ 5 μs		≤ 0.5 μs ⁶⁾	–	≤ 8 μs
Incremental signals	None	~ 1 V _{PP} ²⁾		None	~ 1 V _{PP}
Line counts*	–	512 2048	512	–	16
Cutoff frequency –3 dB	–	512 lines: ≥ 130 kHz; 2048 lines: ≥ 400 kHz		–	≥ 6 kHz
System accuracy	± 20"	± 60"			± 480"
Power supply*	3.6 to 14 V	3.6 to 14 V	5 V ± 5 % or 10 to 30 V	9 to 36 V	5 V ± 5 %
Current consumption without load	≤ 110 mA ⁸⁾	≤ 110 mA ⁸⁾	≤ 100 mA or ≤ 45 mA	≤ 150 mA at 24 V	≤ 160 mA
Electrical connection*	• Flange socket M12, radial • Cable 1 m, with M12 coupling	• Flange socket M23, axial or radial • Cable 1 m/5 m, with or without M23 coupling		Three flange sockets, M12 radial	• Flange socket M23, radial • Cable 1 m, with M23 coupling
Shaft	Solid shaft D = 10 mm				
Mech. permissible speed n	≤ 12000 min ^{–1}				
Starting torque	≤ 0.01 Nm (at 20 °C)				
Moment of inertia of rotor	≤ 2.3 · 10 ^{–6} kgm ²				
Shaft load ⁵⁾	Axial 10 N/radial 20 N at shaft end				
Vibration 55 to 2000 Hz Shock 6 ms/2 ms	≤ 300 m/s ² ; PROFIBUS-DP: ≤ 100 m/s ² (EN 60068-2-6) ≤ 1000 m/s ² /≤ 2000 m/s ² (EN 60068-2-27)				
Max. operating temp. ⁷⁾	100 °C		70 °C		100 °C
Min. operating temp.	Flange socket or fixed cable: –40 °C For frequent flexing: –10 °C		–40 °C		Flange socket or fixed cable: –40 °C For frequent flexing: –10 °C
Protection EN 60529	IP 67 at housing; IP 64 at shaft end ⁴⁾				
Weight	Approx. 0.35 kg				

Bold: These preferred versions are available on short notice

* Please select when ordering

1) Velocity-dependent deviations between the absolute value and incremental signal

2) Restricted tolerances: Signal amplitude 0.8 to 1.2 V_{PP}

3) These functions are programmable

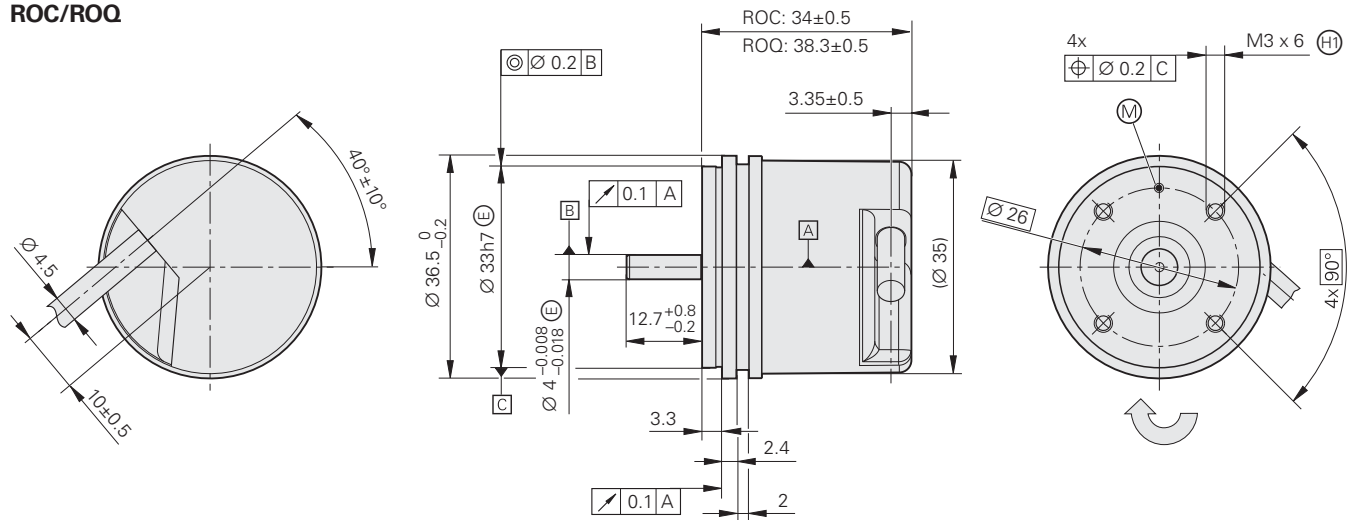
8) Depends on the power supply; see *General Electrical Information*

ROC, ROQ, ROD 1000 Series

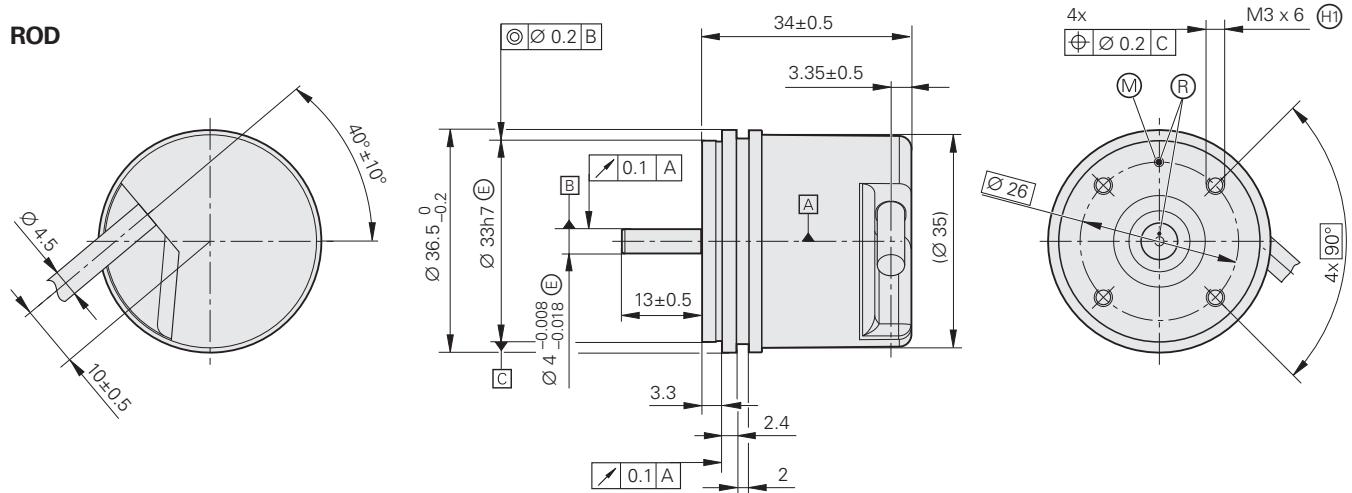
- Rotary encoders for separate shaft coupling
- Compact dimensions
- Synchro flange
- Functional Safety for ROC 1023/ROQ 1035 upon request



ROC/ROQ



ROD



Dimensions in mm



Tolerancing ISO 8015
ISO 2768 - m H
< 6 mm: ±0.2 mm

Cable radial, also usable axially

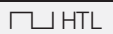
⊠ = Bearing

Ⓜ = Measuring point for operating temperature

Ⓡ = Reference mark position ± 20°

Ⓜ = Threaded mounting hole

↻ Direction of shaft rotation for output signals as per the interface description

	Incremental									
	ROD 1020			ROD 1030			ROD 1080		ROD 1070	
Incremental signals	 TTL			 HTL			 1 V _{PP} ¹⁾		 TTL	
Line counts*	100 1000	200 1024	250 1250	360 1500	400 2000	500 2048	720 2500	900 3600	1000 2500 3600	
Integrated interpolation*	–							5-fold		10-fold
Cutoff frequency –3 dB	–			–			≥ 180 kHz		–	
Scanning frequency	≤ 300 kHz			≤ 160 kHz			–		≤ 100 kHz	
Edge separation a	≥ 0.39 μs			≥ 0.76 μs			–		≥ 0.47 μs	
Power supply	5 V ± 10 %			10 to 30 V			5 V ± 10 %		5 V ± 5 %	
Current consumption without load	≤ 120 mA			≤ 150 mA			≤ 120 mA		≤ 155 mA	
Electrical connection	Cable 1 m/5 m, with or without coupling M23							Cable 5 m without M23 coupling		
Shaft	Solid shaft D = 4 mm									
Mech. permissible speed n	≤ 12 000 min ^{–1}									
Starting torque	≤ 0.001 Nm (at 20 °C)									
Moment of inertia of rotor	≤ 0.5 · 10 ^{–6} kgm ²									
Shaft load	Axial: 5 N Radial: 10 N at shaft end									
Vibration 55 to 2000 Hz	≤ 100 m/s ² (EN 60068-2-6)									
Shock 6 ms	≤ 1 000 m/s ² (EN 60068-2-27)									
Max. operating temp. ²⁾	100 °C			70 °C			100 °C		70 °C	
Min. operating temp.	For fixed cable: –30 °C For frequent flexing: –10 °C									
Protection EN 60529	IP 64									
Weight	Approx. 0.09 kg									

Bold: These preferred versions are available on short notice

* Please select when ordering

¹⁾ Restricted tolerances: Signal amplitude 0.8 to 1.2 V_{PP}

²⁾ For the correlation between the operating temperature and the shaft speed or supply voltage, see *General Mechanical Information*



Absolute
Singleturn
ROC 1023

ROC 1013

Absolute position values	EnDat 2.2	
Ordering designation	EnDat 22	EnDat 01
Positions per revolution	8388608 (23 bits)	8192 (13 bits)
Revolutions	–	
Code	Pure binary	
Elec. permissible speed Deviations ¹⁾	12000 min ⁻¹ (for continuous position value)	4000 min ⁻¹ /12000 min ⁻¹ ± 1 LSB/± 16 LSB
Calculation time t _{cal}	≤ 5 µs	
Incremental signals	–	~ 1 V _{PP} ²⁾
Line count	–	512
Cutoff frequency –3 dB	–	≥ 190 kHz
System accuracy	± 60"	
Power supply Current consumption without load	3.6 V to 14 V ≤ 110 mA ³⁾	
Electrical connection	Cable 1 m, with M12 coupling	Cable 1 m, with M23 coupling
Shaft	Stub shaft Ø 4 mm	
Mech. permissible speed n	12000 min ⁻¹	
Starting torque	≤ 0.001 Nm (at 20 °C)	
Moment of inertia of rotor	Approx. 0.5 · 10 ⁻⁶ kgm ²	
Shaft load	Axial: 5 N Radial: 10 N at shaft end	
Vibration 55 to 2000 Hz Shock 6 ms	≤ 100 m/s ² (EN 60068-2-6) ≤ 1000 m/s ² (EN 60068-2-27)	
Max. operating temp.	100 °C	
Min. operating temp.	For fixed cable: –30 °C For frequent flexing: –10 °C	
Protection EN 60529	IP 64	
Weight	Approx. 0.09 kg	

¹⁾ Velocity-dependent deviations between the absolute and incremental signals

²⁾ Restricted tolerances: Signal amplitude 0.80 to 1.2 V_{PP}

³⁾ Depends on the power supply; see *General Electrical Information*

Multiturn		
ROQ 1035		ROQ 1025
	EnDat 22	EnDat 01
	8388608 (23 bits)	8192 (13 bits)
	4096 (12 bits)	
	12000 min ⁻¹ (for continuous position value)	4000 min ⁻¹ /12000 min ⁻¹ ± 1 LSB/± 16 LSB
	–	~ 1 V _{PP} ²⁾
	–	512
	–	≥ 190 kHz
	3.6 V to 14 V ≤ 140 mA ³⁾	
	Cable 1 m, with M12 coupling	Cable 1 m, with M23 coupling
	≤ 0.002 Nm (at 20 °C)	

Interfaces

Incremental Signals $\sim 1 V_{PP}$

HEIDENHAIN encoders with $\sim 1 V_{PP}$ interface provide voltage signals that can be highly interpolated.

The sinusoidal **incremental signals** A and B are phase-shifted by 90° elec. and have an amplitude of typically $1 V_{PP}$. The illustrated sequence of output signals—with B lagging A—applies for the direction of motion shown in the dimension drawing.

The **reference mark signal** R has a usable component G of approx. $0.5 V$. Next to the reference mark, the output signal can be reduced by up to $1.7 V$ to an idle level H. This must not cause the subsequent electronics to overdrive. Even at the lowered signal level, signal peaks with the amplitude G can also appear.

The data on **signal amplitude** apply when the power supply given in the specifications is connected to the encoder. They refer to a differential measurement at the 120 ohm terminating resistor between the associated outputs. The signal amplitude decreases with increasing frequency. The **cutoff frequency** indicates the scanning frequency at which a certain percentage of the original signal amplitude is maintained:

- $-3 \text{ dB} \triangleq 70 \%$ of the signal amplitude
- $-6 \text{ dB} \triangleq 50 \%$ of the signal amplitude

The data in the signal description apply to motions at up to 20% of the -3 dB cutoff frequency.

Interpolation/resolution/measuring step

The output signals of the $1 V_{PP}$ interface are usually interpolated in the subsequent electronics in order to attain sufficiently high resolutions. For **velocity control**, interpolation factors are commonly over 1000 in order to receive usable velocity information even at low speeds.

Measuring steps for **position measurement** are recommended in the specifications. For special applications, other resolutions are also possible.

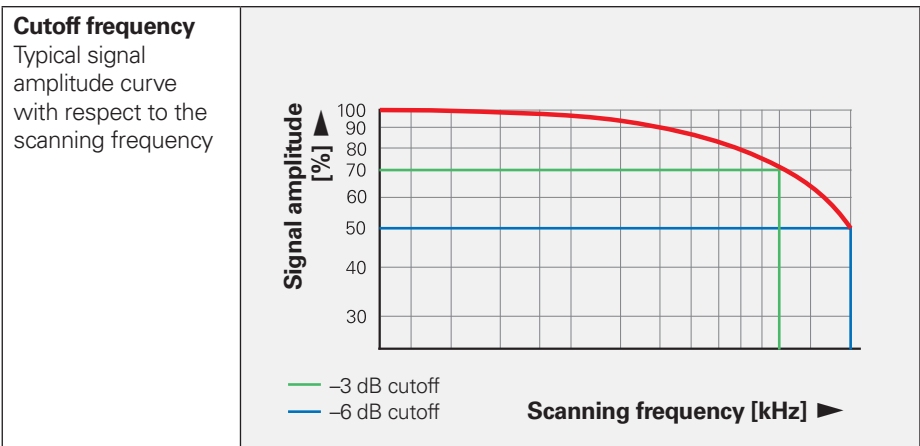
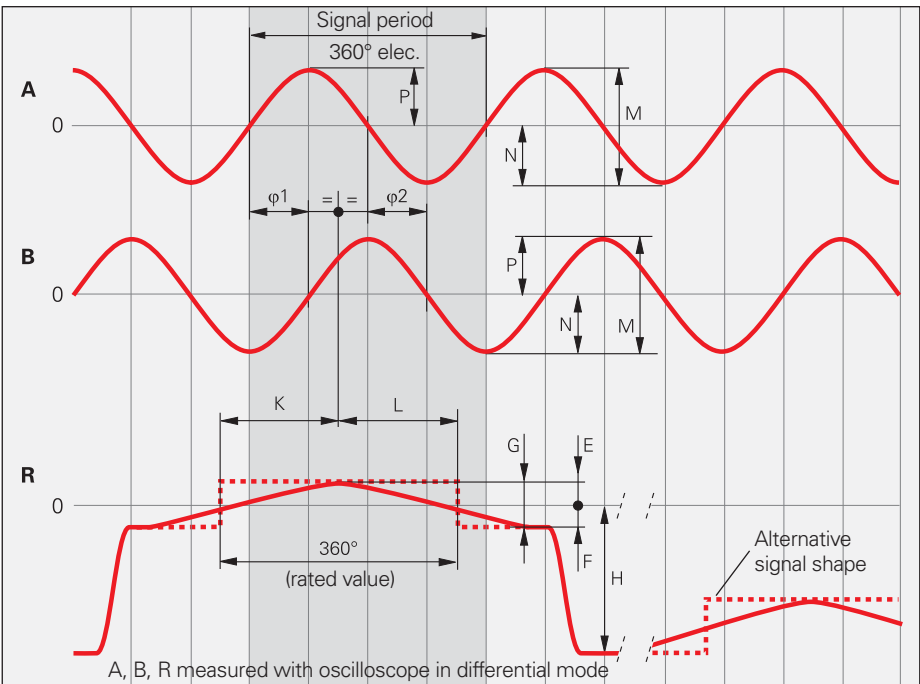
Short-circuit stability

A temporary short circuit of one signal output to $0 V$ or U_P (except encoders with $U_{Pmin} = 3.6 V$) does not cause encoder failure, but it is not a permissible operating condition.

Short circuit at	20 °C	125 °C
One output	< 3 min	< 1 min
All outputs	< 20 s	< 5 s

Interface	Sinusoidal voltage signals $\sim 1 V_{PP}$
Incremental signals	2 nearly sinusoidal signals A and B Signal amplitude M: 0.6 to $1.2 V_{PP}$; typically $1 V_{PP}$ Asymmetry $ P - N /2M$: ≤ 0.065 Amplitude ratio M_A/M_B : 0.8 to 1.25 Phase angle $ \varphi_1 + \varphi_2 /2$: $90^\circ \pm 10^\circ$ elec.
Reference mark signal	One or more signal peaks R Usable component G: $\geq 0.2 V$ Quiescent value H: $\leq 1.7 V$ Switching threshold E, F: 0.04 to $0.68 V$ Zero crossovers K, L: $180^\circ \pm 90^\circ$ elec.
Connecting cables	Shielded HEIDENHAIN cable PUR $[4(2 \times 0.14 \text{ mm}^2) + (4 \times 0.5 \text{ mm}^2)]$ Cable length Propagation time Max. 150 m at 90 pF/m distributed capacitance 6 ns/m

These values can be used for dimensioning of the subsequent electronics. Any limited tolerances in the encoders are listed in the specifications. For encoders without integral bearing, reduced tolerances are recommended for initial servicing (see the mounting instructions).



Input circuitry of the subsequent electronics

Dimensioning

Operational amplifier MC 34074
 $Z_0 = 120\ \Omega$
 $R_1 = 10\ \text{k}\Omega$ and $C_1 = 100\ \text{pF}$
 $R_2 = 34.8\ \text{k}\Omega$ and $C_2 = 10\ \text{pF}$
 $U_B = \pm 15\ \text{V}$
 U_1 approx. U_0

-3dB cutoff frequency of circuitry

Approx. 450 kHz
Approx. 50 kHz with $C_1 = 1000\ \text{pF}$
and $C_2 = 82\ \text{pF}$
The circuit variant for 50 kHz does reduce the bandwidth of the circuit, but in doing so it improves its noise immunity.

Circuit output signals

$U_a = 3.48\ \text{V}_{PP}$ typical
Gain 3.48

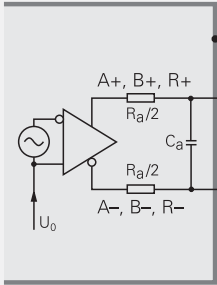
Monitoring of the incremental signals

The following sensitivity levels are recommended for monitoring the signal amplitude M:
Lower threshold: 0.30 V_{PP}
Upper threshold: 1.35 V_{PP}

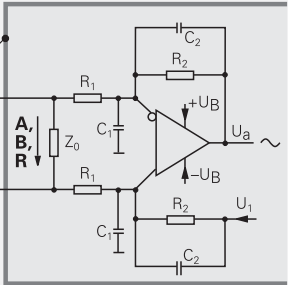
Incremental signals
Reference mark signal

$R_a < 100\ \Omega$,
typically $24\ \Omega$
 $C_a < 50\ \text{pF}$
 $\Sigma I_a < 1\ \text{mA}$
 $U_0 = 2.5\ \text{V} \pm 0.5\ \text{V}$
(relative to 0 V of the power supply)

Encoder



Subsequent electronics



Pin layout

12-pin coupling M23					12-pin M23 connector					15-pin D-sub connector for IK 215 or on encoder				
Power supply					Incremental signals					Other signals				
	12	2	10	11	5	6	8	1	3	4	9	7	/	
	4	12	2	10	1	9	3	11	14	7	5/6/8/15	13	/	
	U_P	Sensor U_P	0 V	Sensor 0 V	A+	A-	B+	B-	R+	R-	Vacant	Vacant	Vacant	
	Brown/ Green	Blue	White/ Green	White	Brown	Green	Gray	Pink	Red	Black	/	Violet	Yellow	

Shield on housing; U_P = Power supply voltage

Sensor: The sensor line is connected in the encoder with the corresponding power line

Interfaces

Incremental Signals TTL

HEIDENHAIN encoders with  TTL interface incorporate electronics that digitize sinusoidal scanning signals with or without interpolation.

The **incremental signals** are transmitted as the square-wave pulse trains U_{a1} and U_{a2} , phase-shifted by 90° elec. The **reference mark signal** consists of one or more reference pulses U_{a0} , which are gated with the incremental signals. In addition, the integrated electronics produce their **inverse signals** $\overline{U_{a1}}$, $\overline{U_{a2}}$ and $\overline{U_{a0}}$ for noise-proof transmission. The illustrated sequence of output signals—with U_{a2} lagging U_{a1} —applies for the direction of motion shown in the dimension drawing.

The **fault-detection signal** $\overline{U_{aS}}$ indicates fault conditions such as breakage of the power line or failure of the light source. It can be used for such purposes as machine shut-off during automated production.

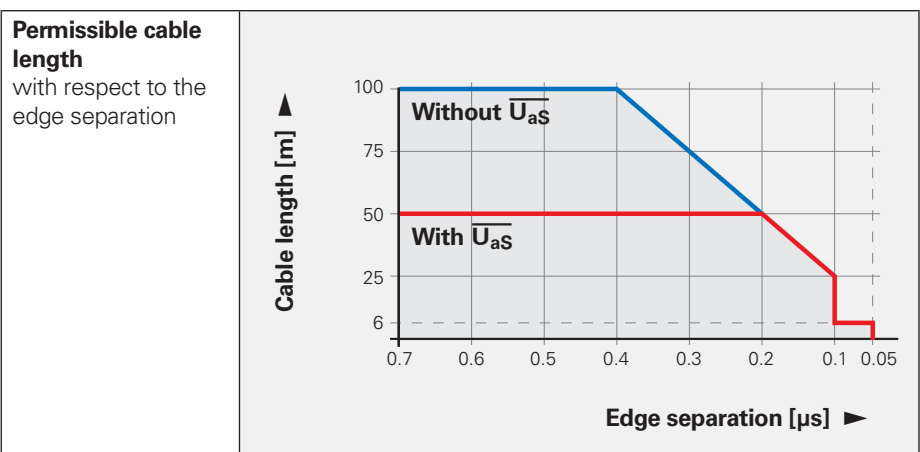
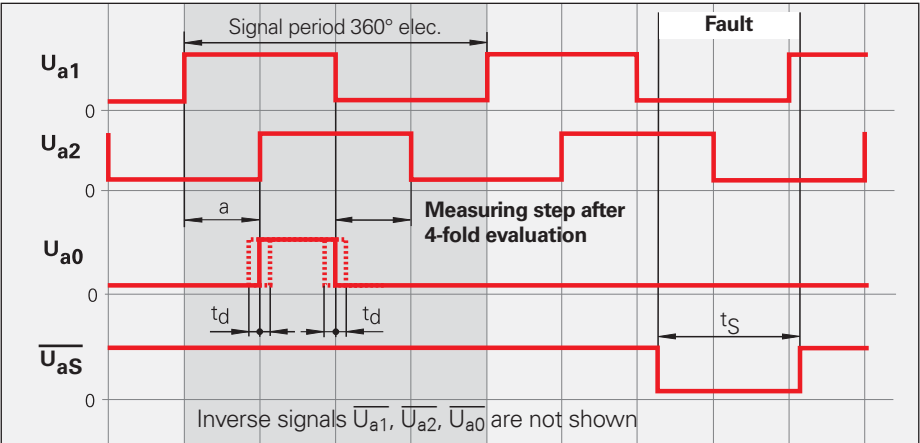
The distance between two successive edges of the incremental signals U_{a1} and U_{a2} through 1-fold, 2-fold or 4-fold evaluation is one **measuring step**.

The subsequent electronics must be designed to detect each edge of the square-wave pulse. The minimum **edge separation a** listed in the *Specifications* applies to the illustrated input circuitry with a cable length of 1 m, and refers to a measurement at the output of the differential line receiver. Propagation-time differences in cables additionally reduce the edge separation by 0.2 ns per meter of cable length. To prevent counting error, design the subsequent electronics to process as little as 90 % of the resulting edge separation.

The max. permissible **shaft speed** or **traversing velocity** must never be exceeded.

The permissible **cable length** for transmission of the TTL square-wave signals to the subsequent electronics depends on the edge separation a . It is max. 100 m, or 50 m for the fault detection signal. This requires, however, that the power supply (see *Specifications*) be ensured at the encoder. The sensor lines can be used to measure the voltage at the encoder and, if required, correct it with an automatic system (remote sense power supply).

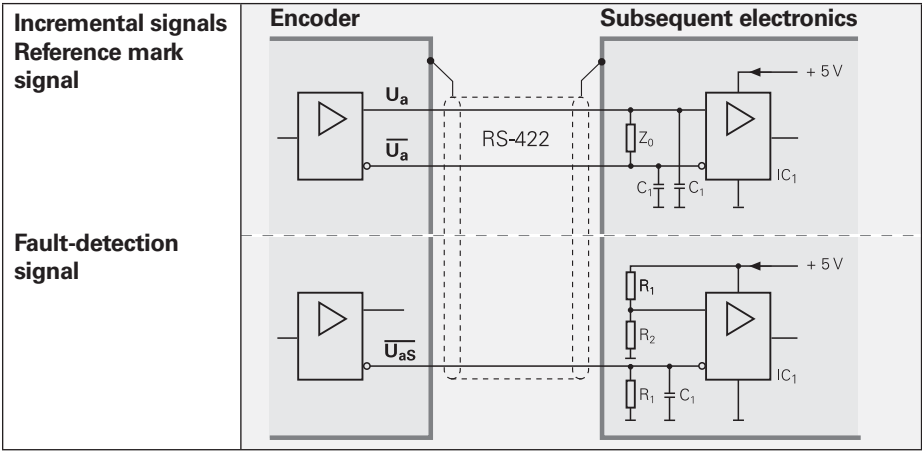
Interface	Square-wave signals 
Incremental signals	2 TTL square-wave signals U_{a1}, U_{a2} and their inverted signals $\overline{U_{a1}}$, $\overline{U_{a2}}$
Reference mark signal Pulse width Delay time	1 or more TTL square-wave pulses U_{a0} and their inverted pulses $\overline{U_{a0}}$ 90° elec. (other widths available on request); LS 323: ungated $ t_d \leq 50$ ns
Fault-detection signal Pulse width	1 TTL square-wave pulse $\overline{U_{aS}}$ Improper function: LOW (upon request: U_{a1}/U_{a2} high impedance) Proper function: HIGH $t_s \geq 20$ ms
Signal amplitude	Differential line driver as per EIA standard RS 422 $U_H \geq 2.5$ V at $-I_H = 20$ mA $U_L \leq 0.5$ V at $I_L = 20$ mA
Permissible load	$Z_0 \geq 100 \Omega$ between associated outputs $ I_L \leq 20$ mA max. load per output $C_{load} \leq 1000$ pF with respect to 0 V Outputs protected against short circuit to 0 V
Switching times (10 % to 90 %)	$t_r / t_f \leq 30$ ns (typically 10 ns) with 1 m cable and recommended input circuitry
Connecting cables Cable length Propagation time	Shielded HEIDENHAIN cable PUR [$4(2 \times 0.14 \text{ mm}^2) + (4 \times 0.5 \text{ mm}^2)$] Max. 100 m ($\overline{U_{aS}}$ max. 50 m) at distributed capacitance 90 pF/m 6 ns/m



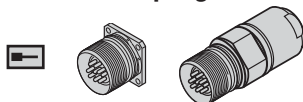
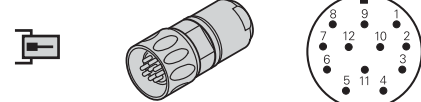
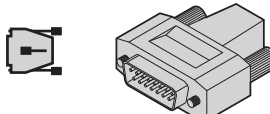
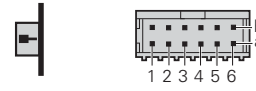
Input circuitry of the subsequent electronics

Dimensioning
IC₁ = Recommended differential line receiver
DS 26 C 32 AT
Only for a > 0.1 µs:
AM 26 LS 32
MC 3486
SN 75 ALS 193

R₁ = 4.7 kΩ
R₂ = 1.8 kΩ
Z₀ = 120 Ω
C₁ = 220 pF (serves to improve noise immunity)



Pin layout

12-pin flange socket or M23 coupling					12-pin M23 connector								
													
15-pin D-sub connector at encoder					12-pin PCB connector								
													
	Power supply				Incremental signals						Other signals		
	12	2	10	11	5	6	8	1	3	4	7	/	9
	4	12	2	10	1	9	3	11	14	7	13	5/6/8	15
	2a	2b	1a	1b	6b	6a	5b	5a	4b	4a	3a	3b	/
	U _P	Sensor U _P	0V	Sensor 0V	U _{a1}	$\overline{U_{a1}}$	U _{a2}	$\overline{U_{a2}}$	U _{a0}	$\overline{U_{a0}}$	$\overline{U_{a5}}$ ¹⁾	Vacant	Vacant ²⁾
	Brown/ Green	Blue	White/ Green	White	Brown	Green	Gray	Pink	Red	Black	Violet	–	Yellow

Shield on housing; **U_P** = Power supply voltage

Sensor: The sensor line is connected in the encoder with the corresponding power line

¹⁾ **LS 323/ERO 14xx:** vacant

²⁾ **Exposed linear encoders:** Switchover TTL/11 µA_{PP} for PWT

Interfaces

Incremental Signals HTL

HEIDENHAIN encoders with  HTL interface incorporate electronics that digitize sinusoidal scanning signals with or without interpolation.

The **incremental signals** are transmitted as the square-wave pulse trains U_{a1} and U_{a2} , phase-shifted by 90° elec. The **reference mark signal** consists of one or more reference pulses U_{a0} , which are gated with the incremental signals. In addition, the integrated electronics produce their **inverse signals** $\overline{U_{a1}}$, $\overline{U_{a2}}$ and $\overline{U_{a0}}$ for noise-proof transmission (not with ERN/ ROD 1x30).
The illustrated sequence of output signals—with U_{a2} lagging U_{a1} —applies for the direction of motion shown in the dimension drawing.

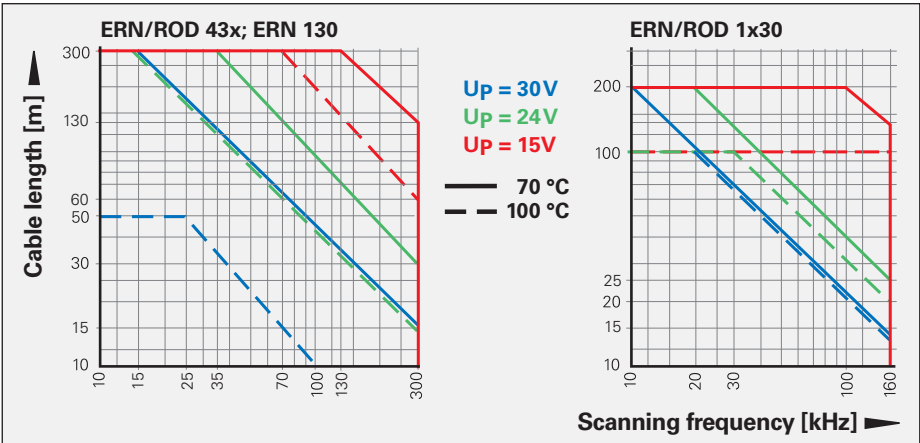
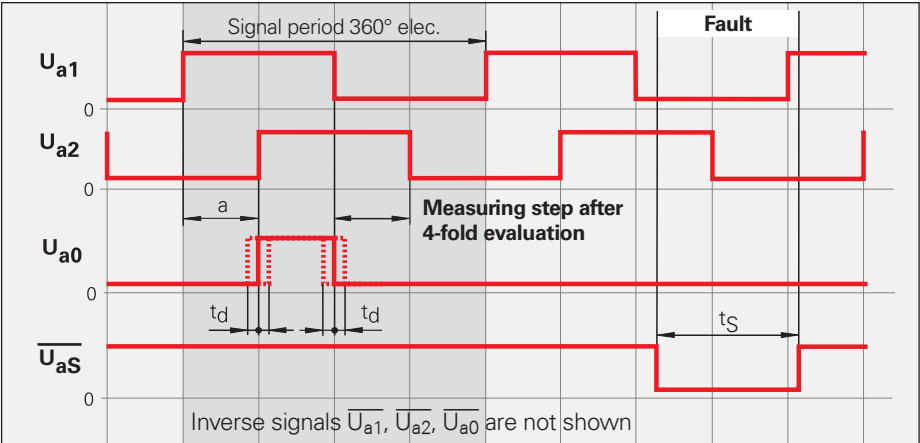
The **fault-detection signal** $\overline{U_{aS}}$ indicates fault conditions such as failure of the light source. It can be used for such purposes as machine shut-off during automated production.

The distance between two successive edges of the incremental signals U_{a1} and U_{a2} through 1-fold, 2-fold or 4-fold evaluation is one **measuring step**.

The subsequent electronics must be designed to detect each edge of the square-wave pulse. The minimum **edge separation a** listed in the *Specifications* refers to a measurement at the output of the given differential input circuitry. To prevent counting error, the subsequent electronics should be designed to process as little as 90 % of the edge separation a. The max. permissible **shaft speed** or **traversing velocity** must never be exceeded.

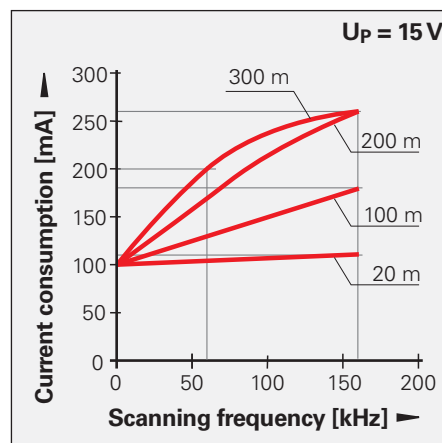
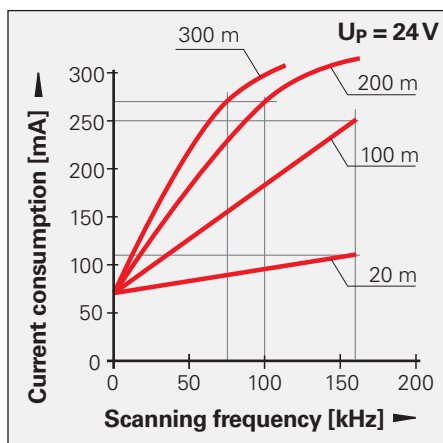
The permissible **cable length** for incremental encoders with HTL signals depends on the scanning frequency, the effective power supply, and the operating temperature of the encoder.

Interface	Square-wave signals  HTL	
Incremental signals	2 HTL square-wave signals U_{a1} , U_{a2} and their inverse signals $\overline{U_{a1}}$, $\overline{U_{a2}}$ (ERN/ROD 1x30 without $\overline{U_{a1}}$, $\overline{U_{a2}}$)	
Reference mark signal	1 or more HTL square-wave pulses U_{a0} and their inverted pulses $\overline{U_{a0}}$ (ERN/ROD 1x30 without $\overline{U_{a0}}$)	
Pulse width	90° elec. (other widths available on request)	
Delay time	$ t_d \leq 50 \text{ ns}$	
Fault-detection signal	1 HTL square-wave pulse $\overline{U_{aS}}$	
Pulse width	Improper function: LOW Proper function: HIGH $t_s \geq 20 \text{ ms}$	
Signal level	$U_H \geq 21 \text{ V}$ at $-I_H = 20 \text{ mA}$ $U_L \leq 2.8 \text{ V}$ at $I_L = 20 \text{ mA}$	With power supply of $U_P = 24 \text{ V}$, without cable
Permissible load	$ I_L \leq 100 \text{ mA}$ $C_{load} \leq 10 \text{ nF}$ Outputs short-circuit proof for max. 1 minute after 0 V and U_P (except $\overline{U_{aS}}$)	Max. load per output, (except $\overline{U_{aS}}$) With respect to 0 V
Switching times	$t_r / t_f \leq 200 \text{ ns}$ (except $\overline{U_{aS}}$) (10 % to 90 %) with 1 m cable and recommended input circuitry	
Connecting cables	HEIDENHAIN cable with shielding PUR $[4(2 \times 0.14 \text{ mm}^2) + (4 \times 0.5 \text{ mm}^2)]$ Max. 300 m (ERN/ROD 1x30 max. 100 m) distributed capacitance 90 pF/m 6 ns/m	
Cable length		
Propagation time		

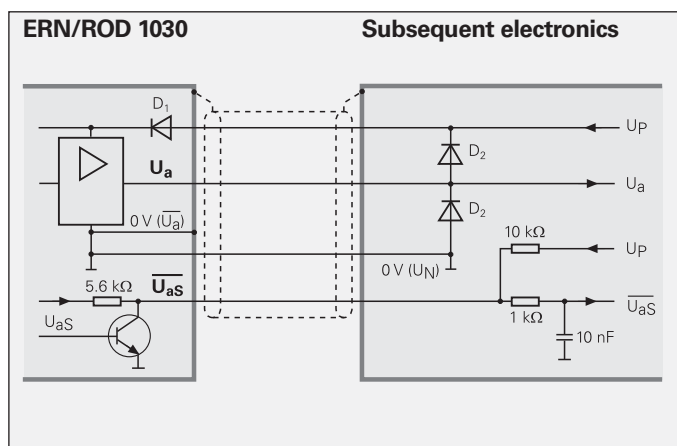
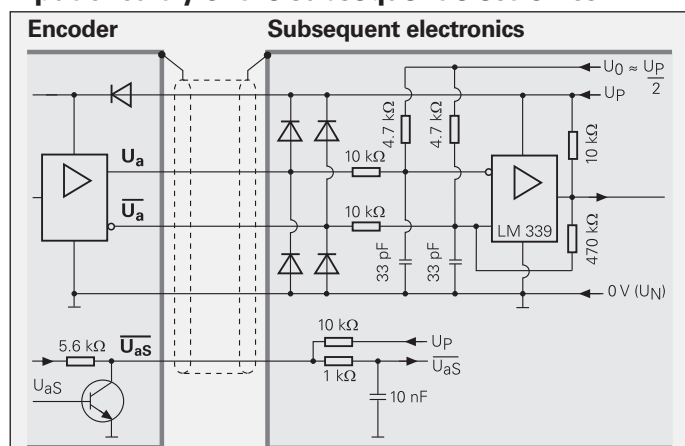


Current consumption

The current consumption for encoders with HTL output signals depends on the output frequency and the cable length to the subsequent electronics. The diagrams show typical curves for push-pull transmission with a 12-line HEIDENHAIN cable. The maximum current consumption may be 50 mA higher.



Input circuitry of the subsequent electronics



Pin layout

12-pin flange socket or M23 coupling					12-pin PCB connector								
Power supply					Incremental signals					Other signals			
12	2	10	11		5	6	8	1	3	4	7	/	9
2a	2b	1a	1b		6b	6a	5b	5a	4b	4a	3a	3b	/
Up	Sensor Up	0 V	Sensor 0 V		Ua1	Ua1-bar	Ua2	Ua2-bar	Ua0	Ua0-bar	UaS	Vacant	Vacant
Brown/ Green	Blue	White/ Green	White		Brown	Green	Gray	Pink	Red	Black	Violet	/	Yellow

Shield on housing; **Up** = Power supply voltage

Sensor: The sensor line is connected in the encoder with the corresponding power line

ERN 1x30, ROD 1030: 0 V instead of inverse signals $\overline{U_{a1}}$, $\overline{U_{a2}}$, $\overline{U_{a0}}$

Interfaces

Absolute Position Values EnDat

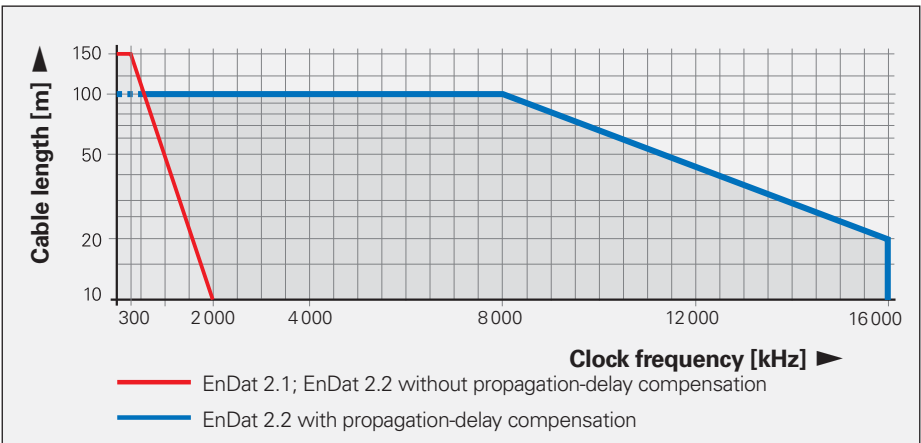
The EnDat interface is a digital, **bidirectional** interface for encoders. It is capable of transmitting **position values** from both absolute and—with EnDat 2.2—incremental encoders, as well as reading and updating information stored in the encoder, or of saving new information. Thanks to the **serial transmission method**, only **four signal lines** are required. The data is transmitted in **synchronism** with the CLOCK signal from the subsequent electronics. The type of transmission (position values, parameters, diagnostics, etc.) is selected through mode commands that the subsequent electronics send to the encoder.

Clock frequency and cable length

Without propagation-delay compensation, the **clock frequency**—depending on the cable length—is variable between **100 kHz** and **2 MHz**.

Because large cable lengths and high clock frequencies increase the signal run time to the point that they can disturb the unambiguous assignment of data, the delay can be measured in a test run and then compensated. With this **propagation-delay compensation** in the subsequent electronics, clock frequencies **up to 16 MHz** at cable lengths up to a maximum of 100 m ($f_{CLK} \leq 8 \text{ MHz}$) are possible. The maximum clock frequency is mainly determined by the cables and connecting elements used. To ensure proper function at clock frequencies above 2 MHz, use only original ready-made HEIDENHAIN cables.

Interface	EnDat serial bidirectional
Data transfer	Absolute position values, parameters and additional information
Data input	Differential line receiver according to EIA standard RS 485 for the signals CLOCK, $\overline{\text{CLOCK}}$, DATA and $\overline{\text{DATA}}$
Data output	Differential line driver according to EIA standard RS 485 for the signals DATA and $\overline{\text{DATA}}$
Code	Pure binary code
Position values	Ascending during traverse in direction of arrow (see dimensions of the encoders)
Incremental signals	$\sim 1 V_{PP}$ (see <i>Incremental Signals 1 V_{PP}</i>) depending on the unit
Connecting cables	Shielded HEIDENHAIN cable With Incremental signals PUR [(4 x 0.14 mm ²) + 4(2 x 0.14 mm ²) + (4 x 0.5 mm ²)] With out signals PUR [(4 x 0.14 mm ²) + (4 x 0.34 mm ²)]
Cable length	max. 150 m
Propagation time	Max. 10 ns; typically 6 ns/m



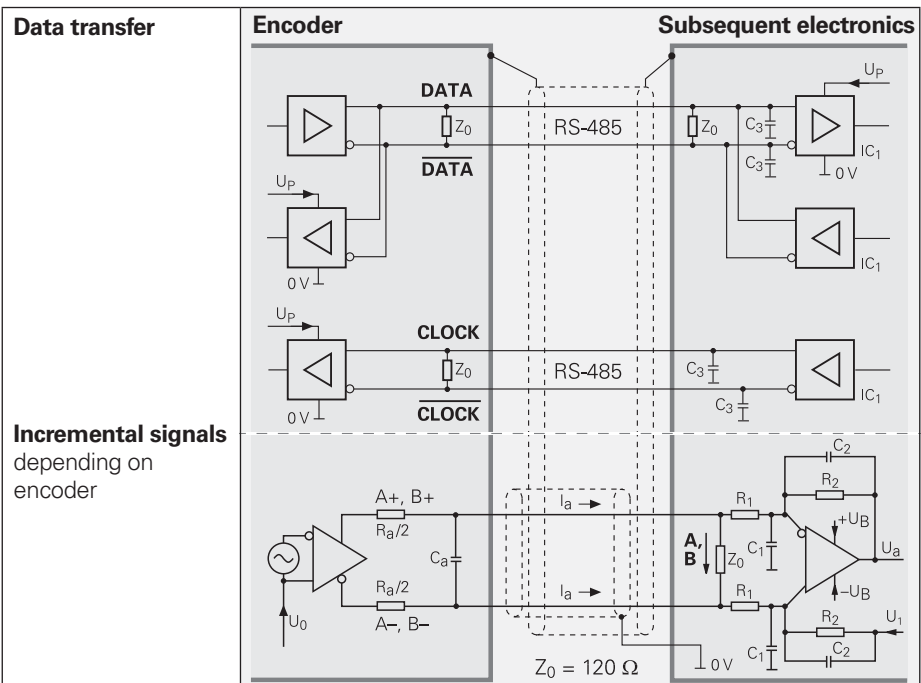
Input circuitry of the subsequent electronics

Dimensioning

IC₁ = RS 485 differential line receiver and driver

$$C_3 = 330 \text{ pF}$$

$$Z_0 = 120 \Omega$$



Benefits of the EnDat Interface

- **Automatic self-configuration:** All information required by the subsequent electronics is already stored in the encoder.
- **High system security** through alarms and messages for monitoring and diagnosis.
- **High transmission reliability** through cyclic redundancy checks.
- **Datum shift** for faster commissioning.

Other benefits of EnDat 2.2

- **A single interface** for all absolute and incremental encoders.
- **Additional information** (limit switch, temperature, acceleration)
- **Quality improvement:** Position value calculation in the encoder permits shorter sampling intervals (25 µs).
- **Online diagnostics** through valuation numbers that indicate the encoder's current functional reserves and make it easier to plan the machine servicing.
- **Safety system** for designing safety-oriented control systems consisting of safe controls and safe encoders based on the IEC 61508 and DIN EN ISO 13849-1 standards.

Advantages of purely serial

transmission specifically for EnDat 2.2 encoders

- Cost optimization through **simple subsequent electronics** with EnDat receiver component and **simple connection technology:** Standard connecting element (M12; 8-pin), single-shielded standard cables and low wiring cost.
- **Minimized transmission times** through **high clock frequencies** up to 16 MHz. Position values available in the subsequent electronics after only approx. 10 µs.
- **Support for state-of-the-art machine designs** e.g. direct drive technology.

Ordering designation	Command set	Incremental signals	Clock frequency	Power supply
EnDat 01	EnDat 2.1 or EnDat 2.2	With	≤ 2 MHz	See specifications of the encoder
EnDat 21		Without		
EnDat 02	EnDat 2.2	With	≤ 2 MHz	Extended range 3.6 to 5.25 V or 14 V
EnDat 22	EnDat 2.2	Without	≤ 16 MHz	

Specification of the EnDat interface (bold print indicates standard versions)

Versions

The extended EnDat 2.2 interface version is compatible in its communication, command set and time conditions with version 2.1, but also offers significant advantages. It makes it possible, for example, to transfer additional information with the position value without sending a separate request for it. The interface protocol was expanded and the time conditions (clock frequency, processing time, recovery time) were optimized.

Ordering designation

Indicated on the ID label and can be read out via parameter.

Command set

The command set is the sum of all available mode commands. (See "Selecting the transmission type"). The EnDat 2.2 command set includes EnDat 2.1 mode commands. When a mode command from the EnDat 2.2 command set is transmitted to EnDat-01 subsequent electronics, the encoder or the subsequent electronics may generate an error message.

Incremental signals

EnDat 2.1 and EnDat 2.2 are both available with or without incremental signals. EnDat 2.2 encoders feature a high internal resolution. Therefore, depending on the control technology being used, interrogation of the incremental signals is not necessary. To increase the resolution of EnDat 2.1 encoders, the incremental signals are interpolated and evaluated in the subsequent electronics.

Power supply

Encoders with ordering designations EnDat 02 and EnDat 22 have an extended power supply range.

Functions

The EnDat interface transmits absolute position values or additional physical quantities (only EnDat 2.2) in an unambiguous time sequence and serves to read from and write to the encoder's internal memory. Some functions are available only with EnDat 2.2 mode commands.

Position values can be transmitted with or without additional information. The additional information types are selectable via the Memory Range Select (MRS) code. Other functions such as *Read parameter* and *Write parameter* can also be called after the memory area and address have been selected. Through simultaneous transmission with the position value, additional information can also be requested of axes in the feedback loop, and functions executed with them.

Parameter reading and writing is possible both as a separate function and in connection with the position value. Parameters can be read or written after the memory area and address is selected.

Reset functions serve to reset the encoder in case of malfunction. Reset is possible instead of or during position value transmission.

Servicing diagnosis makes it possible to inspect the position value even at standstill. A test command has the encoder transmit the required test values.

You can find more information on the EnDat 2.2 on the Internet under www.endat.de or in the *Technical Information EnDat 2.2*.

Selecting the Transmission Type

Transmitted data are identified as either position values, position values with additional information, or parameters. The type of information to be transmitted is selected by mode commands. **Mode commands** define the content of the transmitted information. Every mode command consists of 3 bits. To ensure reliable transmission, every bit is transmitted redundantly (inverted or double). The EnDat 2.2 interface can also transfer parameter values in the additional information together with the position value. This makes the current position values constantly available for the control loop, even during a parameter request.

Control cycles for transfer of position values

The transmission cycle begins with the first falling **clock edge**. The measured values are saved and the position value is calculated. After two clock pulses (2T), to **select the type of transmission** the subsequent electronics transmit the mode command "Encoder transmit position value" (with/without additional information). The subsequent electronics continue to transmit clock pulses and observe the data line to detect the start bit. The **start bit** starts data transmission from the encoder to the subsequent electronics. Time t_{cal} is the smallest time duration after which the position value can be read by the encoder. The subsequent **error messages**, error 1 and error 2 (only with EnDat 2.2 commands), are group signals for all monitored functions and serve as failure monitors.

Beginning with the LSB, the encoder then transmits the absolute **position value** as a complete data word. Its length varies depending on which encoder is being used. The number of required clock pulses for transmission of a position value is saved in the parameters of the encoder manufacturer. The data transmission of the position value is completed with the **Cyclic Redundancy Check (CRC)**.

In EnDat 2.2, this is followed by additional information 1 and 2, each also concluded with a CRC. With the end of the data word, the clock must be set to HIGH. After 10 to 30 μ s or 1.25 to 3.75 μ s (with EnDat 2.2 parameterizable recovery time t_m) the data line falls back to LOW. Then a **new data transmission** can begin by starting the clock.

Mode commands

- Encoder transmit position value - Selection of memory area - Encoder to receive parameters - Encoder to send parameters - Encoder receive reset¹⁾ - Encoder to send test values - Encoder to receive test command	EnDat 2.1	EnDat 2.2
- Encoder transmit position value with additional information - Encoder transmit position value and receive selection of memory area²⁾ - Encoder transmit position value and receive parameters²⁾ - Encoder transmit position value and transmit parameters²⁾ - Encoder transmit position value and receive error reset²⁾ - Encoder transmit position value and receive test command²⁾ - Encoder receive communication command³⁾		

¹⁾ Same reaction as from switching the power supply off and on

²⁾ Selected additional information is also transmitted

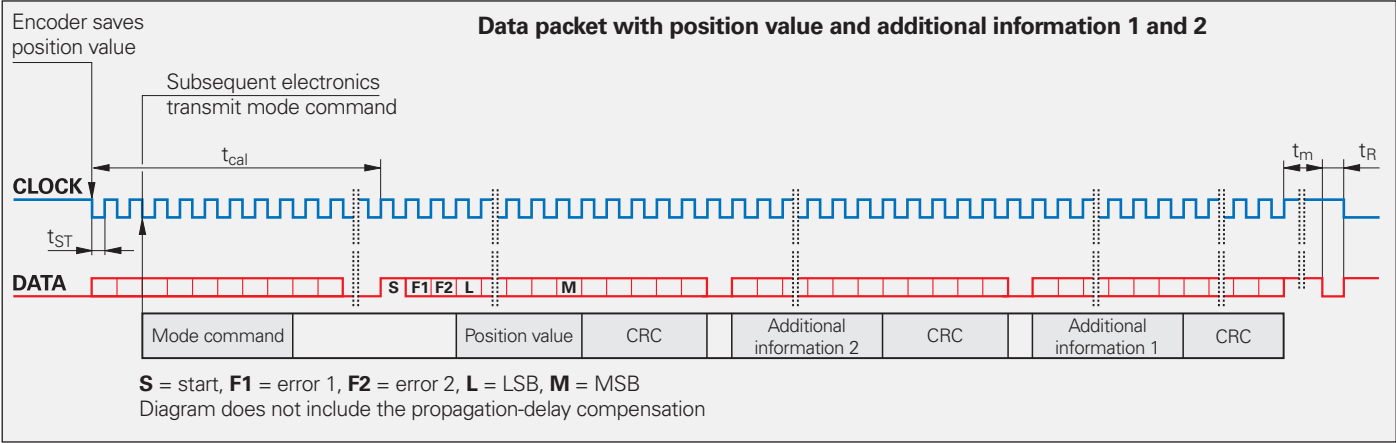
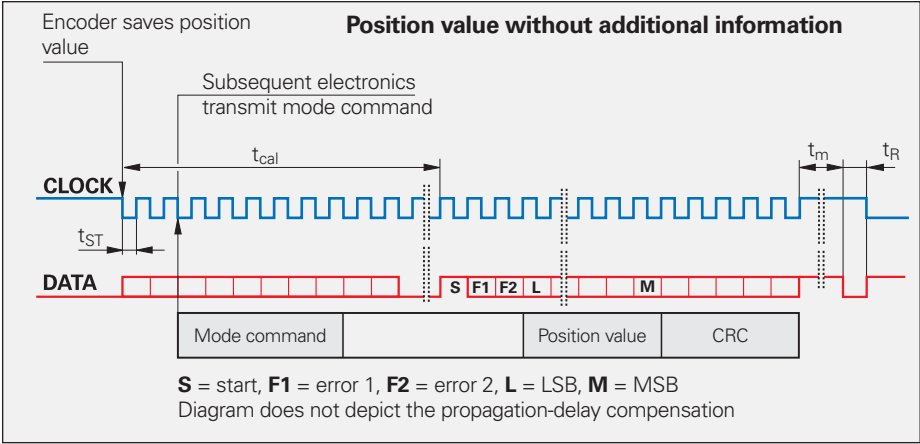
³⁾ Reserved for encoders that do not support the safety system

The time absolute linear encoders need for calculating the position values t_{cal} differs depending on whether EnDat-2.1 or EnDat-2.2 mode commands are transmitted (see catalog: *Linear Encoders for Numerically Controlled Machine Tools – Specifications*). If the incremental signals are evaluated for axis control, then the EnDat 2.1 mode commands should be used. Only in this manner can an active error message be transmitted synchronously with the currently requested position value. EnDat 2.1 mode commands should not be used for pure serial position-value transfer for axis control.

		Without delay compensation	With delay compensation
Clock frequency	f_c	100 kHz ... 2 MHz	100 kHz ... 16 MHz
Calculation time for Position value Parameters	t_{cal} t_{ac}	See <i>Specifications</i> Max. 12 ms	
Recovery time	t_m	EnDat 2.1: 10 to 30 μ s EnDat 2.2: 10 to 30 μ s or 1.25 to 3.75 μ s ($f_c \geq 1$ MHz) (parameterizable)	
	t_R	Max. 500 ns	
	t_{ST}	–	2 to 10 μ s
Data delay time	t_D	(0.2 + 0.01 x cable length in m) μ s	
Pulse width	t_{HI}	0.2 to 10 μ s	Pulse width fluctuation HIGH to LOW max. 10 %
	t_{LO}	0.2 to 50 ms/30 μ s (with LC)	

EnDat 2.2 –Transmission of Position Values

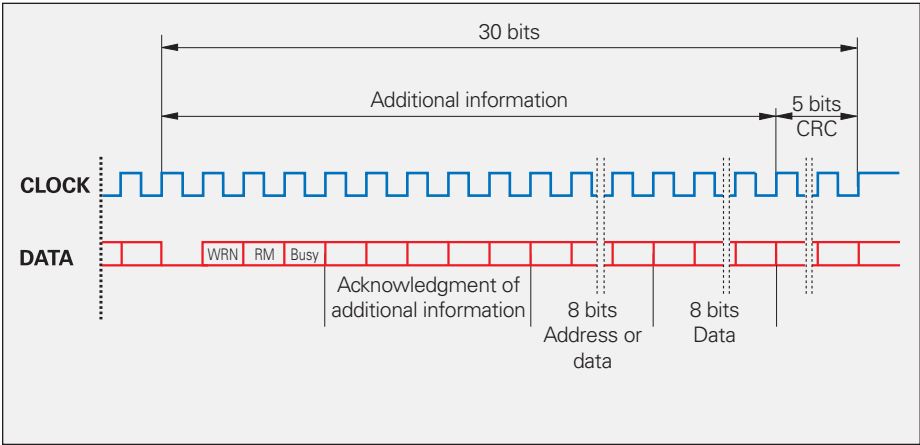
EnDat 2.2 can transmit position values with or without additional information.



Additional information

With EnDat 2.2, one or two pieces of additional information can be appended to the position value. Each additional datum is 30 bits long with LOW as first bit, and ends with a CRC check. The additional information supported by the respective encoder is saved in the encoder parameters.

The content of the additional information is determined by the MRS code and is transmitted in the next sampling cycle for additional information. This information is then transmitted with every sample until a selection of a new memory area changes the content.



The additional information always begins with:

Status data
Warning – WRN
Reference mark – RM
Parameter request – busy
Acknowledgment of additional information

The additional information can contain the following data:

Additional information 1
Diagnosis (valuation numbers)
Position value 2
Memory parameters
MRS-code acknowledgment
Test values
Encoder temperature
External temperature sensors
Sensor data

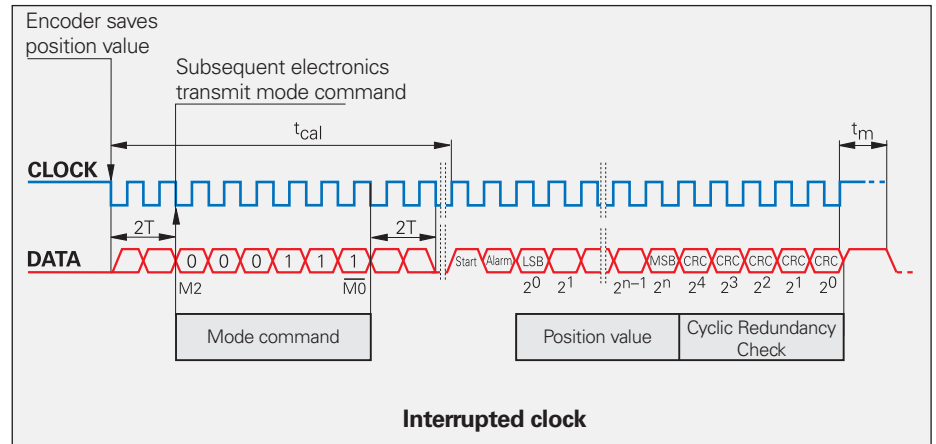
Additional information 2
Commutation
Acceleration
Limit position signals
Operating status error sources

EnDat 2.1 – Transmission of Position Values

EnDat 2.1 can transmit position values with interrupted clock pulse (as in EnDat 2.2) or continuous clock pulse.

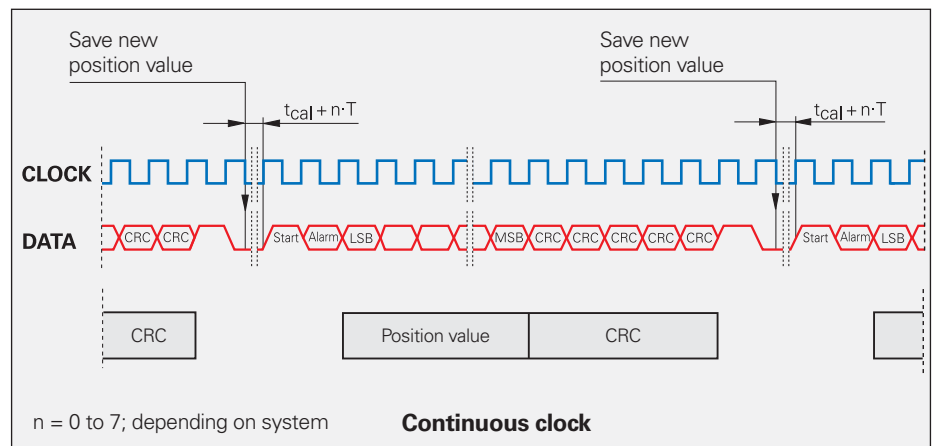
Interrupted clock

The interrupted clock is intended particularly for time-clocked systems such as closed control loops. At the end of the data word, the clock signal is set to HIGH level. After 10 to 30 μs (t_m) the data line falls back to LOW. Then a new data transmission can begin by starting the clock.



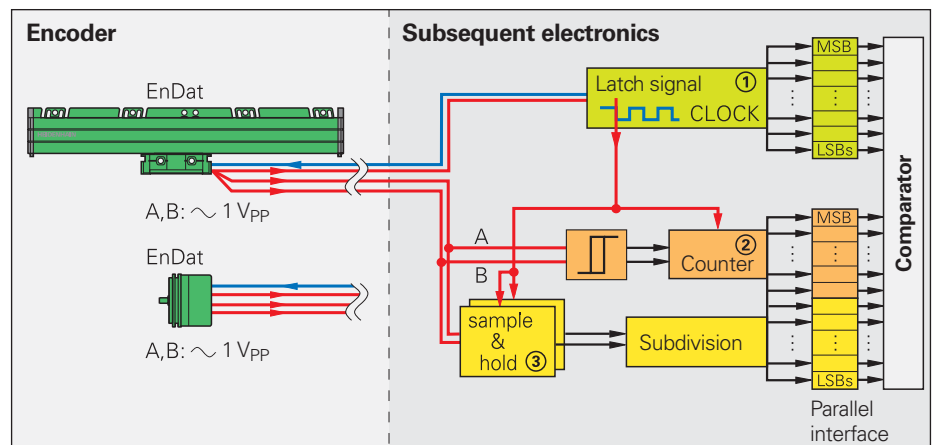
Continuous clock

For applications that require fast acquisition of the measured value, the EnDat interface can have the clock run continuously. Immediately after the last CRC bit has been sent, the data line is switched to HIGH for one clock cycle, and then to LOW. The new position value is saved with the very next falling edge of the clock and is output in synchronism with the clock signal immediately after the start bit and alarm bit. Because the mode command *Encoder transmit position value* is needed only before the first data transmission, the continuous-clock transfer mode reduces the length of the clock-pulse group by 10 periods per position value.



Synchronization of the serially transmitted code value with the incremental signal

Absolute encoders with EnDat interface can exactly synchronize serially transmitted absolute position values with incremental values. With the first falling edge (latch signal) of the **CLOCK** signal from the subsequent electronics, the scanning signals of the individual tracks in the encoder and counter are frozen, as are the A/D converters for subdividing the sinusoidal incremental signals in the subsequent electronics.



The code value transmitted over the serial interface unambiguously identifies one incremental signal period. The position value is absolute within one sinusoidal period of the incremental signal. The subdivided incremental signal can therefore be appended in the subsequent electronics to the serially transmitted code value.

After switch-on and initial transmission of position values, two redundant position values are available in the subsequent electronics. Since encoders with EnDat interface guarantee a precise synchronization—regardless of cable length—of the serially transmitted code value with the incremental signals, the two

values can be compared in the subsequent electronics. This monitoring is possible even at high shaft speeds thanks to the EnDat interface's short transmission times of less than 50 μs . This capability is a prerequisite for modern machine design and safety systems.

Parameters and Memory Areas

The encoder provides several memory areas for parameters. These can be read from by the subsequent electronics, and some can be written to by the encoder manufacturer, the OEM, or even the end user. Certain memory areas can be write-protected.

➔ The parameters, which in most cases are set by the OEM, largely define the function of the encoder and the EnDat interface. When the encoder is exchanged, it is therefore essential that its parameter settings are correct. Attempts to configure machines without including OEM data can result in malfunctions. If there is any doubt as to the correct parameter settings, the OEM should be consulted.

Parameters of the encoder manufacturer

This write-protected memory area contains all **information specific to the encoder**, such as encoder type (linear/angular, singleturn/multiturn, etc.), signal periods, position values per revolution, transmission format of position values, direction of rotation, maximum speed, accuracy dependent on shaft speeds, warnings and alarms, part number and serial number. This information forms the basis for **automatic configuration**. A separate memory area contains the parameters typical for EnDat 2.2: Status of additional information, temperature, acceleration, support of diagnostic and error messages, etc.

Parameters of the OEM

In this freely definable memory area, the OEM can store his information, e.g. the "electronic ID label" of the motor in which the encoder is integrated, indicating the motor model, maximum current rating, etc.

Operating parameters

This area is available for a **datum shift** and the configuration of diagnostics. It can be protected against overwriting.

Operating status

This memory area provides detailed alarms or warnings for diagnostic purposes. Here it is also possible to activate write protection for the OEM parameter and operating parameter memory areas, and to interrogate their status. Once activated, **the write protection** can be reversed only by HEIDENHAIN service personnel.

Monitoring and Diagnostic Functions

The EnDat interface enables comprehensive monitoring of the encoder without requiring an additional transmission line. The alarms and warnings supported by the respective encoder are saved in the "parameters of the encoder manufacturer" memory area.

Error message

An error message becomes active if a **malfunction of the encoder** might result in incorrect position values. The exact cause of the disturbance is saved in the encoder's "operating status" memory. It is also possible to interrogate over the additional information "operating status error sources." Here the EnDat interface transmits the error bits—error 1 and error 2 (only with EnDat 2.2 commands). These are group signals for all monitored functions and serve for failure monitoring. The two error messages are generated independently from each other.

Warning

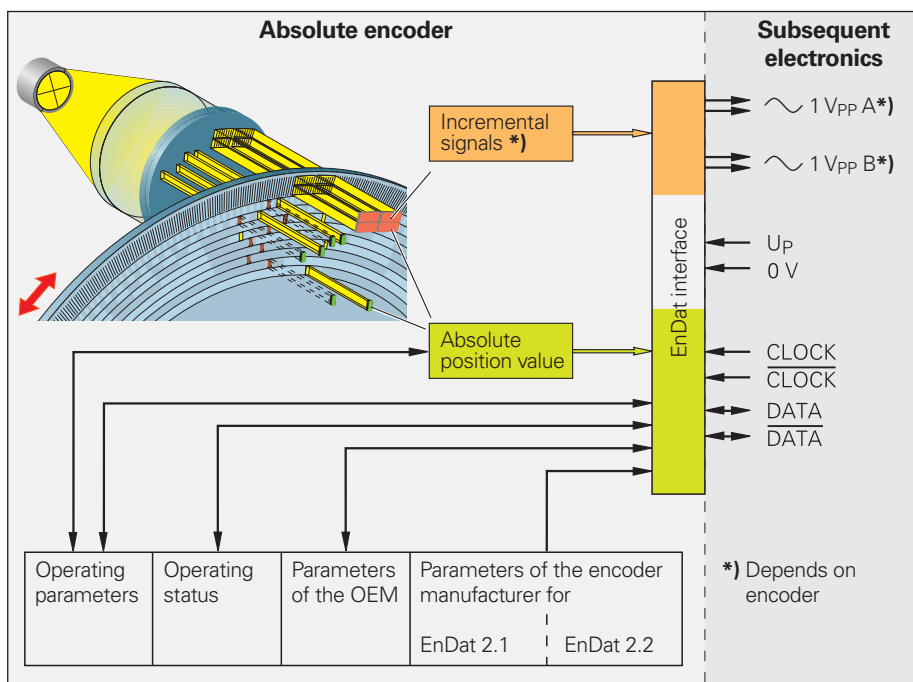
This collective bit is transmitted in the status data of the additional information. It indicates that certain **tolerance limits of the encoder** have been reached or exceeded—such as shaft speed or the limit of light source intensity compensation through voltage regulation—without implying that the measured position values are incorrect. This function makes it possible to issue preventive warnings in order to minimize idle time.

Online diagnostics

Encoders with purely serial interfaces do not provide incremental signals for evaluation of encoder function. EnDat 2.2 encoders can therefore cyclically transmit so-called valuation numbers from the encoder. The valuation numbers provide the current state of the encoder and ascertain the encoder's "functional reserves." The identical scale for all HEIDENHAIN encoders allows uniform valuation. This makes it easier to plan machine use and servicing.

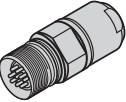
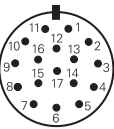
Cyclic Redundancy Check

To ensure **reliability of data transfer**, a cyclic redundancy check (CRC) is performed through the logical processing of the individual bit values of a data word. This 5 bit long CRC concludes every transmission. The CRC is decoded in the receiver electronics and compared with the data word. This largely eliminates errors caused by disturbances during data transfer.



Pin Layout



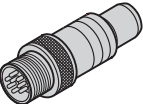
17-pin coupling M23													
													
	Power supply					Incremental signals ¹⁾				Absolute position values			
	7	1	10	4	11	15	16	12	13	14	17	8	9
	U _P	Sensor U _P	0 V	Sensor 0 V	Internal shield	A+	A–	B+	B–	DATA	<u>DATA</u>	CLOCK	<u>CLOCK</u>
	Brown/ Green	Blue	White/ Green	White	/	Green/ Black	Yellow/ Black	Blue/ Black	Red/ Black	Gray	Pink	Violet	Yellow

Cable shield connected to housing; **U_P** = power supply voltage

Sensor: The sensor line is connected in the encoder with the corresponding power line

Vacant pins or wires must not be used!

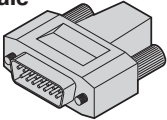
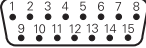
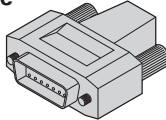
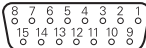
¹⁾ Only with ordering designations EnDat 01 and EnDat 02

8-pin coupling M12								
								
	Power supply				Absolute position values			
	8	2	5	1	3	4	7	6
	U _P	U _P ¹⁾	0 V	0 V ¹⁾	DATA	DATA	CLOCK	CLOCK
	Brown/Green	Blue	White/Green	White	Gray	Pink	Violet	Yellow

Cable shield connected to housing; **U_P** = power supply voltage

Vacant pins or wires must not be used!

¹⁾ The parallel-configured power line is connected in the encoder with the corresponding power supply

15-pin D-sub connector, male for IK 115/IK 215						15-pin D-sub connector, female for HEIDENHAIN controls and IK 220									
															
	Power supply						Incremental signals ¹⁾				Absolute position values				
	4	12	2	10	6	1	9	3	11	5	13	8	15		
	1	9	2	11	13	3	4	6	7	5	8	14	15		
	U _P	Sensor U _P	0V	Sensor 0V	Internal shield	A+	A–	B+	B–	DATA	DATA	CLOCK	CLOCK		
	Brown/ Green	Blue	White/ Green	White	/	Green/ Black	Yellow/ Black	Blue/ Black	Red/ Black	Gray	Pink	Violet	Yellow		

Cable shield connected to housing; **U_P** = power supply voltage

Sensor: The sensor line is connected in the encoder with the corresponding power line

Vacant pins or wires must not be used!

¹⁾ Only with ordering designations EnDat 01 and EnDat 02

Interface

PROFIBUS-DP Absolute Position Values



PROFIBUS DP

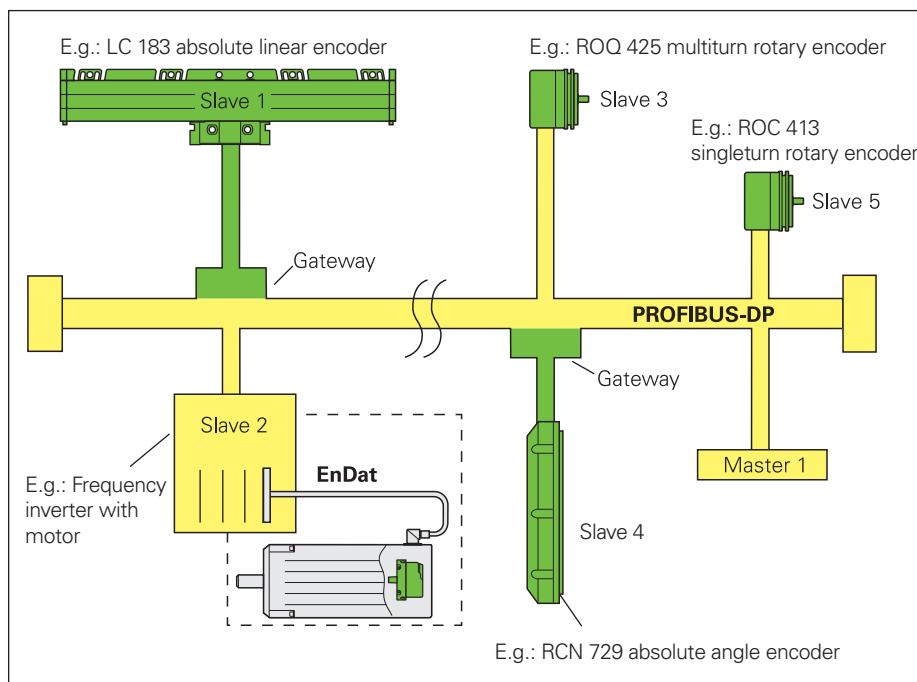
PROFIBUS is a nonproprietary, open field bus in accordance with the international EN 50 170 standard. The connecting of sensors through field bus systems minimizes the cost of cabling and reduces the number of lines between encoder and subsequent electronics.

Topology and bus assignment

The PROFIBUS-DP is designed as a linear structure. It permits transfer rates up to 12 Mbps. Both mono-master and multi master systems are possible. Each master can serve only its own slaves (polling). The slaves are polled cyclically by the master. Slaves are, for example, sensors such as absolute rotary encoders, linear encoders, or also control devices such as motor frequency inverters.

Physical characteristics

The electrical features of the PROFIBUS-DP comply with the RS-485 standard. The bus connection is a shielded, twisted two-wire cable with active bus terminations at each end.



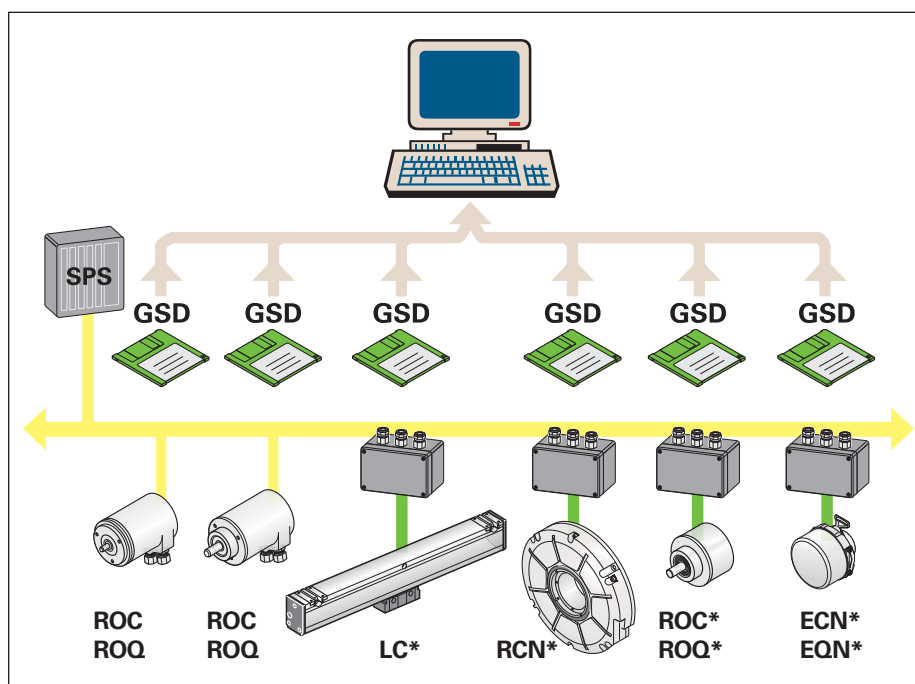
Bus structure of PROFIBUS-DP

Self-configuration

The characteristics of the HEIDENHAIN encoders required for system configuration are included as "electronic data sheets"—also called device identification records (GSD)—in the gateway. These device identification records (GSD) completely and clearly describe the characteristics of a unit in an exactly defined format. This makes it possible to integrate the encoders into the bus system in a simple and application-friendly way.

Configuration

PROFIBUS-DP devices can be configured and the parameters assigned to fit the requirements of the user. Once these settings are made in the configuration tool with the aid of the GSD file, they are saved in the master. It then configures the PROFIBUS devices every time the network starts up. This simplifies exchanging the devices: there is no need to edit or reenter the configuration data.



* With EnDat interface

PROFIBUS-DP profile

The PNO (PROFIBUS user organization) has defined a standard, nonproprietary profile for the connection of absolute encoders to the PROFIBUS-DP, thus ensuring high flexibility and simple configuration on all systems that use this standardized profile.

You can request the profile for absolute encoders from the PNO in Karlsruhe, Germany, under the order number 3.062. There are two classes defined in the profile, whereby class 1 provides minimum support, and class 2 allows additional, in part optional functions.

Supported functions

Particularly important in decentralized field bus systems are the **diagnostic functions** (e.g. warnings and alarms), and the **electronic ID label** with information on the type of encoder, resolution, and measuring range. But also programming functions such as counting direction reversal, **preset/zero shift** and **changing the resolution (scaling)** are possible. The **operating time** of the encoder can also be recorded.

Feature	Class	ECN 113 ¹⁾ ECN 413 ¹⁾ ROC 413	EQN 425 ¹⁾ ROQ 425	ROC 415 ¹⁾ ROC 417 ¹⁾	LC 483 ¹⁾ LC 183 ¹⁾
Position value in pure binary code	1, 2	✓	✓	✓	✓
Data word length	1, 2	16	32	32	32
Scaling function					
Measuring steps/rev	2	✓	✓	✓ ²⁾	–
Total resolution	2	✓	✓	–	–
Reversal of counting direction	1, 2	✓	✓	✓	–
Preset/Datum shift	2	✓	✓	✓	–
Diagnostic functions					
Warnings and alarms	2	✓	✓	✓	✓
Operating time recording	2	✓	✓	✓	✓
Profile version	2	✓	✓	✓	✓
Serial number	2	✓	✓	✓	✓

1) Connectible with EnDat Interface over gateway to PROFIBUS-DP

2) Scaling factor in binary steps

Encoders with EnDat interface for connection via gateway

All absolute encoders from HEIDENHAIN with **EnDat interface** are suitable for PROFIBUS-DP. The encoder is electrically connected through a **gateway**.

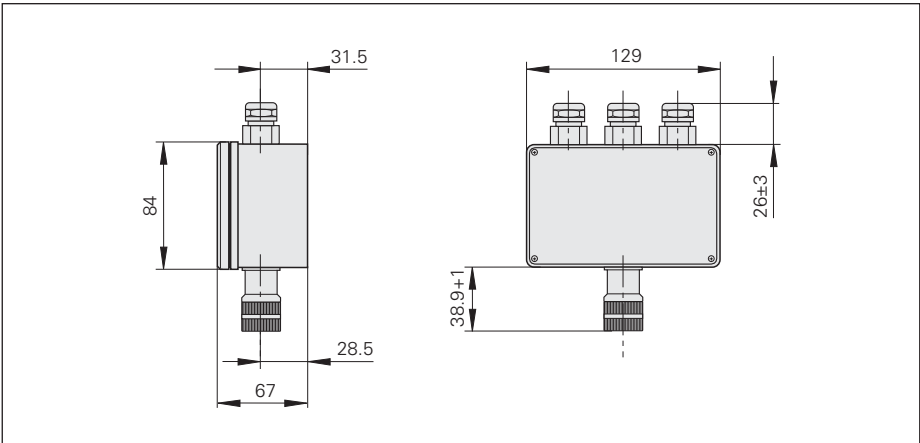
The complete interface electronics are integrated in the gateway, as well as a voltage converter for supplying EnDat encoders with 5 V ± 5 %. This offers a number of benefits:

- Simple connection of the field bus cable, since the terminals are easily accessible.
- Encoder dimensions remain small.
- No temperature restrictions for the encoder. All temperature-sensitive components are in the gateway.
- No bus interruption when an encoder is exchanged.

Besides the EnDat encoder connector, the gateway provides connections for the PROFIBUS and the power supply. In the gateway there are coding switches for addressing and selecting the terminating resistor. Since the gateway is connected directly to the bus lines, the cable to the encoder is not a stub line, although it can be up to 150 meters long.



	Gateway
Power supply	10 to 30 V Max. 400 mA
Protection	IP 67
Operating temperature	–40 °C to 80 °C
Electrical connection	Flange socket 17-pin EnDat PROFIBUS DP Terminations, PG9 cable outlet
ID	325771-01



Encoders with PROFIBUS-DP

The absolute rotary encoders with **integrated PROFIBUS-DP interface** are connected directly to the PROFIBUS. LEDs on the rear of the encoder display the power supply and bus status **operating states**.

The coding switches for the addressing (0 to 99) and for selecting the terminating resistor are easily accessible under the bus housing. The terminating resistor is to be activated if the rotary encoder is the last participant on the PROFIBUS-DP and the external terminating resistor is not used.

Accessory:

Adapter M12 (male), 4-pin, B-coded
Fits 5-pin bus output, with PROFIBUS terminating resistor. Required for last participant if the encoder's internal terminating resistor is not to be used.
ID 584 217-01

Connection

PROFIBUS-DP and the power supply are connected via the M12 connecting elements. The necessary mating connectors are:

Bus input:

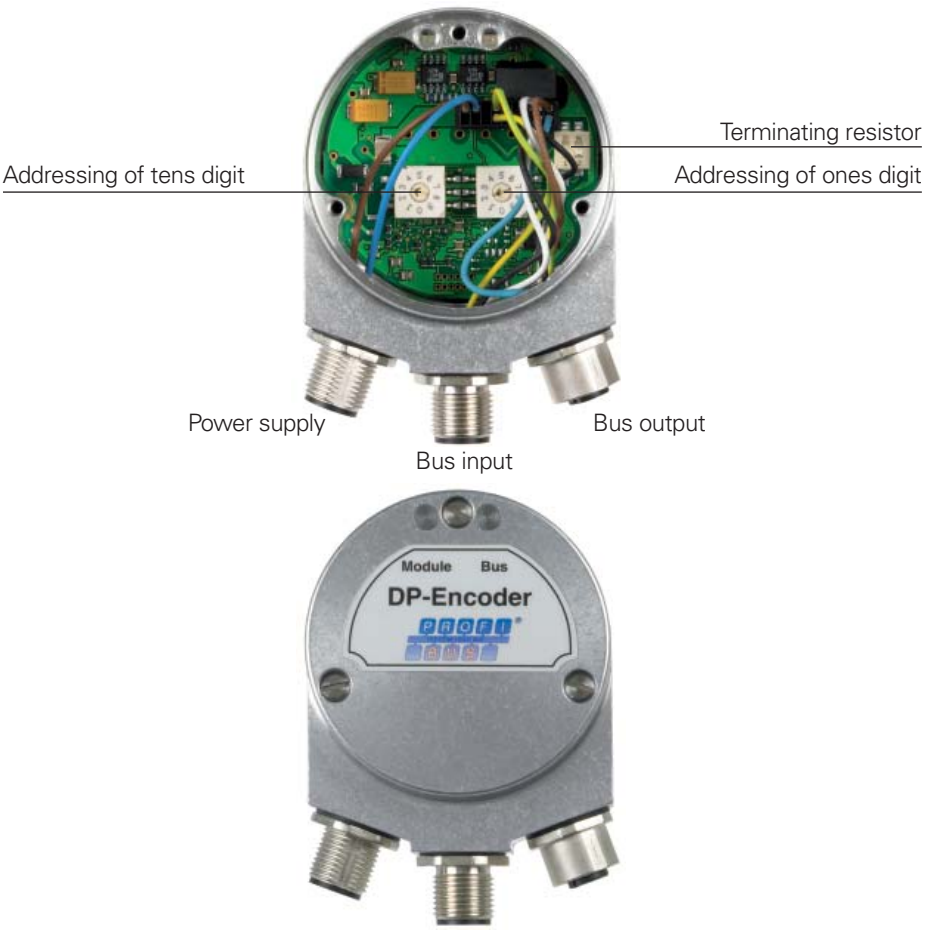
M12 connector (female), 5-pin, B-coded

Bus output:

M12 coupling (male), 5-pin, B-coded

Power supply:

M12 connector, 4-pin, A-coded



Pin layout

Bus input, 5-pin coupling (male) M12 B-coded					Bus output, 5-pin connector (female) M12 B-coded	
	Power supply				Absolute position values	
	1	3	5	Housing	2	4
BUS in	/	/	Shield	Shield	DATA (A)	DATA (B)
BUS out	U ¹⁾	0V ¹⁾	Shield	Shield	DATA (A)	DATA (B)

¹⁾ For supplying the external terminating resistor

Power supply, 4-pin coupling (male) M12 A-coded				
	1	3	2	4
	U _P	0V	Vacant	Vacant

Interfaces

SSI Absolute Position Values

The **absolute position value** beginning with the Most Significant Bit (MSB first) is transferred on the DATA lines in synchronism with a CLOCK signal transmitted by the control. The SSI standard data word length for singleturn absolute encoders is 13 bits, and for multiturn absolute encoders 25 bits. In addition to the absolute position values, sinusoidal **incremental signals** with 1- V_{PP} levels are transmitted. For signal description see *Incremental signals 1 V_{PP}* .

For the ECN/EQN 4xx and ROC/ROQ 4xx rotary encoders, the following **functions** can be activated via the programming inputs of the interfaces by applying the supply voltage U_P :

- Direction of rotation**
 Continuous application of a HIGH level to pin 2 reverses the direction of rotation for ascending position values.
- Zeroing**
 Applying a positive edge ($t_{min} > 1\text{ ms}$) to pin 5 sets the current position to zero.

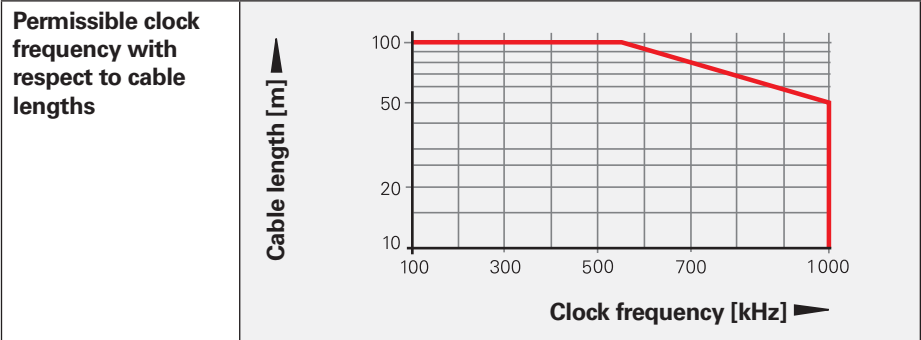
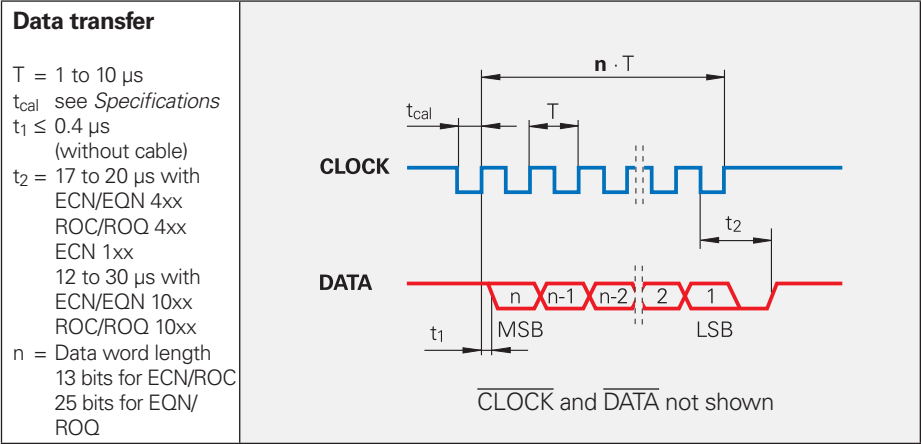
Note: The programming inputs must always be terminated with a resistor (see input circuitry of the subsequent electronics).

Control cycle for complete data format
 When not transmitting, the clock and data lines are on high level. The current position value is stored on the first falling edge of the clock. The stored data is then clocked out on the first rising edge.

After transmission of a complete data word, the data line remains low for a period of time (t_2) until the encoder is ready for interrogation of a new value. If another data-output request (CLOCK) is received within this time, the same data will be output once again.

If the data output is interrupted (CLOCK = high for $t \geq t_2$), a new position value will be stored on the next falling edge of the clock, and on the subsequent rising edge it will be clocked out to the subsequent electronics.

Interface	SSI serial
Ordering designation	<i>Singleturn:</i> 39r1, 39n1 <i>Multiturn:</i> 41r1
Data transfer	Absolute position values
Data input	Differential line receiver according to EIA standard RS 485 for the CLOCK and $\overline{\text{CLOCK}}$ signals
Data output	Differential line driver according to EIA standard RS 485 for the signals DATA and $\overline{\text{DATA}}$
Code	Gray
Ascending position values	With clockwise rotation viewed from flange side (can be switched via interface)
Incremental signals	$\sim 1\text{ V}_{PP}$ (see <i>Incremental signals 1 V_{PP}</i>)
Programming inputs	Direction of rotation and zero reset (for ECN/EQN 4xx, ROC/ROQ 4xx)
Inactive	
Active	
Switching time	LOW < $0.25 \times U_P$ HIGH > $0.6 \times U_P$ $t_{min} > 1\text{ ms}$
Connecting cables	Shielded HEIDENHAIN cable PUR [(4 x 0.14 mm ²) + 4(2 x 0.14 mm ²) + (4 x 0.5 mm ²)] Max. 150 m at 90 pF/m distributed capacitance 6 ns/m
Cable length	
Propagation time	



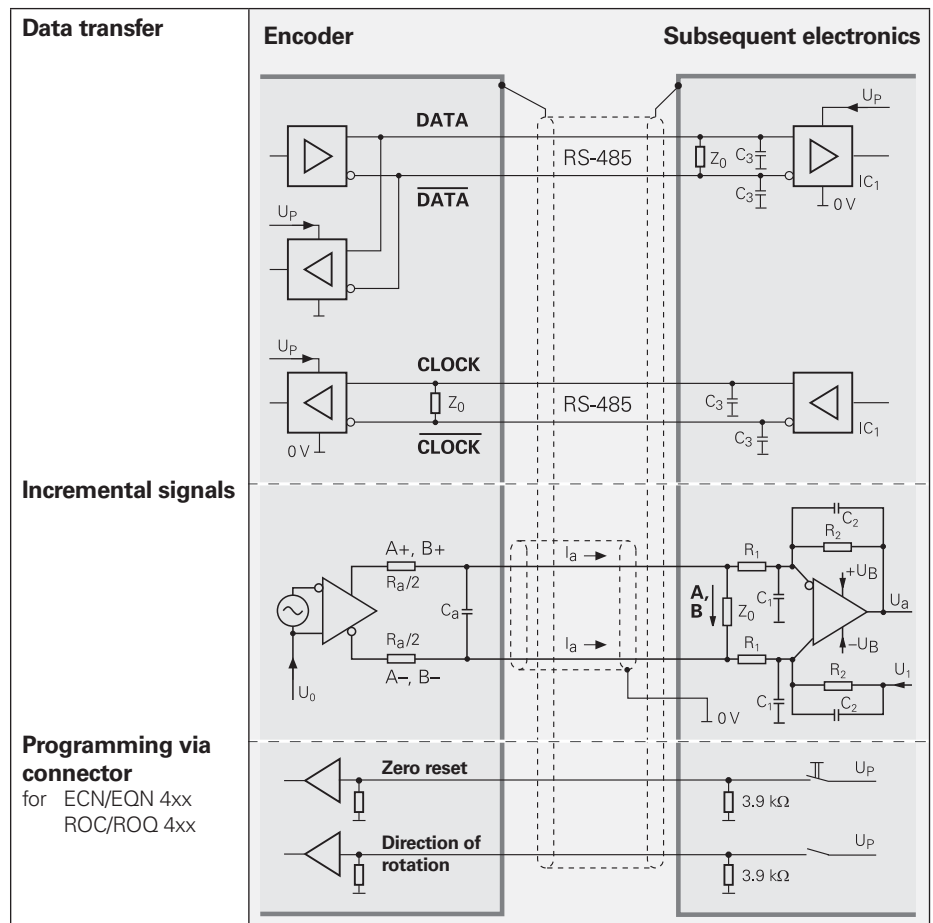
Input circuitry of the subsequent electronics

Dimensioning

IC₁ = Differential line receiver and driver
E.g. SN 65 LBC 176
LT 485

$$Z_0 = 120 \, \Omega$$

C₃ = 330 pF (serves to improve noise immunity)



Pin layout

17-pin coupling M23															
	Power supply					Incremental signals				Absolute position values				Other signals	
	7	1	10	4	11	15	16	12	13	14	17	8	9	2	5
	U _P	Sensor U _P	0 V	Sensor 0 V	Internal shield	A+	A-	B+	B-	DATA	DATA	CLOCK	CLOCK	Direction of rotation ¹⁾	Zero reset ¹⁾
	Brown/ Green	Blue	White/ Green	White	/	Green/ Black	Yellow/ Black	Blue/ Black	Red/ Black	Gray	Pink	Violet	Yellow	Black	Green

Shield on housing; **U_P** = Power supply voltage

Sensor: With a 5 V supply voltage, the sensor line is connected in the encoder with the corresponding power line.

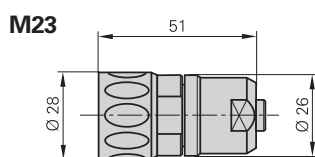
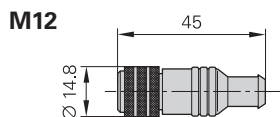
¹⁾ Vacant on ECN/EQN 10xx and ROC/ROQ 10xx

Connecting Elements and Cables

General Information

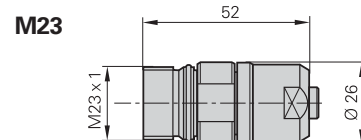
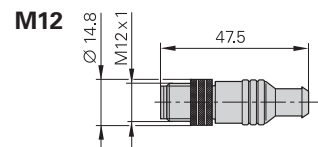
Connector (insulated): A connecting element with a coupling ring. Available with male or female contacts.

Symbols

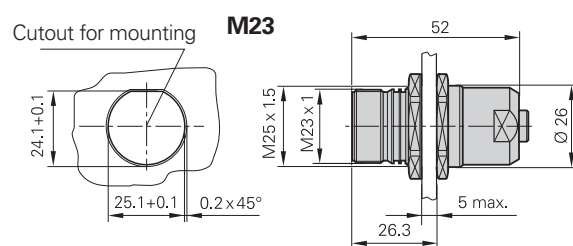


Coupling (insulated): Connecting element with external thread; available with male or female contacts.

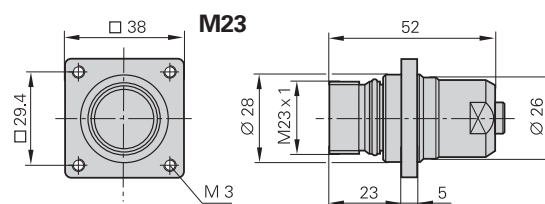
Symbols



Mounted coupling with central fastening

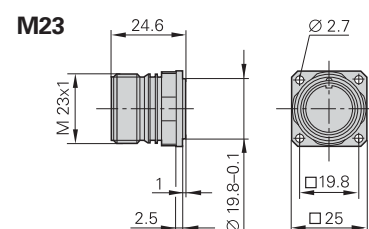


Mounted coupling with flange



Flange socket: Permanently mounted on a housing, with external thread (like the coupling), and available with male or female contacts.

Symbols



The pins on connectors are **numbered** in the direction opposite to those on couplings or flange sockets, regardless of whether the connecting elements are

male contacts or



female contacts.



Accessories for flange sockets and M23 mounted couplings

Bell seal

ID 266526-01

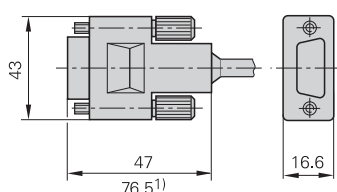
Threaded metal dust cap

ID 219926-01

When engaged, the connections are **protected** to IP 67 (D-sub connector: IP 50; EN 60529). When not engaged, there is no protection.

D-sub connector: For HEIDENHAIN controls, counters and IK absolute value cards.

Symbols



¹⁾ With integrated interpolation electronics

Connecting Cables

8-pin 12-pin 17-pin
M12 M23 M23

		For EnDat without incremental signals	For $\sim 1V_{PP}$ TTL	For EnDat with incremental signals SSI
PUR connecting cables	8-pin: $[(4 \times 0.14 \text{ mm}^2) + (4 \times 0.34 \text{ mm}^2)]$ 12-pin: $[4(2 \times 0.14 \text{ mm}^2) + (4 \times 0.5 \text{ mm}^2)]$ 17-pin: $[(4 \times 0.14 \text{ mm}^2) + 4(2 \times 0.14 \text{ mm}^2) + (4 \times 0.5 \text{ mm}^2)]$		Ø 6 mm Ø 8 mm Ø 8 mm	
Complete with connector (female) and coupling (male)		368330-xx	298401-xx	323897-xx
Complete with connector (female) and connector (male)		–	298399-xx	–
Complete with connector (female) and D-sub connector (female) for IK 220		533627-xx	310199-xx	332115-xx
Complete with connector (female) and D-sub connector (male) for IK 115/IK 215		524599-xx	310196-xx	324544-xx
With one connector (female)		634265-xx	309777-xx	309778-xx
Cable without connectors , Ø 8 mm		–	244957-01	266306-01
Mating element on connecting cable to connector on encoder cable	Connector (female) for cable Ø 8 mm 	–	291697-05	291697-26
Connector on connecting cable for connection to subsequent electronics	Connector (male) for cable Ø 8 mm Ø 6 mm 	–	291697-08 291697-07	291697-27
Coupling on connecting cable	Coupling (male) for cable Ø 4.5 mm Ø 6 mm Ø 8 mm 	–	291698-14 291698-03 291698-04	291698-25 291698-26 291698-27
Flange socket for mounting on subsequent electronics	Flange socket (female) 	–	315892-08	315892-10
Mounted couplings	With flange (female) Ø 6 mm Ø 8 mm 	–	291698-17 291698-07	291698-35
	With flange (male) Ø 6 mm Ø 8 mm 	–	291698-08 291698-31	291698-41 291698-29
	With central fastening (male) Ø 6 mm 	–	291698-33	291698-37
Adapter $\sim 1V_{PP}/11 \mu A_{PP}$ For converting the $1V_{PP}$ signals to $11 \mu A_{PP}$; M23 connector (female) 12-pin and M23 connector (male) 9-pin		–	364914-01	–

HEIDENHAIN Measuring Equipment

For Incremental Encoders

The **PWM 9** is a universal measuring device for checking and adjusting HEIDENHAIN incremental encoders. There are different expansion modules available for checking the different encoder signals. The values can be read on an LCD monitor. Soft keys provide ease of operation.



	PWM 9
Inputs	Expansion modules (interface boards) for 11 μ App; 1 V _{PP} ; TTL; HTL; EnDat*/SSI*/commutation signals *No display of position values or parameters
Functions	• Measures signal amplitudes, current consumption, operating voltage, scanning frequency • Graphically displays incremental signals (amplitudes, phase angle and on-off ratio) and the reference-mark signal (width and position) • Displays symbols for the reference mark, fault detection signal, counting direction • Universal counter, interpolation selectable from single to 1024-fold • Adjustment support for exposed linear encoders
Outputs	• Inputs are connected through to the subsequent electronics • BNC sockets for connection to an oscilloscope
Power supply	10 to 30 V, max. 15 W
Dimensions	150 mm x 205 mm x 96 mm

For Absolute Encoders

HEIDENHAIN offers an adjusting and testing package for diagnosis and adjustment of HEIDENHAIN encoders with absolute interface.

- PC expansion board **IK 215**
- **ATS** adjusting and testing software



	IK 215
Encoder input	• EnDat 2.1 or EnDat 2.2 (absolute value with/without incremental signals) • FANUC serial interface • Mitsubishi high speed serial interface • SSI
Interface	PCI bus Rev. 2.1
System requirements	• Operating system: Windows XP (Vista upon request) • Approx. 20 MB free space on the hard disk
Signal subdivision for incremental signals	Up to 65536-fold
Dimensions	100 mm x 190 mm

	ATS
Languages	Choice between English or German
Functions	• Position display • Connection dialog • Diagnostics • Mounting wizard for ECI/EQI • Additional functions (if supported by the encoder) • Memory contents

Evaluation Electronics

IK 220

Universal PC counter card

The IK 220 is an expansion board for PCs for recording the measured values of two incremental or absolute linear or angle encoders. The subdivision and counting electronics subdivide the sinusoidal input signals up to 4096-fold. A driver software package is included in delivery.



For more information, see the *IK 220* Product Information document as well as the Product Overview of *Interface Electronics*.

	IK 220			
Input signals (switchable)	~ 1 V _{pp}	~ 11 µA _{pp}	EnDat 2.1	SSI
Encoder inputs	Two D-sub connections (15-pin), male			
Input frequency	≤ 500 kHz	≤ 33 kHz	–	
Cable length	≤ 60 m		≤ 50 m	≤ 10 m
Signal subdivision (signal period : meas. step)	Up to 4096-fold			
Data register for measured values (per channel)	48 bits (44 bits used)			
Internal memory	For 8192 position values			
Interface	PCI bus			
Driver software and demonstration program	For Windows 98/NT/2000/XP in VISUAL C++, VISUAL BASIC and BORLAND DELPHI			
Dimensions	Approx. 190 mm × 100 mm			

General Electrical Information

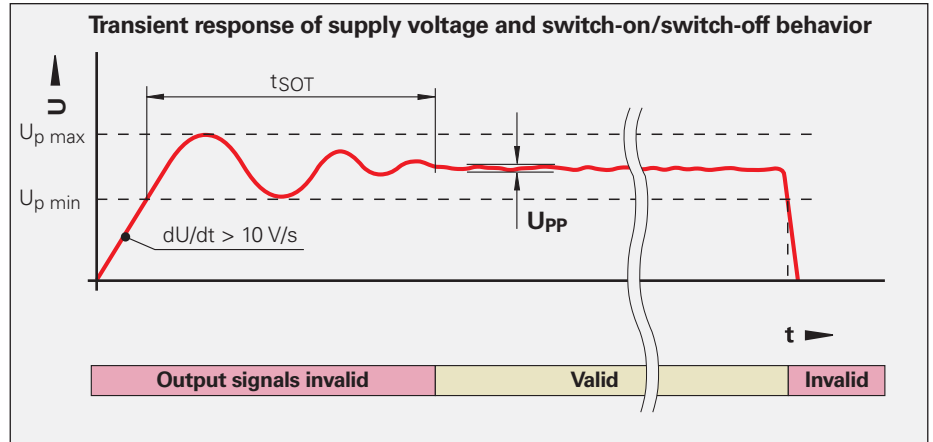
Power Supply

Connect HEIDENHAIN encoders only to subsequent electronics whose power supply is generated from PELV systems (**EN 50178**). In addition, in safety-related applications, overcurrent protection and sometimes overvoltage protection are required.

The encoders require a **stabilized DC voltage U_p** as power supply. The respective *Specifications* state the required power supply and the current consumption. The permissible ripple content of the DC voltage is:

- High frequency interference
 $U_{PP} < 250 \text{ mV}$ with $dU/dt > 5 \text{ V}/\mu\text{s}$
- Low frequency fundamental ripple
 $U_{PP} < 100 \text{ mV}$

The values apply as measured at the encoder, i.e., without cable influences. The voltage can be monitored and adjusted with the encoder's **sensor lines**. If a controllable power supply is not available, the voltage drop can be halved by switching the sensor lines parallel to the corresponding power lines.



Calculation of the **voltage drop**:

$$\Delta U = 2 \cdot 10^{-3} \cdot \frac{L_C \cdot I}{56 \cdot A_P}$$

where ΔU : Voltage attenuation in V

L_C : Cable length in m

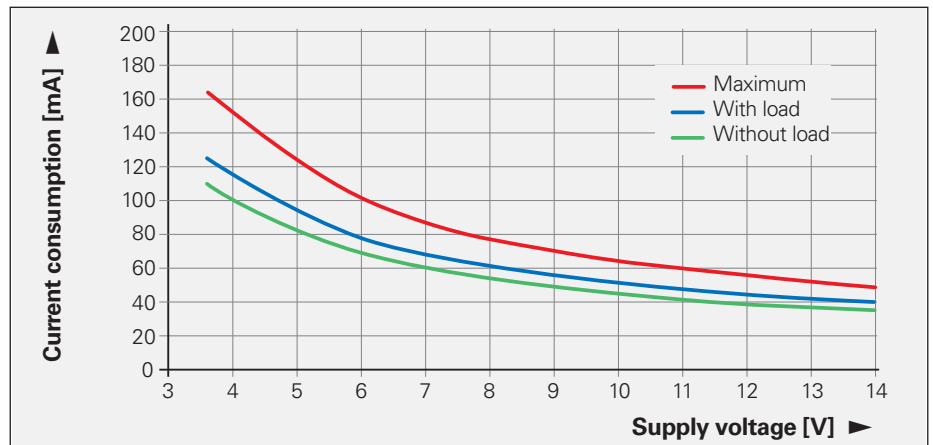
I : Current consumption in mA

A_P : Cross section of power lines in mm^2

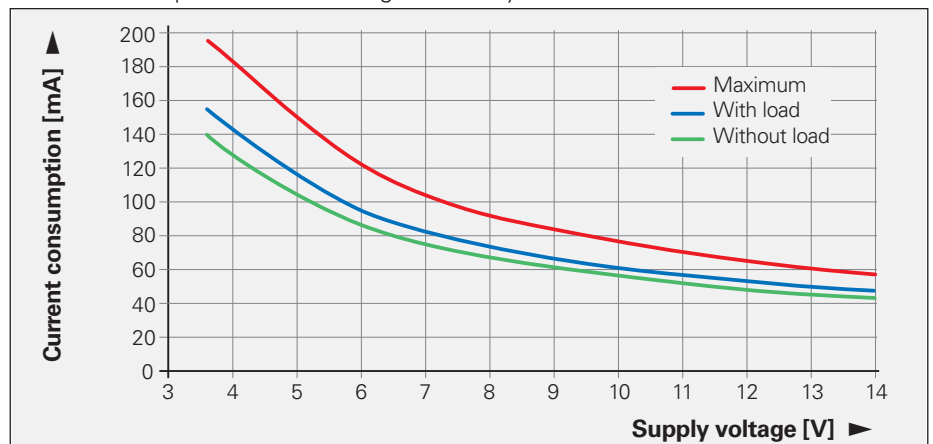
Current consumption

For encoders with extended voltage range (3.6 V to 14 V), the current consumption depends on the actual supply voltage. In the specifications it is stated for a 3.6 V supply (without load). These diagrams show the actual current consumption for different operating situations depending on the respective supply voltage values:

- Without load (only power supply connected)
- With load (normal operation with position value request at 8 MHz)
- Maximum value (additionally with aging and possible temperature influences)



Current consumption of the ECN singleturn rotary encoders



Current consumption of the EQN multiturn rotary encoders

Switch-on/off behavior of the encoders

The output signals are valid no sooner than after switch-on time $t_{SOT} = 1.3 \text{ s}$ (2 s for PROFIBUS-DP) (see diagram). During time t_{SOT} they can have any levels up to 5.5 V (with HTL encoders up to U_{Pmax}). If an interpolation electronics unit is inserted between the encoder and the power supply, the unit's switch-on/off characteristics must also be considered. If the power supply is switched off, or when the supply voltage falls below U_{min} , the output signals are also invalid. During restart, the signal level must remain below 1 V for the time t_{SOT} before power up. These data apply to the encoders listed in the catalog—customer-specific interfaces are not included.

Encoders with new features and increased performance range may take longer to switch on (longer time t_{SOT}). If you are responsible for developing subsequent electronics, please contact HEIDENHAIN in good time.

Isolation

The encoder housings are isolated against internal circuits.

Rated surge voltage: 500 V

(preferred value as per VDE 0110 Part 1, overvoltage category II, contamination level 2)

Cable

HEIDENHAIN cables with connectors are mandatory for use in safety-related applications.

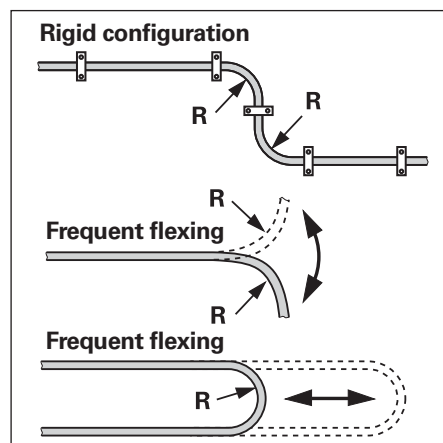
Versions

The cables or almost all HEIDENHAIN encoders and all adapter and connecting cables are sheathed in **polyurethane (PUR cable)**. Most adapter cables for within motors and a few cables on encoders are sheathed in a **special elastomer (EPG cable)**. These cables are identified in the specifications or in the cable tables with "EPG."

Durability

PUR cables resistant to oil and hydrolysis in accordance with **VDE 0472** (Part 803/test type B) and resistant to microbes in accordance with **VDE 0282** (Part 10). They are free of PVC and silicone and comply with UL safety directives. The **UL certification** AWM STYLE 20963 80 °C 30 V E63216 is documented on the cable.

EPG cables are resistant to oil in accordance with **VDE 0472** (Part 803/test type B) and to hydrolysis in accordance with **VDE 0282** (Part 10). They are free of silicone and halogens. In comparison with PUR cables, they are only conditionally resistant to media, frequent flexing and continuous torsion.



Temperature range

HEIDENHAIN cables can be used for
rigid configuration (PUR) –40 to 80 °C
rigid configuration (EPG) –40 to 120 °C
frequent flexing (PUR) –10 to 80 °C

PUR cables with limited resistance to hydrolysis and microbes are rated for up to 100 °C. If required, please ask for assistance from HEIDENHAIN Traunreut.

Lengths

The **cable lengths** listed in the *Specifications* apply only for HEIDENHAIN cables and the recommended input circuitry of subsequent electronics.

Cable	Cross section of power supply lines A_p				Bend radius R	
	1 V _{PP} /TTL/HTL	11 μ App	EnDat/SSI 17-pin	EnDat ⁵⁾ 8-pin	Rigid configuration	Frequent flexing
Ø 3.7 mm	0.05 mm ²	–	–	–	≥ 8 mm	≥ 40 mm
Ø 4.3 mm	0.24 mm ²	–	–	–	≥ 10 mm	≥ 50 mm
Ø 4.5 mm EPG	0.05 mm ²	–	0.05 mm ²	0.09 mm ²	≥ 18 mm	–
Ø 4.5 mm Ø 5.1 mm	0.14/0.09 ²⁾ mm ² 0.05 ^{2), 3)} mm ²	0.05 mm ²	0.05 mm ²	0.14 mm ²	≥ 10 mm	≥ 50 mm
Ø 6 mm Ø 10 mm ¹⁾	0.19/0.14 ^{2), 4)} mm ²	–	0.08 mm ²	0.34 mm ²	≥ 20 mm ≥ 35 mm	≥ 75 mm ≥ 75 mm
Ø 8 mm Ø 14 mm ¹⁾	0.5 mm ²	1 mm ²	0.5 mm ²	1 mm ²	≥ 40 mm ≥ 100 mm	≥ 100 mm ≥ 100 mm

¹⁾ Metal armor

²⁾ Rotary encoders

³⁾ Length gauges

⁴⁾ LIDA 400

⁵⁾ Also Fanuc, Mitsubishi

Electrically Permissible Speed/ Traversing Speed

The maximum permissible shaft speed or traversing velocity of an encoder is derived from

- the **mechanically** permissible shaft speed/traversing velocity (if listed in *Specifications*) and the **electrically** permissible shaft speed/traversing velocity.

For encoders with **sinusoidal output signals**, the electrically permissible shaft speed/traversing velocity is limited by the –3dB/ –6dB cutoff frequency or the permissible input frequency of the subsequent electronics.

For encoders with **square-wave signals**, the electrically permissible shaft speed/traversing velocity is limited by

- the maximum permissible scanning frequency $f_{\max}$ of the encoder and
- the minimum permissible edge separation a for the subsequent electronics.

For angular or rotary encoders

$$n_{\max} = \frac{f_{\max}}{z} \cdot 60 \cdot 10^3$$

For linear encoders

$$v_{\max} = f_{\max} \cdot SP \cdot 60 \cdot 10^{-3}$$

Where:

$n_{\max}$: Elec. permissible speed in min^{-1}

$v_{\max}$: Elec. permissible traversing velocity in m/min

$f_{\max}$: Max. scanning/output frequency of encoder or input frequency of subsequent electronics in kHz

z : Line count of the angle or rotary encoder per 360°

SP : Signal period of the linear encoder in μm

Noise-Free Signal Transmission

Electromagnetic compatibility/CE compliance

When properly installed, and when HEIDENHAIN connecting cables and cable assemblies are used, HEIDENHAIN encoders fulfill the requirements for electromagnetic compatibility according to 2004/108/EC with respect to the generic standards for:

• Noise EN 61000-6-2:

Specifically:

- ESD EN 61000-4-2
- Electromagnetic fields EN 61000-4-3
- Burst EN 61000-4-4
- Surge EN 61000-4-5
- Conducted disturbances EN 61000-4-6
- Power frequency magnetic fields EN 61000-4-8
- Pulse magnetic fields EN 61000-4-9

• Interference EN 61000-6-4:

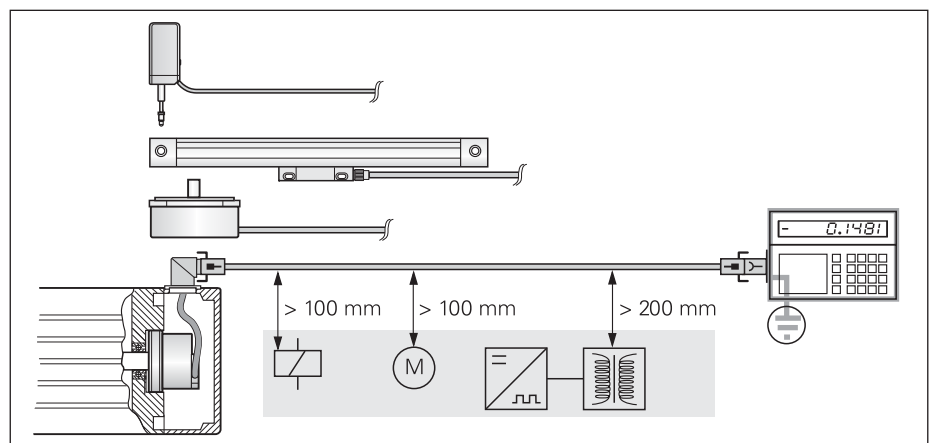
Specifically:

- For industrial, scientific and medical equipment (ISM) EN 55011
- For information technology equipment EN 55022

Transmission of measuring signals— electrical noise immunity

Noise voltages arise mainly through capacitive or inductive transfer. Electrical noise can be introduced into the system over signal lines and input or output terminals. Possible sources of noise include:

- Strong magnetic fields from transformers, brakes and electric motors
- Relays, contactors and solenoid valves
- High-frequency equipment, pulse devices, and stray magnetic fields from switch-mode power supplies
- AC power lines and supply lines to the above devices



Minimum distance from sources of interference

Protection against electrical noise

The following measures must be taken to ensure disturbance-free operation:

- Use only original HEIDENHAIN cables. Consider the voltage attenuation on supply lines.
- Use connecting elements (such as connectors or terminal boxes) with metal housings. Only the signals of and power supply for the connected encoder may be routed through these elements. Applications in which additional signals are led through the connecting element require specific measures regarding electrical safety and EMC.
- Connect the housings of the encoder, connecting elements and subsequent electronics through the shield of the cable. Ensure that the shield has complete contact over the entire surface (360°). For encoders with more than one electrical connection, refer to the documentation for the respective product.
- For cables with multiple shields, the inner shields must be led separately from the outer shield. Connect inner shield to 0 V of the subsequent electronics. Do not connect the inner shields with the outer shield, neither in the encoder nor in the cable.
- Connect the shield to protective ground as per the mounting instructions.
- Prevent contact of the shield (e.g. connector housing) with other metal surfaces. Pay attention to this when installing cables.
- Do not install signal cables in the direct vicinity of interference sources (inductive consumers such as contacts, motors, frequency inverters, solenoids, etc.).
 - Sufficient decoupling from interference-signal-conducting cables can usually be achieved by an air clearance of 100 mm or, when cables are in metal ducts, by a grounded partition.
 - A minimum spacing of 200 mm to inductors in switch-mode power supplies is required.
- If compensating currents are to be expected within the overall system, a separate equipotential bonding conductor must be provided. The shield does not have the function of an equipotential bonding conductor.
- Only provide power to position encoders from PELV systems (**EN 50 178**). Provide high-frequency grounding with low impedance (**EN 60204-1 Chap. EMC**).
- For encoders with 11 µAPP interface: For extension cables, use only HEIDENHAIN cable ID 244.955-01. Overall length max. 30 m.

Sales and Service

For More Information

Other devices for angular measurement from HEIDENHAIN include rotary encoders, which are used primarily on electrical motors, for elevator control and for potentially explosive atmospheres. Angle encoders from HEIDENHAIN serve for high-accuracy position acquisition of angular movements.



Brochure ***Encoders for Servo Drives***

Contents:
Rotary encoders
Angle encoders
Linear encoders



Product Overview ***Rotary Encoders for the Elevator Industry***



Brochure ***Angle Encoders with Integral Bearing***

Contents:
Absolute angle encoders
RCN
Incremental angle encoders
RON, RPN, ROD



Product Overview ***Rotary Encoders for Potentially Explosive Atmospheres***



Brochure ***Angle Encoders without Integral Bearing***

Contents:
Incremental angle encoders
ERA, ERP



Product Information ***Magnetic Modular Encoders***

Further HEIDENHAIN products

- Linear encoders
- Length gauges
- Measuring systems for machine tool inspection and acceptance testing
- Subsequent electronics
- NC controls for machine tools
- Touch probes

HEIDENHAIN on the Internet

Visit our home page at www.heidenhain.com for up-to-date information on:

- The company
- The products

Also included:

- Technical articles
- Press releases
- Addresses
- CAD drawings

Addresses in Germany

HEIDENHAIN is represented in Germany and all other important industrial nations as well. In addition to the addresses listed on the back page, there are many service agencies located worldwide. For their addresses, please refer to the Internet or contact HEIDENHAIN Traunreut.

Germany – Technical Information

HEIDENHAIN Technisches Büro Nord

Rhinstraße 134
12681 Berlin, Deutschland
☎ 030 54705-240
FAX 030 54705-200
E-Mail: tbn@heidenhain.de

HEIDENHAIN Technisches Büro West

Revierstraße 19
44379 Dortmund, Deutschland
☎ 0231 618083-0
FAX 0231 618083-29
E-Mail: tbw@heidenhain.de

HEIDENHAIN Technisches Büro Südost

Dr.-Johannes-Heidenhain-Straße 5
83301 Traunreut, Deutschland
☎ 08669 311345
FAX 08669 5061
E-Mail: tbso@heidenhain.de

HEIDENHAIN Technisches Büro Mitte

Kaltes Feld 22
08468 Heinsdorfergrund, Deutschland
☎ 03765 69544
FAX 03765 69628
E-Mail: tbn@heidenhain.de

HEIDENHAIN Technisches Büro Südwest

Ebene 6
Gutenbergstraße 17
70771 Leinfelden-Echterdingen, Deutschland
☎ 0711 993395-0
FAX 0711 993395-28
E-Mail: tbsw@heidenhain.de

Germany – Information and Sales

KLEIN & MISTELE GmbH

Im Hegen 14a
22113 Oststeinbek
☎ 040 7148672-0
E-Mail: info@klein-mistele.de

RHEINWERKZEUG GmbH & Co.KG

Gablonzstraße 8
38114 Braunschweig
☎ 0531 25659-0
E-Mail: braunschweig@rheinwerkzeug.de

FRIEDRICH STRACK Maschinen GmbH

Buchenhofener Straße 19
42329 Wuppertal
☎ 0202 385-0
E-Mail: info@strack-maschinen.de

Walter BAUTZ GmbH

Mess- und Spanntechnik
Mühlenweg 8
64347 Griesheim
☎ 06155 8422-0
E-Mail: info@walterbautz-gmbh.de

BRAUN Werkzeugmaschinen Vertrieb und Service GmbH

Industriestraße 41
72585 Riederich
☎ 07123 9343-0
E-Mail: hh@braun-werkzeugmaschinen.de

HAAS Werkzeugmaschinen GmbH

Heinrich-Hertz-Straße 16
78052 VS-Villingen
☎ 07721 9559-0
E-Mail: info@haas-wzm.de

BRAUN Werkzeugmaschinen Vertrieb und Service GmbH

Anton-Pendele-Straße 3
82275 Emmering
☎ 08141 9714
E-Mail: info@braunem.de

TEDI Technische Dienste GmbH

Werkstraße 113
19061 Schwerin
☎ 0385 61721-0
E-Mail: schwerin-jh@tedi-online.de

TEDI Technische Dienste GmbH

Liebknechtstraße 65
39110 Magdeburg
☎ 0391 732529-0
E-Mail: magdeburg-jh@tedi-online.de

MOSER Industrie-Elektronik GmbH

Geneststraße 5
10829 Berlin
☎ 030 7515737
E-Mail: mosergmbh.berlin@t-online.de

TEDI Technische Dienste GmbH

Großenhainer Straße 99
01127 Dresden
☎ 0351 4278020
E-Mail: dresden-jh@tedi-online.de

WWZ-Vertrieb GmbH

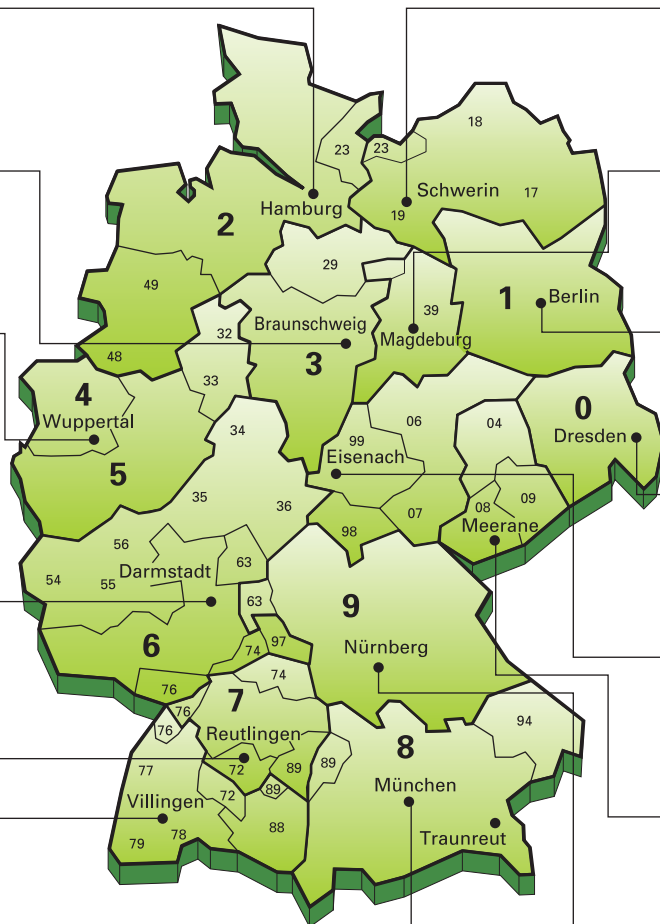
Werkzeugmaschinen
An der Allee 9
99848 Wutha-Farnroda
☎ 036921 23-0
E-Mail: J.Wellendorf@wwz-vertrieb.de

HEMPEL Werkzeugmaschinen

Pestalozzistraße 58
08393 Meerane
☎ 03764 3064
E-Mail: info@hempel-wzm.de

KL Messtechnik & Service GmbH & Co. KG

Im Gewerbegebiet 4
91093 Heßdorf
☎ 09135 73609-0
E-Mail: info@kl-messtechnik.de



HEIDENHAIN

DR. JOHANNES HEIDENHAIN GmbH

Dr.-Johannes-Heidenhain-Straße 5
83301 Traunreut, Germany
 ☎ +49 8669 31-0
 ☎ +49 8669 5061
 E-mail: info@heidenhain.de

www.heidenhain.de

Vollständige und weitere Adressen siehe www.heidenhain.de
 For complete and further addresses see www.heidenhain.de

DE	HEIDENHAIN Technisches Büro Nord 12681 Berlin, Deutschland ☎ 030 54705-240	ES	FARRESA ELECTRONICA S.A. 08028 Barcelona, Spain www.farresa.es	PH	Machinebanks` Corporation Quezon City, Philippines 1113 E-mail: info@machinebanks.com
	HEIDENHAIN Technisches Büro Mitte 08468 Heinsdorfergrund, Deutschland ☎ 03765 69544	FI	HEIDENHAIN Scandinavia AB 02770 Espoo, Finland www.heidenhain.fi	PL	APS 02-489 Warszawa, Poland www.apservis.com.pl
	HEIDENHAIN Technisches Büro West 44379 Dortmund, Deutschland ☎ 0231 618083-0	FR	HEIDENHAIN FRANCE sarl 92310 Sèvres, France www.heidenhain.fr	PT	FARRESA ELECTRÓNICA, LDA. 4470 - 177 Maia, Portugal www.farresa.pt
	HEIDENHAIN Technisches Büro Südwest 70771 Leinfelden-Echterdingen, Deutschland ☎ 0711 993395-0	GB	HEIDENHAIN (G.B.) Limited Burgess Hill RH15 9RD, United Kingdom www.heidenhain.co.uk	RO	HEIDENHAIN Reprezentantă Romania Braşov, 500338, Romania www.heidenhain.ro
	HEIDENHAIN Technisches Büro Südost 83301 Traunreut, Deutschland ☎ 08669 31-1345	GR	MB Milionis Vassilis 17341 Athens, Greece www.heidenhain.gr	RS	Serbia → BG
		HK	HEIDENHAIN LTD Kowloon, Hong Kong E-mail: service@heidenhain.com.hk	RU	OOO HEIDENHAIN 125315 Moscow, Russia www.heidenhain.ru
AR	NAKASE SRL. B1653AOX Villa Ballester, Argentina www.heidenhain.com.ar	HR	Croatia → SL	SE	HEIDENHAIN Scandinavia AB 12739 Skärholmen, Sweden www.heidenhain.se
AT	HEIDENHAIN Techn. Büro Österreich 83301 Traunreut, Germany www.heidenhain.de	HU	HEIDENHAIN Kereskedelmi Képviselő 1239 Budapest, Hungary www.heidenhain.hu	SG	HEIDENHAIN PACIFIC PTE LTD. Singapore 408593 www.heidenhain.com.sg
AU	FCR Motion Technology Pty. Ltd Laverton North 3026, Australia E-mail: vicsales@fcrmotion.com	ID	PT Servitama Era Toolsindo Jakarta 13930, Indonesia E-mail: ptset@group.gts.co.id	SK	KOPRETINA TN s.r.o. 91101 Trenčín, Slovakia www.kopretina.sk
BA	Bosnia and Herzegovina → SL	IL	NEUMO VARGUS MARKETING LTD. Tel Aviv 61570, Israel E-mail: neumo@neumo-vargus.co.il	SL	Posredništvo HEIDENHAIN NAVO d.o.o. 2000 Maribor, Slovenia www.heidenhain-hubl.si
BE	HEIDENHAIN NV/SA 1760 Roosdaal, Belgium www.heidenhain.be	IN	HEIDENHAIN Optics & Electronics India Private Limited Chennai – 600 031, India www.heidenhain.in	TH	HEIDENHAIN (THAILAND) LTD Bangkok 10250, Thailand www.heidenhain.co.th
BG	ESD Bulgaria Ltd. Sofia 1172, Bulgaria www.esd.bg	IT	HEIDENHAIN ITALIANA S.r.l. 20128 Milano, Italy www.heidenhain.it	TR	T&M Mühendislik San. ve Tic. LTD. ŞTİ. 34728 Ümraniye-Istanbul, Turkey www.heidenhain.com.tr
BR	DIADUR Indústria e Comércio Ltda. 04763-070 – São Paulo – SP, Brazil www.heidenhain.com.br	JP	HEIDENHAIN K.K. Tokyo 194-0215, Japan www.heidenhain.co.jp	TW	HEIDENHAIN Co., Ltd. Taichung 40768, Taiwan R.O.C. www.heidenhain.com.tw
BY	Belarus GERTNER Service GmbH 50354 Huerth, Germany www.gertner.biz	KR	HEIDENHAIN Korea LTD. Gasam-Dong, Seoul, Korea 153-782 www.heidenhain.co.kr	UA	Gertner Service GmbH Büro Kiev 01133 Kiev, Ukraine www.gertner.biz
CA	HEIDENHAIN CORPORATION Mississauga, Ontario L5T2N2, Canada www.heidenhain.com	ME	Montenegro → SL	US	HEIDENHAIN CORPORATION Schaumburg, IL 60173-5337, USA www.heidenhain.com
CH	HEIDENHAIN (SCHWEIZ) AG 8603 Schwerzenbach, Switzerland www.heidenhain.ch	MK	Macedonia → BG	VE	Maquinaria Diekmann S.A. Caracas, 1040-A, Venezuela E-mail: purchase@diekmann.com.ve
CN	DR. JOHANNES HEIDENHAIN (CHINA) Co., Ltd. Beijing 101312, China www.heidenhain.com.cn	MX	HEIDENHAIN CORPORATION MEXICO 20235 Aguascalientes, Ags., Mexico E-mail: info@heidenhain.com	VN	AMS Advanced Manufacturing Solutions Pte Ltd HCM City, Việt Nam E-mail: davidgoh@amsvn.com
CZ	HEIDENHAIN s.r.o. 102 00 Praha 10, Czech Republic www.heidenhain.cz	MY	ISOSERVE Sdn. Bhd 56100 Kuala Lumpur, Malaysia E-mail: isoserve@po.jaring.my	ZA	MAFEMA SALES SERVICES C.C. Midrand 1685, South Africa www.heidenhain.co.za
DK	TPTEKNIKA/S 2670 Greve, Denmark www.tp-gruppen.dk	NL	HEIDENHAIN NEDERLAND B.V. 6716 BM Ede, Netherlands www.heidenhain.nl		
		NO	HEIDENHAIN Scandinavia AB 7300 Orkanger, Norway www.heidenhain.no		

